Reproducing the LHS 1140 b helium line with a solved wind: how far EXHALE gets, and what is missing

Updated 2026-10-04: the current answer is in Sects. 1.–3.. They report the LHS 1140 b equivalent-width campaign of 2026-10-03/04 (`md/PLAN_20261003_lhs1140b_ew.md`, sections 8 and 9; `md/Update_EXHALE_stage3.md` sections 63–92). Every EXHALE state, equivalent width, crossing and escape rate in Sects. 4., 5., 9., 10., 11., 12., 13., 14., 15., 16. and 17. is of the physics of 2026-09-16 and is superseded. That text is kept unchanged as the record, and each of those sections opens with a note saying what in it no longer holds. Four conclusions changed sign. (1) On the photochemical column the equivalent width rose with He/H in 2026-09 (1.4765 % Å at $\mathrm{He/H} = 2.09$ to 2.2289 at 12.01, unbroadened); it now falls as He/H rises. (2) At $\mathrm{He/H} = 2.09$ the line was 33 per cent too strong and is now 34 per cent too weak (0.7334, unbroadened), so where every 2026-09 solution overshot the measurement, no He/H now reaches it. (3) The line-forming gas was collisional ($\mathrm{Kn} \leq 0.012$ at 1.1–3 R_p) and the critical point lay outside the domain; now 31–40 per cent of the equivalent width forms beyond $\mathrm{Kn_{bulk}} = 0.1$ and the sonic point lies inside the domain. (4) The solution's own composition was unusable above $4\,R_p$ (a factor 13 between its equivalent width and the advection-corrected one); with the ionization stages transported the two agree to 1 per cent. What still stands: the width demand, $f \approx 22\,\mathrm{km\,s^{-1}}$, and the convention of Sect. 12.2 that measures it, which is now the adopted rule for every model line compared with an observation; the reproduction of the paper's p-winds model; the literature sections; the inputs (Sect. 6.).

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $\mathrm{H^+ + H}$ and $\mathrm{He^+ + He}$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn(H_2)}$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

Earlier banner (2026-09-25, kept as the record). Stale since 2026-09-24. Every EXHALE state, equivalent width and composition in this memo is of the physics before the atomic-data audit of 2026-09-24 (`md/Update_EXHALE_stage3.md` sections 17 and 18). The largest correction for this line is the $2^3\mathrm{S} \rightarrow 1^1\mathrm{S}$ electron-impact de-excitation, the detailed-balance reverse of the excitation the network carried, which the metastable network had left out: it lowers the He 10830 equivalent width of the certified molecular states by about 20 per cent after a re-solve, so the compositions this memo reads from the line are low. The p-winds side is unaffected. The EXHALE side is being re-solved, the molecular $K_{zz} = 10^9$ family first.

1. Goal, and the answer of 2026-10

STALE (2026-10-07). Every energy-coupled molecular state of this section and of section 1.'s sequels (the He/H scan, the EUV match of Figs. 5–7, the ascent to $\mathrm{He/H} = 10^5$) was computed with a code defect found on 2026-10-07: with the oxygen chemistry off, the transported H_2 row kept the loss $O + H_2 \rightarrow OH + H$ without its production, reverse or reaction heat (`diffusive_photochemistry.f90`, `oxygen_carrier_rows` called without the oxygen-chemistry guard). It made 72–95% of the H_2 destruction, and the resulting $H + H + M$ recombination heat was 63–69% of the wind's heating (`md/AN_matching_surface_diagnosis_20261007.md`). The numbers below are kept as the record of those runs and are being recomputed with the corrected code; they are not results.

The question is whether a wind that is solved, rather than assumed, reproduces the measured He I 10830 Å transit line of LHS 1140 b (Cherubim et al. 2026, hereafter C26). The model is EXHALE, a one-dimensional calculation of photoionization-driven escape in which the temperature, the velocity and the composition come out of the solution. The measured quantity is the window equivalent width, 1.1084 ± 0.0289 % Å (vacuum 10832.60–10834.20 Å, the observation's samples, trapezoid weights; quoted as 1.108 ± 0.030 in the older sections).

Answer. With the present equation set no helium abundance reproduces the line. The width-matched equivalent width rises as He/H falls, but it stays below the measurement: the largest converged model, $\mathrm{He/H} = 0.65$, gives 0.8529 % Å, 8.8σ below. Fits over the four models with $\mathrm{He/H} \leq 1.2$, continued to $\mathrm{He/H} = 0$, give 0.877 (linear) and 0.905 (quadratic). These are extrapolations, and they are upper estimates of the maximum only if the slope does not steepen below 0.65 (an assumption, since the equivalent width must vanish at $\mathrm{He/H} = 0$). Even the bound stays 7–8σ below the measurement. There is no crossing in He/H. At $\mathrm{He/H} = 0.65$ the deficit survives every numerical and physical item that was measured (Table 2): all the items that raise the equivalent width together reach at most 0.870, 8.2σ below. The velocity spread that the line asks for is unchanged from 2026-09: $f = 21.9$–$22.3\,\mathrm{km\,s^{-1}}$ FWHM on every state.

The σ counts here use the independent-sample error of the window equivalent width; the released spectrum does not carry the covariance or the continuum-fit uncertainty that a complete significance would need. The missing amount is 0.2555 % Å, 30 per cent of the $\mathrm{He/H} = 0.65$ value. As a scale of the missing opacity, and not as a model, a uniform factor 1.30 on the stored $n(2\,^3S)$ of that state, with every other field held fixed, gives a width-matched equivalent width of 1.0957, inside the observed interval (factor 1.50: 1.2545; `md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md` section 1, update log section 93). Those scaled profiles are not atmospheric solutions.

All numbers in Sects. 1.–3. are MEASURED and read from the run records named with them unless marked READ (taken from a publication or dataset) or DERIVED. Equivalent widths are in % Å. Every wind and equivalent-width verdict of phases 5 and 6 was made with the binary `LHS1140b/models/.P2/EXHALE_99a76cb0.x` (md5 `99a76cb05e56ad3a6ee80e6c433814fd`).

1.1 The equation set

The states of the campaign carry the following physics (the `input.inp` and the coupling line of the accepted state; plan `md/PLAN_20261003_lhs1140b_ew.md`, section 9):

- Hydrodynamics: HLLC fluxes, the well-balanced reconstruction, the stationary residual assembled with WENO3, the JFNK stationary solve, a spherical domain of 500 cells with its outer face at $29.27\,R_p$ (decision 3

of the plan).

- Heat conduction with the conductivity of the gas mixture (`md/Update_EXHALE_stage3.md` section 41), the bath below the matching level at the profile's temperature (section 70), and the conducted flux written into the lower atmosphere (section 76).
- The lower atmosphere: a Photochem column handed over as a profile at 1 μbar, closed with the wind on the elemental fluxes of H and He (tolerance 0.05) and coupled in energy (the column's energy balance under the conducted flux, matching-level temperature tolerance 6 K; sections 76–80). Since update log section 93 the acceptance also holds the conducted heat flux to the same tolerance as the elemental fluxes; every stored converged closure meets it (0.0001–0.0375), so no verdict in this memo changes. The column balance uses the published vibrational-translational rates, the NH_3 ν_2 + H_2 rate from the CD_4 bending analog (no measurement exists), and the band transfer `cooling_to_space`, which includes the absorption by the overlying column (Dickinson 1972, Eq. 33, as cited in section 89).
- Ionization transport of the H and He stages, with their enthalpy flux (sections 67 and 72), and He($2\,^3S$) transport, the metastable carried as a fourth helium state on the helium nucleus flux (section 81).
- Molecular chemistry (H_2, H_2^+, H_3^+, HeH^+) with H_2 carrier transport; He/H element diffusion with K_{zz} from the profile; the element and carrier enthalpy fluxes; `Secondary_ionization: Immediate`.
- The stellar spectrum `lhs1140_sed_gj1132_at_b.txt`, the GJ 1132 proxy of C26.
- The line: `EXHALE_transit.py` at $R = 68{,}000$, and the width-matching Gaussian kernel of Sect. 12.2 (section 87). The kernel's FWHM f is solved for each model so that the three-Gaussian extractor returns the observed red-pair FWHM, 0.841 Å; the equivalent width is then the window equivalent width of the broadened curve. The kernel stands for a velocity field that the one-dimensional model does not contain. It is not a computed physics term, and every result that uses it reports f.

1.2 The He/H scan

Each He/H was solved as a full energy-coupled closure (climate, chemistry, wind and spectrum), started by continuation in He/H from a neighboring certified state. Table 1 and Fig. 1 give the result (run notes `phase5/NOTES.md` under `LHS1140b/models/.P2/ew_fit/`, sections 2, 7 and 8, `phase5/lowHeH_numbers.txt`; update log sections 86, 89 and 90).

Stale (2026-10-07). The Knudsen numbers, $Kn_{bulk} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $Kn_{bulk} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $Kn(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

The equivalent width rises as He/H falls, by shrinking steps: 0.07–0.21 per unit He/H between 2.09 and 1.2, and 0.04–0.06 per unit He/H below 1.2. Reaching 1.108 would need about 0.39 per unit He/H below 0.65, seven to ten times the measured slope. Three forms with EW(0) = 0 and a maximum fit the four low models with an rms of 0.008–

Table 1: The He/H scan of 2026-10 (deep He/H; observed 1.1084 ± 0.0289). EW_f: width-matched window EW of the solution product, with its kernel FWHM f; EW_0: the same without the kernel. T_{match}: the temperature at the 1 μbar matching level; F_{cond}: the heat conducted into the column [erg cm^{-2} s^{-1}]; $\dot{M}$ at cell $N-20$; r_{Kn}: the radius where $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$; r_{s}: the sonic point. The states at 2.09 and 1.8 carry the optically thin column treatment; the cooling-to-space transfer would raise their T_{match} by 2.05 and 2.40 K (update log section 90). The rows in brackets are not converged closures and are diagnostic.

He/H	EW_f	f [km s^{-1}]	EW_0	$\log_{10}\dot{M}$ [g s^{-1}]	T_{match} [K]	F_{cond}	r_{Kn} [R_p]	r_{s} [R_p]
(3.0, k01)	(0.6746)	(21.92)	(0.6977)	(7.8246)	(350.9)	(6.70)	(5.06)	(24.54)
2.09	0.7091	22.02	0.7334	7.8854	349.88	9.63	5.69	25.69
1.8	0.7710	22.05	0.7975	7.9359	359.55	11.65	6.11	26.26
1.5	0.8050	22.09	0.8327	7.9770	375.98	13.87	6.55	26.83
1.2	0.8308	22.14	0.8593	8.0218	386.37	17.16	7.08	27.45
1.0	0.8368	22.17	0.8655	8.0531	399.39	19.82	7.50	27.89
0.8	0.8444	22.21	0.8734	8.0930	410.53	22.95	8.04	28.40
0.65	0.8529	22.24	0.8822	8.1325	412.59	25.87	8.57	28.79
(0.5, k00)	(0.8482)	(22.28)	(0.8773)	(8.179)	(398.51)	(29.51)	(9.24)	none

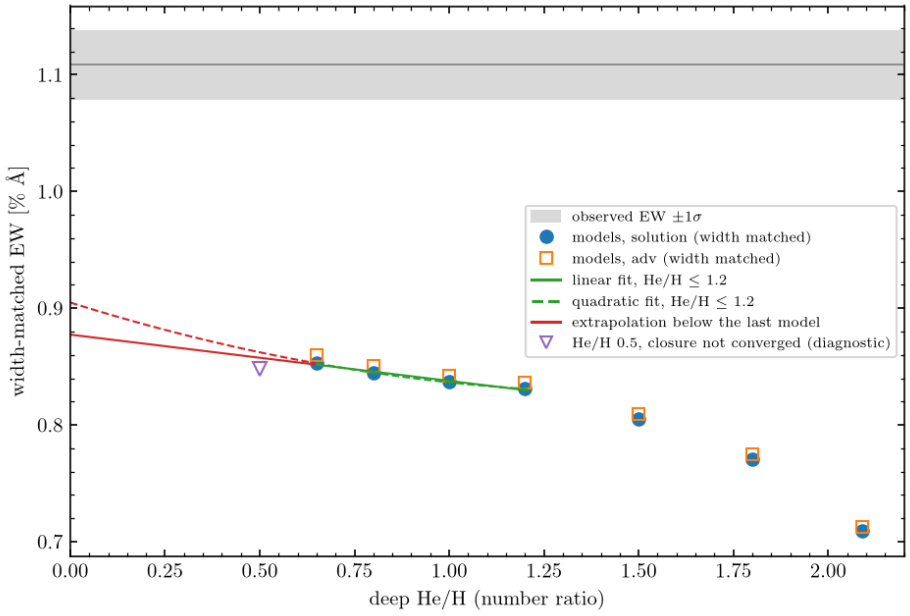


Figure 1: Width-matched equivalent width against the deep He/H, solution (filled) and advection-corrected (open) products, with the linear and quadratic fits over $\mathrm{He/H} \leq 1.2$ and their continuation to $\mathrm{He/H} = 0$ (extrapolation, red). The triangle is the unconverged $\mathrm{He/H} = 0.5$ closure. Grey band: the observed equivalent width and its 1σ interval. Source: `phase5/fig_ew_vs_heh.png`.

0.010, five times their scatter: the models show no turnover down to He/H = 0.65. The advection-corrected product sits 0.003–0.007 above the solution product throughout.

He/H = 0.5 did not converge in four attempts, all kept. The last two stopped after the full 60 outer passes on the He(2^3S) carrier balance at the first gated cell ($r = 1.248\,R_p$; 1.1×10^{-4} and 1.6×10^{-4} against 10^{-5}), and no tolerance was loosened. Its first closure iterate gives 0.8482 and has no sonic point inside the domain: the sonic point moves outward as He/H falls and leaves the $30\,R_p$ domain between 0.65 and 0.5. He/H = 3.0 stopped at its second iterate unconverged (the evaluate of that time refused the mass row at its rounding floor, the defect fixed in update log section 88) and has not been resumed.

Along the scan the escape rate rises by 0.25 dex and the heat conducted into the column by a factor 2.7, which warms the matching level from 350 to 413 K. Lower He/H also moves the $\mathrm{Kn_{bulk}} = 0.1$ radius outward, from 5.69 to $8.57\,R_p$. The column's optically thin treatment is past its validity limit at every He/H (the absorbed share of the upward band emission is 0.279–0.377 against the limit 0.10 on He/H = 1.2–0.65; update log section 90), which is why the cooling-to-space transfer is used. What that transfer changes is small: +4.4 to +6.2 K in T_{match} at He/H = 1.0–0.65, and 0.0009 in the width-matched equivalent width at 1.5.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

1.3 The line against the observation

Figure 2 sets the He/H = 0.65 line against the released C26 spectrum (`phase6/NOTES.md` section 2; update log section 92). Measured with the three-Gaussian extractor used on the data, the width-matched model ($f = 22.24\,\mathrm{km\,s^{-1}}$) has red-pair depth 0.985 against the observed 1.254 %, blue depth 0.128 against 0.198 %, and red/blue 7.67 against 6.35; its FWHM equals the observed 0.841 Å by construction. Without the kernel the red pair is 3.306 % deep and 0.251 Å wide. The window χ^2 is 136 for 44 samples, against 1962 for a zero model. The deficit sits in the red pair: the mean residual is 0.224 ± 0.025 % over 10832.9–10833.7 Å and 0.051 ± 0.031 % over the blue component. In 2026-09 the same convolution put depth, blue depth and red/blue inside the observed 1σ interval at once, but there the equivalent width had been matched by choosing He/H. Now, with the equivalent width short, the shape is right and the line is 21 per cent too shallow.

The kernel width is the same on every state, 21.92–22.28 $\mathrm{km\,s^{-1}}$ (Table 1; 22.22 on the 2026-09 solutions). Broadening lowers the window equivalent width by 3.3 per cent, because the wings carry absorption out of the window, so the kernel moves every model further below the measurement (`phase5/nonthermal_kernel/NOTES.md`; update log section 87).

The rest frame. The equivalent-width operator assumes that the observation is in the planet rest frame and that the model carries no line-of-sight velocity. The model line has none ($-0.04\,\mathrm{km\,s^{-1}}$; the spherical outflow is symmetric). The observed line is redshifted. The three-Gaussian fit gives $+2.85\,\mathrm{km\,s^{-1}}$, and fitting the model shape in shift gives $+2.24\,\mathrm{km\,s^{-1}}$ under independent errors. The published fit with correlated noise gives $2.0^{+2.0}_{-2.2}\,\mathrm{km\,s^{-1}}$ (READ, the dataset README), consistent with zero at 1σ. The window is not symmetric about the red pair, so a shift moves the equivalent

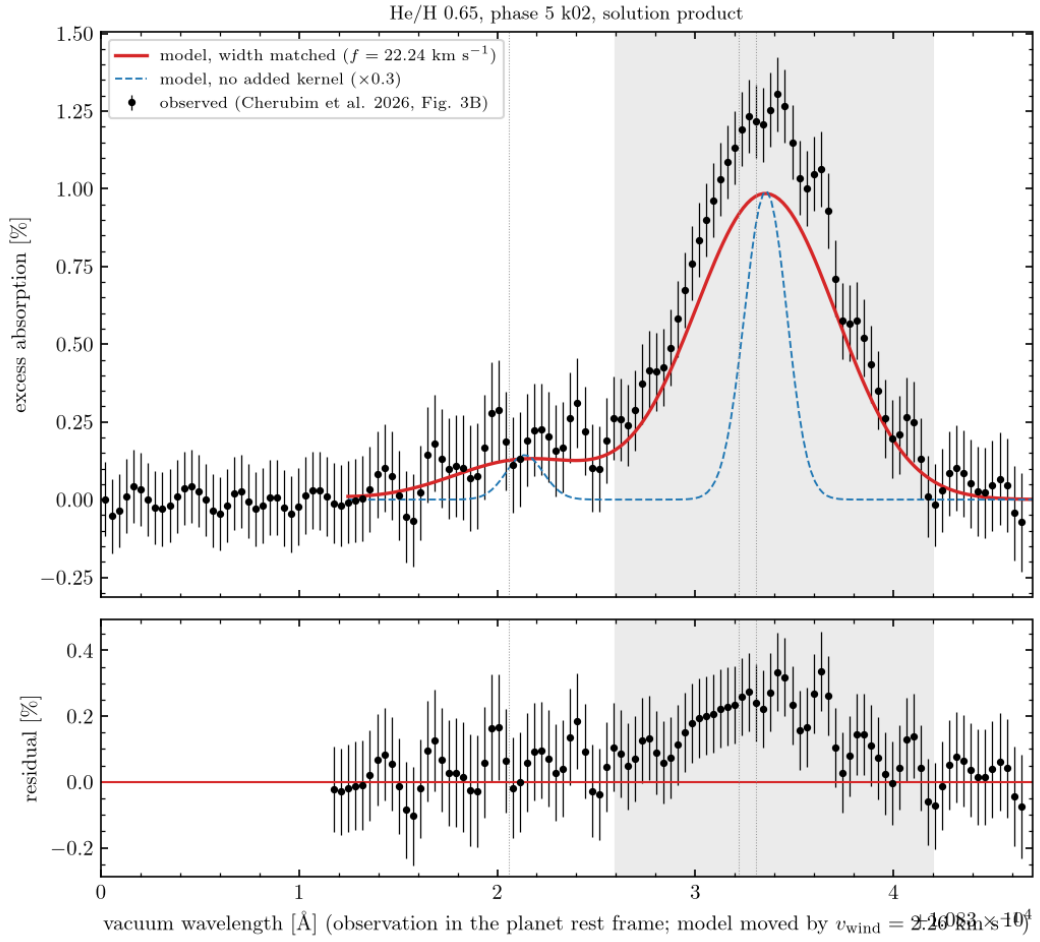


Figure 2: The He/H = 0.65 solution line, width matched ($f = 22.24\,\mathrm{km\,s^{-1}}$, red) and without the kernel (blue dashed, scaled by 0.3), against the C26 spectrum (points; vacuum wavelength, planet rest frame). As in every earlier figure of this memo, the model curves are moved by the fitted $v_{\rm wind} = +2.26\,\mathrm{km\,s^{-1}}$ for the overlay only (the model carries no line-of-sight bulk velocity); every number is computed from the unshifted spectra. Grey: the equivalent-width window. Lower panel: observed minus the moved model. Source: `phase6/observation/HeH0.65_k02_solution.png`.

width: the model moved by +0.081 to +0.104 Å gains 0.0115 to 0.0137. This is carried as a systematic in Table 2.

1.4 Robustness at He/H 0.65, and the limits of validity

Phase 6 tested the model with the largest equivalent width, against a numerical target of 0.0058 for each test (Table 2; `phase6/NOTES.md` sections 1 and 3; update log section 92). The reference is 0.852918.

Table 2: Changes of the width-matched equivalent width at He/H = 0.65 (% Å). Target 0.0058 for each test. The items are separate measurements and are not combined into one interval; any interval formed from them is conditional on their independence.

item	change	/ target
grid 500 → 1000 cells (certified)	–0.00726	1.25 (not met)
grid 1000 → 2000 cells (diagnostic)	–0.00404	0.70
grid 500 cells → continuum (Richardson, DERIVED, not established)	–0.0164	2.8
JFNK target $4 \times 10^{-6} \rightarrow 10^{-8}$	0	0
closure 0.05/6 K → 0.01/2 K	+0.00038	0.06
domain $30\,R_p$ → L1 ($37.6\,R_p$)	+0.00195	0.34
NH_3 ν_2 + H_2 rate, water analog / no H_2 term	–0.00037 / +0.00105	0.06 / 0.18
ion-neutral drift of He I, cases A / B	+0.000037 / +0.000034	0.006
tide, Roche-consistent flow and synthesis (diagnostic)	–0.0324	5.6
frame, observed +2.2–2.9 km s^{-1} shift	+0.0115 to +0.0137	2.0–2.4

- *Grid.* Halving every spacing (outer face kept) lowers the equivalent width by 0.00726, 1.25 times the target, so the grid test fails. A further halving gives –0.00404 (that state is not certified: the He($2\,^3S$) carrier row at the first gated cell stays at 4.7×10^{-5} after 300 passes, while the equivalent width does not move between pass 150 and 300). Taken at face value the three grids give an order of 0.85 and a Richardson continuum of 0.8366, 0.016 below the 500-cell value. Because the 2000-cell state, the solve and its continuation alike, ended uncertified (solver info 1 with exit status zero), the continuum value and the order are not established. Every refinement moves the equivalent width down, away from the observation. The error enters through the steep base layer, which sets $\dot{M}$: $\dot{M}$ falls by 0.8 and then 0.4 per cent with each halving.
- *Solver and closure.* The JFNK reaches its rounding floor ($\|R\| = 1.2$–1.6×10^{-8}) at every target, and the certification gates are compile-time, so tightening the target changes nothing. The tight closure converged at the fixed point $F_{\rm H} = 6.4803 \times 10^7$, $F_{\rm He} = 7.0914 \times 10^7$ g s^{-1}, $T_{\rm match} = 415.86$ K, $\log_{10}\dot{M} = 8.130$, with 25.83 erg cm^{-2} s^{-1} conducted into the column.
- *Domain and tide.* Extending the domain to L1 leaves the sonic point (28.80 against $28.79\,R_p$) and $\dot{M}$ unchanged. The Roche-consistent re-solve with a Roche synthesis gives 0.8205, so the geometric interval is $[0.8205, 0.8529]$; this value is diagnostic, because the evaluate of the reloaded Roche state refuses it on the mass row of cell 1 (1.05 times its rounding anchor).

- *Rates and drift.* At $\mathrm{He/H} = 0.65$ the equivalent width is nearly insensitive to T_{match} (+39 K moves it by +0.001), so the NH_3 $\nu_2 + H_2$ bracket is 15–40 times smaller than at 2.09. The ion-neutral drift, estimated at fixed background, changes $n(2\,^3S)$ by +0.1 per cent at 5–10 R_p and by −1 to −4 per cent at 20–29 R_p.

Validity limits. 31.4 per cent of the width-matched equivalent width forms beyond $\mathrm{Kn_{bulk}} = 0.1$ (8.57 R_p), 9.1 per cent beyond $R_*/R_p = 12.87$, and 0.01 per cent beyond the exobase (28.63 R_p), which lies just below the sonic point (28.79 R_p). The largest Kn in the heating and acceleration region is 1.3. A third of the line therefore forms where the continuum description is past Kn = 0.1. That is the dominant validity limit, and no test here removes it (at $\mathrm{He/H} = 2.09$ the share was 39.8 per cent of the unbroadened equivalent width, phase 4). The single-velocity stage transport neglects an ion-neutral drift that reaches 0.2–1.5 of the helium velocity beyond 5 R_p; its measured effect is in Table 2, but where the drift is comparable to the velocity it is itself outside the first Chapman–Enskog approximation. The column's optically thin treatment is flagged (absorbed share 0.377), and the exchange inside the warmed layer that the cooling-to-space transfer neglects moves T_{match} by 0.64 K. The NH_3 $\nu_2 + H_2$ rate is unmeasured and only bracketed. The tide is an interval, not resolved. The width kernel is not a computed term.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

1.5 What changed from the 2026-09 answer

On the 2026-09 physics the flux-consistent solution at $\mathrm{He/H} = 2.0924$ gave 1.4765 (unbroadened). The present state at 2.09 gives 0.7334, half of it. Between the two, the code gained the following, each recorded with its own measured effect:

- the $2\,^3S \rightarrow 1\,^1S$ electron-impact de-excitation of the 2026-09-24 atomic-data audit (about −20 per cent on the certified molecular states; the earlier banner, update log sections 17–18);
- heat conduction in the model. Without it, the $\mathrm{He/H} = 2.09$ state of that time gave 1.50 (update log section 66), and on three earlier fiducial cases conduction lowered the equivalent width to 0.18–0.60 of its value (plan, phase 5 item 1);
- the energy coupling to Photochem, which raises it again: 0.5793 → 0.7255 at 2.09 (update log section 79);
- carried ionization stages, and $\mathrm{He}(2\,^3S)$ transport (0.7255 → 0.7334; update log section 81).

The sign of $\mathrm{d\,EW/d(He/H)}$ reversed over the range where the two sequences overlap. Which of these changes reversed it was measured at the two ends of the present scan (Sect. 2., last paragraph): with heat conduction in the wind removed the equivalent width falls again as He/H falls. Section 2. measures what sets the present sign. The high-He/H side (> 2.09) has not been solved with the present equations (Sect. 3.).

2. Why the line strengthens as He/H falls

The explanation put forward first was that lower He/H gives a denser, cooler outer wind with more electrons and more He^+, and that $n(2\,^3S)$ follows $n_e\,n(He^+)$. It was tested by decomposing the change of the width-matched equivalent width between the four certified states He/H = 2.09, 1.8, 1.2 and 0.65 (`phase5/dEW_decomposition/NOTES.md`; update log section 91). The stored states were only read.

2.1 Method

The He I 10830 branch of `EXHALE_transit.py` was restated as a function of (r, T, v, n_3, R_0), with $n_3 \equiv n(2\,^3S)$. On the four stored states it reproduces the stored instrument-convolved curve to 4.4×10^{-16}, and the equivalent width, the width-matched equivalent width and f of the records. On three profiles with substituted fields, `EXHALE_transit.py` itself agrees with it to 2×10^{-14}. The kernel FWHM f is solved again for every synthesized profile (22.02–22.24 km s^{-1}). The metastable density is written as $n_3 = C\,n_{3,\mathrm{loc}}$, where $n_{3,\mathrm{loc}}$ is the local root of the code's own level balance (the `tr_triplet_row` terms plus the H_2 sink), evaluated with the code's rate functions from `Cool_coeff.f90`, and C carries the transport. $C = 0.9955$–1.0072 over 1.2–8 R_p and up to 1.18 above 8 R_p. The $2\,^3S$ photoionization rate P_3, which the code does not write, was measured from the carrier-row terms of each state; it is the same in the four states to 0.3 per cent. The level is fed by recombination alone: the collisional feed from $1\,^1S$ is below 10^{-14} of the gain wherever n_3 matters.

The change is attributed with Shapley values over nine factor fields, radial profiles swapped whole between two states: n_H (the density), the local $y = n_\mathrm{He}/n_\mathrm{H}$, the helium ionization fractions, the hydrogen fractions, n_e, T (rates and thermal width), v, the transport (C and P_3), and R_0. That takes 512 syntheses for each pair of states. The values do not depend on the order of substitution and sum to the change exactly (residual 3×10^{-16} on every pair). A second Shapley decomposition runs over the terms of the balance,

$$n_3 = C\,\frac{n_e\,n_\mathrm{HeII}\,\alpha_\mathrm{eff}(T)}{L_P + L_e + L_r}, \tag{1}$$

with $L_P = n_\mathrm{HI}\,Q_{31}$ the destruction by neutral hydrogen ($He(2\,^3S)$ + H → He + H^+ + e, and the associative channel to HeH^+; called Penning below), L_e the electron collisions, and L_r the sum of A_{31}, P_3 and the H_2 sink. The factors interact: single substitutions "alone from A" and "last into B" differ by up to 0.55 for y. That is why the Shapley values, not single substitutions, carry the attribution, and why the radial split in Fig. 3 (right) is a first-order one.

2.2 Contributions of the gas fields

The change is a small difference of large terms (Table 3, Fig. 3). Between 2.09 and 0.65 the denser gas, its electrons and its lower temperature add +0.92. Two things remove −0.77: the lower helium abundance of the flow, since diffusion lowers y at the top of the domain from 0.59 to 0.28, and the lower He^+ share of the helium. By radius (Shapley over zones) the rise comes from beyond 6 R_p (+0.177). Inside 3.5 R_p the line weakens (−0.059): the n_3 peak falls from 52.5 cm^{-3} at 1.40 R_p to 22.5 cm^{-3} at 1.67 R_p. Absorbers inside 6 R_p give 63 per cent of the width-matched equivalent width at He/H = 2.09 and 48 per cent at 0.65.

Table 3: Shapley contributions to the change of the width-matched equivalent width (% Å) between the deep He/H values named in the header (`phase5/dEW_decomposition/dEW_table.txt`).

factor	2.09 → 1.8	1.8 → 1.2	1.2 → 0.65	2.09 → 0.65
n_{H} (density)	+0.0624	+0.1353	+0.1510	+0.3524
$y = n_{\mathrm{He}}/n_{\mathrm{H}}$, local	−0.0566	−0.2223	−0.3428	−0.6261
He ionization fractions	−0.0257	−0.0489	−0.0652	−0.1406
H fractions	−0.0005	−0.0016	−0.0025	−0.0046
n_e	+0.0645	+0.1273	+0.1761	+0.3687
T	+0.0189	+0.0708	+0.1030	+0.1946
v	−0.0000	−0.0001	−0.0001	−0.0002
transport (C, P_3)	−0.0019	−0.0040	−0.0043	−0.0112
R_0	+0.0009	+0.0032	+0.0069	+0.0109
ΔEW	+0.0619	+0.0598	+0.0221	+0.1438

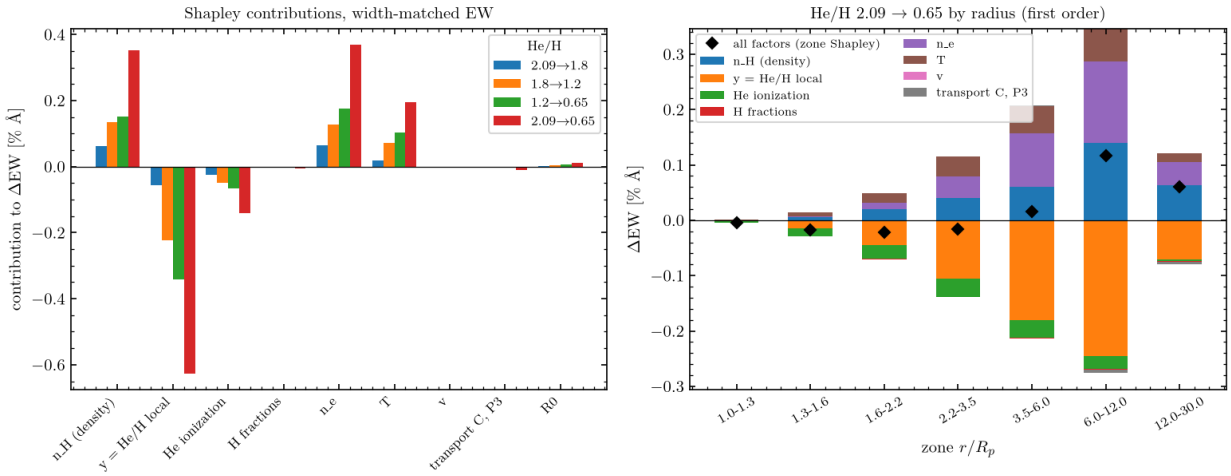


Figure 3: Left: Shapley contributions of the nine factor fields to the change of the width-matched equivalent width, for the three adjacent pairs and for 2.09 → 0.65. Right: 2.09 → 0.65 split by radial zone, first order for each factor (bars), with the zone Shapley value of all factors together (diamonds). Source: `phase5/dEW_decomposition/fig_dEW_factors.png`.

2.3 Does $n(2^3S)$ follow $n_e\, n(\mathrm{He}^+)$?

Only in part (Table 4, Fig. 4). The product $n_e\, n_{\mathrm{HeII}}$ is the largest positive term, +0.703 from 2.09 to 0.65. The Penning loss by neutral hydrogen cancels 95 per cent of it (−0.666). Over 1.5–8 R_p, where most of the line forms, He(2^3S) + H takes 50–70 per cent of the loss rate at He/H = 2.09 and 73–93 per cent at 0.65; A_{31} carries more than half of the loss only above 9.9 R_p (2.09) to 13.8 R_p (0.65). At fixed r, between 2.09 and 0.65, $n_e\, n_{\mathrm{HeII}}$ rises by 1.2–2.9 times but n_{HI} by 3.3–4.6 times. So $n_3/(n_e\, n_{\mathrm{HeII}})$ at 0.65 is 0.41–0.51 of its 2.09 value over 2–8 R_p and reaches it only above about 20 R_p.

Table 4: Shapley contributions of the terms of Eq. (1) to the change of the width-matched equivalent width (% Å).

term	2.09 → 1.8	1.8 → 1.2	1.2 → 0.65	2.09 → 0.65
$n_e\, n_{\mathrm{HeII}}$ (recombination product)	+0.1475	+0.2444	+0.2920	+0.7034
$\alpha_{\mathrm{eff}}(T)$	+0.0137	+0.0538	+0.0844	+0.1557
$L_P = n_{\mathrm{HI}}\, Q_{31}$ (Penning)	−0.0850	−0.2229	−0.3388	−0.6656
L_e (electron collisions)	−0.0129	−0.0133	−0.0160	−0.0448
L_r (A_{31}, P_3, H_2) and C	−0.0019	−0.0040	−0.0042	−0.0109
T in the thermal width	−0.0004	−0.0014	−0.0021	−0.0050
v	−0.0000	−0.0001	−0.0001	−0.0002
R_0	+0.0009	+0.0032	+0.0069	+0.0113
ΔEW	+0.0619	+0.0598	+0.0221	+0.1438

The net rise, +0.144, is therefore carried by two things. One is the cooler gas, whose larger recombination coefficient gives +0.156. The other is the outer wind beyond about 10–14 R_p, where A_{31}, which does not grow with the density, replaces the hydrogen loss. The velocity contributes nothing (−0.0002) and the thermal width −0.005. The first explanation is right that lower He/H gives a denser, cooler wind with a larger $n_e\, n_{\mathrm{HeII}}$. It is not right that n_3 tracks that product in the line-forming region (2–12 R_p), because the same denser, more neutral hydrogen quenches the level.

2.4 What sets the density and $\dot{M}$

From 2.09 to 0.65, $\dot{M}$ rises from 7.68×10^7 to 1.36×10^8 g s^{-1} (`mdot_drivers.txt`). In all four states the net heating minus the conducted loss equals the energy the flow carries out to 0.04–0.62 per cent, at 1.95–1.98 $\times 10^{12}$ erg g^{-1} against GM_p/R_0 = 1.91–1.92 $\times 10^{12}$. $\dot{M}$ is therefore set by the heat that is not conducted down. The heat deposited below 1.02 R_p, mostly the molecular-chemistry channel, is close to the conducted loss (1.57 against 1.63×10^{20} erg s^{-1} at 2.09; 4.63 against 4.43×10^{20} at 0.65). The conducted loss rises by a factor 2.7 but does not drive the wind. The heat deposited above 1.02 R_p rises by a factor 1.41, and $\dot{M}$ for each unit of it by 1.26, so $\dot{M}$ rises by 1.77. The heating rate per gram in the wind is nearly unchanged. The heat is deposited higher instead: the radius that halves it moves from 1.35 to 1.53 R_p, the mean mass per particle at $T_{\max}$ falls from 2.62 to 1.91 m_{H}, and the scale height there rises from 0.174 to 0.215 R_p. Beyond 2 R_p the mass density is 2.0–2.2 times larger at 0.65 than at 2.09.

Not verified (interpretation). The chain "lighter gas, larger scale height, heat deposited higher over more mass, larger $\dot{M}$ at the same energy per gram" fits every number above, but it is an interpretation. The code does not write the absorbed stellar XUV power, so it was not measured whether the total absorbed power changes or only its radial distribution,

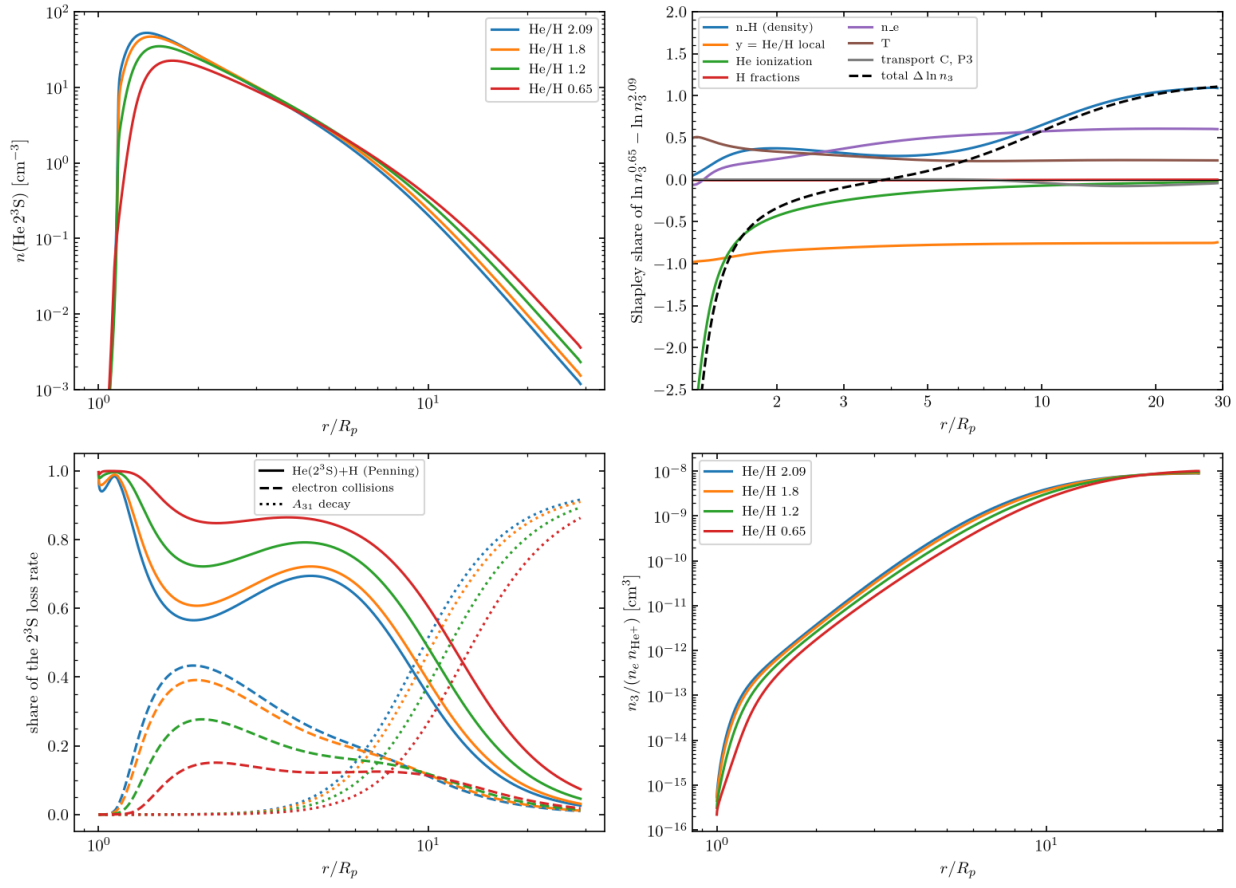


Figure 4: The He($2\,^3S$) balance of the four states. Top left: n_3. Top right: Shapley split of $\ln n_3^{0.65} - \ln n_3^{2.09}$ at each radius. Bottom left: share of the $2\,^3S$ loss rate taken by He($2\,^3S$) + H (solid), electron collisions (dashed) and A_{31} (dotted). Bottom right: $n_3/(n_e\, n_{\mathrm{He}^+})$. Source: `phase5/dEW_decomposition/fig_n3_balance.png`.

and no solve that changes the mean molecular weight alone was run. The rate coefficients of the decomposition were evaluated from the working tree, not from the build that solved the states; that they agree is inferred from the smooth, state-independent measured P_3.

Which physics makes $\dot{M}$ depend on composition. Measured by solving both He/H with one term removed at a time, everything else held (`mdot_attribution/NOTES.md`; update log section 100): B, the uncoupled column (no conducted heat into it); C, no heat conduction in the wind; D, conduction on without the element, carrier and stage enthalpy fluxes. $\dot{M}(0.65)/\dot{M}(2.09)$ is 1.76 (full), 2.11 (B), 1.31 (C) and 1.99 (D). Heat conduction in the wind is what makes $\dot{M}$ depend strongly on composition: it lowers $\log_{10}\dot{M}$ by 0.247 at 2.09 and by 0.038 at 0.65, and $\dot{M}$ follows the net heat of the gas above $1.02\,R_p$ (B/C 0.59 at 2.09, 0.93 at 0.65), which at 2.09 loses heating and gains H_3^+ infrared cooling in a cool layer the conduction keeps above $1.02\,R_p$. The energy coupling works the other way (−0.079 dex on the ratio) and the enthalpy fluxes add +0.026 dex. Without conduction the width-matched equivalent width falls as He/H falls (1.351 → 1.076, ΔEW = −0.275): no factor of the decomposition changes sign, but the helium-abundance term doubles (−1.14 against −0.47) and the electron term halves (+0.20 against +0.42), because the 2.09 wind then carries more helium (escaping He/H 0.73 against 0.50). *Not verified:* the winds without conduction stay subsonic inside the $30\,R_p$ domain (Mach at most 0.68), so their $\dot{M}$ depends on the outer boundary, which was held fixed; the remainder 1.31 without conduction is consistent with the lighter-gas chain above but was not isolated, and the cool-layer mechanism is an interpretation of the measured budgets.

3. Open points of the 2026-10 campaign

1. *NH_3 and CH_4 in the overlap region* (`md/NH3_CH4_ISSUES_20261004.md`; TO_BE_DONE (AI)). The region just above the 1 μbar matching level is described twice, and the two descriptions disagree. The column energy balance absorbs the band emission of the warmed layer in the Photochem NH_3 and CH_4 above the match: 86–93 per cent of the gas that absorbs that emission lies in the wind's domain (measured on the He/H = 1.8 profile), and 87–90 per cent of the thin absorbed share comes from it. The wind, which carries C, N and O as atoms, neither emits nor absorbs in those bands. A resolution must first decide whether NH_3 and CH_4 survive there at the wind's temperature and radiation field, and then use one description of the region for both codes. The problem grows toward low He/H. Three corrections to what had been said about it (review `md/LHS1140b_NH3_CH4_EW_REVIEW_20261004.md`; update log section 93). First, the molecular emission estimate quoted for the wind's lowest cells (5.5×10^{-6} to 3.8×10^{-5} erg cm^{-3} s^{-1}, phase 1) was computed on the older He/H = 9.0103 state and sums six molecules (CH_4, NH_3, H_2O, C_2H_2, CO, CO_2), with fixed abundances in optically thin LTE. It is neither an NH_3+CH_4 rate nor a measurement on a current state, and emission alone does not establish net cooling. Second, the lower-column tests (the NH_3 ν_2 + H_2 bracket of Table 2) move the equivalent width by at most 0.41 per cent of the missing amount; they say nothing about the cooling missing in the wind. Third, the sign of a wind response to NH_3/CH_4 cannot be assumed. Cooling changes the recombination coefficient, n_e, the helium ionization, the hydrogen quenching of $2\,^3S$, the scale height, the conduction and $\dot{M}$ together. Section 2. shows the competition: the cooler, denser wind of low He/H has a smaller inner n_3 peak. A

cooler profile does not by itself predict a larger equivalent width.

2. *The He(2^3S) carrier row at* $1.2R_p$. The carrier balance at the first gated cell decided every hard ending of the campaign: He/H = 0.5, the closure iterate at the tight tolerance, the first 1000-cell run, the 2000-cell run, and the first jump to 0.8. The accepted state passed it at 9.65×10^{-6} against 10^{-5}. The stall is not diagnosed, and no tolerance was loosened.

3. *Grid error.* The 500-cell scan fails the grid target (–0.00726, 1.25 times it), and every refinement widens the deficit. The 2000-cell solve and its continuation both ended uncertified, so the order (0.85 at face value) and the continuum value (0.8366, DERIVED) are not established. The scan was not re-solved on a finer grid.

4. *The Roche-mode evaluate* refuses the reloaded state that the solve certified (mass row of cell 1), so the Roche-consistent equivalent width is diagnostic. In spherical mode the evaluate reproduces the solve's verdict.

5. *Next routes* (`md/PLAN_20261004_lhs1140b_heh_high_and_xuv.md`). Route A: He/H above 2.09, first closing 3.0 and then 5, 10, 30, 100, to test whether the equivalent width passes a minimum and rises where the gas is mostly helium; the 2026-09 sequence rose there, but on other physics. Route B, step 1 (run; update log section 95, `LHS1140b/models/.P2/ew_fit/route_xuv/NOTES.md`): the GJ 699 proxy spectrum (F_{XUV} 169.93 against 33.45 erg cm^{-2} s^{-1} at the orbit) gives a width-matched equivalent width of 3.3–3.7 at every He/H tried (0.65–10), three times the measurement and weakly dependent on He/H; the measurement lies between the two proxies. Route B, step 2 (run; update log section 98, `LHS1140b/models/.P2/ew_fit/route_xuv_scaled/NOTES.md`): at He/H = 2.09, the GJ 1132 spectrum with its flux at $\lambda \leq 911.75$ Å (X-ray and hydrogen-ionizing EUV) multiplied by s and the rest by one factor that holds the bolometric flux, so that the He(2^3S) photoionizing band (911.75–2600 Å) keeps its shape. This is a sensitivity construction, not a stellar model. Every closure converged and certified (s = 1.3, 1.36, 1.4, 1.5, 2, 4). The width-matched equivalent width rises monotonically with s: 0.711 (s = 1), 1.045 (1.3), 1.107 (1.36), 1.147 (1.4), 1.244 (1.5), 1.670 (2), 2.792 (4), with $\log_{10}\dot{M}$ from 7.886 to 8.607 and f from 22.0 to 21.7 km s^{-1}. The measurement 1.108 ± 0.029 is matched for s = 1.33–1.39, i.e. F_{EUV}(100–911 Å) = 13.4–14.0 erg cm^{-2} s^{-1} at the orbit (crossing at s = 1.36, 13.7, which is 0.195 of the GJ 699 value 70.2; $\log_{10}\dot{M}$ = 8.065 there). The converged s = 1.36 state lies 0.0013 from the measurement; its width-matched red peak is 1.278 per cent against the observed 1.275 (expected once the equivalent width and the FWHM are both matched), its red/blue ratio 7.51, and its peak 0.15 Å blueward of the observed one, as at every s. The interval carries the observational 1σ only, not the grid error or the other model uncertainties, and has been computed at He/H = 2.09 only. A moderate rise of the EUV over the GJ 1132 reconstruction, well inside the factor of seven between the two proxies, appears sufficient to reproduce the measured equivalent width on the present equations (interpretation; a sensitivity result, not a measurement of the star). Route B, step 3 (run; update log section 99, the same NOTES, section 6): the same search at He/H = 0.65, 1.2 and 5, seeded from the converged GJ 1132 states of the same He/H (14 closures, all converged and certified, sonic point inside the domain in every state). The measurement is crossed at s = 1.300, 1.280, 1.361 and 1.336 for He/H = 0.65, 1.2, 2.09 and 5, i.e. F_{EUV}(100–911 Å) = 13.07, 12.86, 13.68 and 13.43 erg cm^{-2} s^{-1}, with 1σ intervals 12.71–13.43, 12.55–13.17, 13.40–13.97 and 13.16–13.70 (Fig. 5); every He/H has a converged state within 0.005 of the measurement. The four crossings scatter by ±3 per cent about 13.3, comparable to the ±2–3 per cent 1σ half-width of each, and

do not vary monotonically with He/H (lowest at 1.2). Along this line $\log_{10}\dot{M}$ falls from 8.24 to 8.00 and the peak temperature rises from 3630 to 6190 K as He/H rises from 0.65 to 5. The matched lines are nearly identical (Fig. 7): red/blue depth ratio 7.60, 7.56, 7.51, 7.46 against the observed 6.35 (same extractor; 16–84 percentiles 5.25–7.89 from a Monte Carlo of the tabulated errors treated as independent), and the red peak 0.14–0.15 Å blueward of the observed one at every He/H. Interpretation: on the present equations the measured equivalent width fixes F_{EUV}(100–911 Å) at about 13.3 ± 0.7 erg cm^{-2} s^{-1} at the orbit and leaves the deep He/H undetermined over 0.65–5; neither unfitted line property separates the He/H. Not computed for these states: the column checks, the grid error and the Roche-mode evaluate. (The route-B records of sections 95 and 98 quoted an observed red/blue of 7.6; the observed value from the extractor is 6.35, corrected in the records.) Route B, step 4 (run;

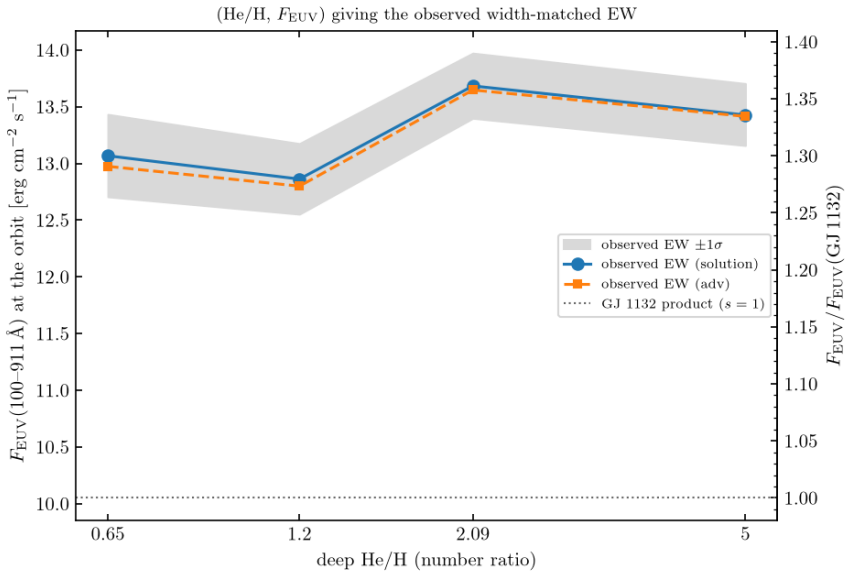


Figure 5: F_{EUV}(100–911 Å) at the orbit at which the width-matched equivalent width equals the measurement, against the deep He/H: the GJ 1132 spectrum with its flux at $\lambda \leq 911.75$ Å multiplied by s (bolometric flux held; a sensitivity construction, not a stellar model). Linear interpolation in $\ln s$ between bracketing converged states; solution (circles) and advection-corrected (squares) products; grey band, the 1σ of the measurement; dotted, the unscaled GJ 1132 product; right axis, the factor over the GJ 1132 value (s). Update log sections 98 and 99.

update log sections 102 and 108, NOTES section 7): the same search at He/H = 0.1, 0.2, 0.3, 0.45, 14, 100, 10^3, 10^4 and 10^5 (every closure converged and certified; a converged state within 0.0058 of the measurement at every He/H). The crossings are at $s = 2.532$, 1.656, 1.448, 1.339, 1.271, 1.331, 1.253, 1.143 and 1.015, i.e. F_{EUV}(100–911 Å) = 25.45, 16.65, 14.56, 13.46, 12.77, 13.38, 12.60, 11.49 and 10.20 erg cm^{-2} s^{-1} (Fig. 6). The required F_{EUV} exceeds the GJ 1132 value (10.05) at every He/H, by 1.5 per cent at 10^5, 14 per cent at 10^4 and by a factor 2.5 at He/H = 0.1; it is nearly flat (12.6–13.7) over He/H = 0.45–10^3, rises steeply below 0.45 (the $s = 1$ equivalent width falls below He/H = 0.65, so the same EUV gives less line) and falls above 10^3 (where the $s = 1$ equivalent width has turned up, update log section 101; at 10^5 the unscaled GJ 1132 spectrum is within 2 per cent of the measured equivalent width). The 1σ intervals carry the observed error only. At He/H = 0.1

the cooling-to-space column transfer reached its stated validity bound (the exchange it neglects shifts $T_{\rm match}$ by 1.003 K against 1 K); the transfer option that solves that exchange was added (update log section 102) and the He/H = 0.1 closure runs with it. The ascent above 10^4 (2×10^4 to 10^5 at $s = 1.14$, then an s scan at 10^5) was run with the binary that carries the corrected reservoir composition of the lower ghost cells (update log section 105); the equivalent widths at $s = 1.14$ are 1.129, 1.161, 1.208 and 1.243 at He/H = 2×10^4, 4×10^4, 7×10^4 and 10^5.

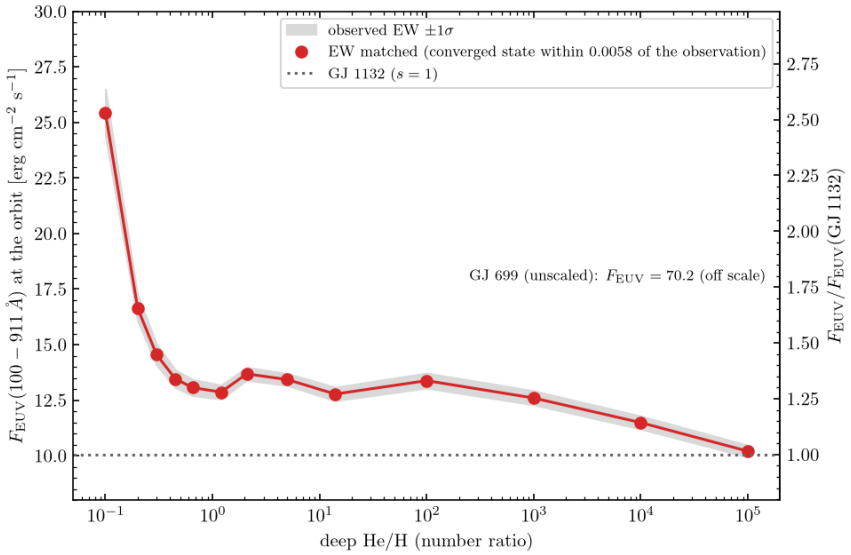


Figure 6: As Fig. 5 for deep He/H = 0.1 to 10^5 (thirteen He/H; solution product; a point is an open circle only if no converged state lies within 0.0058 of the measurement): F_{EUV}(100–911 Å) at which the width-matched equivalent width equals the measurement; grey band, the 1σ of the measurement; dotted, the unscaled GJ 1132 product; right axis, the factor over the GJ 1132 value (s). GJ 699 (unscaled) has 70.2, off the scale. Update log sections 102 and 108.

6. *The region beyond* $\mathrm{Kn_{bulk}} = 0.1$, which carries a third of the line, needs a kinetic treatment that is outside the plan.

4. The answer of 2026-09-16 (superseded)

Note added 2026-10-04. This section is the answer as of 2026-09-16 and is superseded by Sect. 1.. Its EXHALE numbers predate heat conduction in the model, the energy coupling to Photochem, the carried ionization stages, He(2^3S) transport and the 2026-09-24 atomic-data audit. No longer true: that the solved photochemical column overshoots the measurement and that its equivalent width rises with He/H (it now falls and stays below); that the line-forming gas is collisional and the critical point outside the domain; that the solution's own composition is unusable above $4R_p$. Still true: the width demand of about 22 km s^{-1} and everything said here about the p-winds model.

The question is whether a wind that is solved, rather than assumed, reproduces the measured He I 10830 Å transit line of LHS 1140 b. The model is EXHALE, a one-dimensional calculation of photoionization-driven escape in which the temperature, the velocity and the composition all come out of the solution instead of being imposed. The answer is that the strength of the absorption can be matched, but only by choosing a helium abundance, and that the shape of the

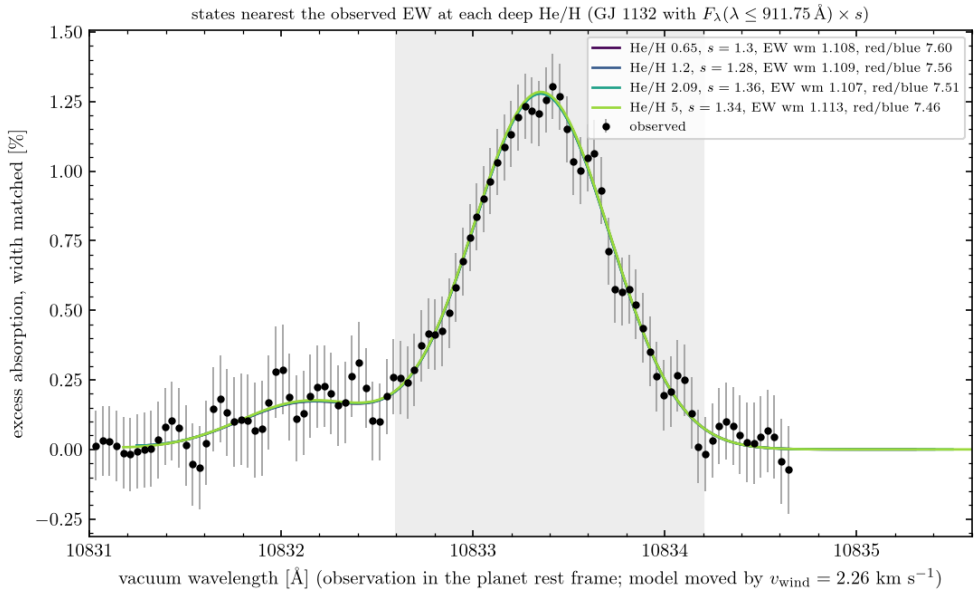


Figure 7: Width-matched excess absorption of the converged state nearest the measured equivalent width at each deep He/H (route B, steps 2 and 3), over the observed spectrum (vacuum wavelength, planet rest frame; grey band, the equivalent-width window). The model curves are moved by the fitted $v_{\text{wind}} = +2.26\,\text{km s}^{-1}$ for the overlay only, as in the earlier figures; every equivalent width and metric is from the unshifted spectra. The kernel FWHM f is chosen for each curve to reproduce the observed FWHM; it stands for a velocity field the one-dimensional model does not contain.

line cannot be matched at all. At the abundance that matches the strength the modeled line is 3.3 times too deep and 3.3 times too narrow, and it lacks $22\,\text{km s}^{-1}$ of velocity spread, the same number for every solution tried. To match the observation something would have to supply that spread without adding absorption, and neither the gas temperature, nor the outflow speed, nor any composition in these solutions supplies it.

Everything below was re-solved on 2026-09-16, with one build of the code; the build and the files are listed in Appendix A. The results of 2026-08-30 that earlier revisions of this memo carried are superseded twice over. Those states were never stationary below $1.1\,R_p$ once the conservation equations are measured the way the code now measures them. The campaign that replaced them on 2026-09-14 was itself solved with a lower boundary that decided its entropy branch on the velocity in the first cell rather than on the mass flux through its face, so that a run labeled "226 K base" in fact sat at 418 K in that cell. Both are recorded in Sect. 5., with how far each number moved. Every EXHALE number below is read from the set of models kept as the record for this study. A number that is still the archived one is marked *archived 2026-08-30* where it appears, and one from the superseded campaign *preL21*.

Result. The *amount* of absorption can be matched over a prescribed metal-free base whose composition is a single number, at a helium abundance that depends on what is assumed about the stellar spectrum and about mixing. Over a lower atmosphere solved as a photochemical column and handed over as a full profile, with C, N and O included, it can no longer be matched from above. The smallest helium reservoir that the base-flux iteration returns, $\text{He}/\text{H} = 2.09$, already makes an equivalent width 33 per cent larger than the measured one, and the sequence of solutions rises from there. The composition such a base would need lies below every solution obtained and has not been solved. The *width*

is not matched over either base, and the velocity it asks for is the same number every time.

Two of the changes of Sect. 5. act together on the branch with the solved column. The line cooling by C, N and O, in the six ions that the photochemical column brings into the base, used to be evaluated in the coronal limit, that is, above the critical density of the levels it excites. It is now taken from CHIANTI statistical equilibrium at the electron density of each cell. The closed-form fits it replaces also mixed theoretical with observed level energies, which at 2000 K is a factor 14 in $\exp(-\Delta E/kT)$ for O II. The metals in the base therefore no longer act as a thermostat. We measured this at one composition, He/H = 2.13 over the scalar base, with and without the column's C, N and O reservoirs: $T_{\max}$ went 5730 → 5566 K and the red-pair equivalent width 1.3749 → 1.3721 % Å, a change of 0.20 per cent. The factor 2.6 that the 2026-08-30 draft carried for this step is gone.

(i) The amount is always matchable, so what the line returns is a pair of conditional numbers rather than a measurement. Take the prescribed base first. When that base is a single state carrying no metals, every combination of assumed spectrum and assumed mixing reaches the observed red-pair equivalent width at some helium abundance, and the conditions stack on top of each other. In the well-mixed limit the crossing is He/H = 0.4044 (0.3877–0.4237 on the measurement's 1σ interval) on the GJ 1132 proxy spectrum and 0.0526 (0.0510–0.0543), below solar, on the GJ 699 one: the second stand-in for LHS 1140's unobserved XUV, 5.1 times brighter, which the paper also uses. When helium and hydrogen are allowed to separate by diffusion, helium settles out of the base unless an eddy term lifts it, and the crossing rises again. It is 3.10 wherever $K_{zz} \leq 10^5$ cm^2 s^{-1} does nothing against the molecular coefficient, 1.516 at the adopted $K_{zz} = 10^9$ and 0.893 at 10^{11} (Sect. 12.4). The dependence on the eddy coefficient is weak and monotonic, as $K_{zz}^{-0.115}$ over the four decades where the eddy term acts at all (a least-squares fit to the four crossings at 10^8 to 10^{11}); the dependence on the assumed spectrum is a factor 7.7.

The branch with the solved column no longer brackets the measurement. Replacing the simple base with a photochemical column (Photochem, an external photochemistry code, run with its climate step solved and handed to EXHALE as a profile matched at 1 μbar, at the same $K_{zz} = 10^9$ and with the same diffusion active) brings C, N and O into the base. Nine reservoirs were made consistent with the elemental fluxes the wind carries, and the equivalent width they return runs from 1.4765 % Å at He/H = 2.0924 to 2.2289 % Å at 12.0137. Every one of them stands above the measured 1.108 ± 0.030, the lowest by 33 per cent (Sect. 15.7). There is no crossing in this sequence. Read against its own local slope, $\mathrm{d}\ln EW/\mathrm{d}\ln(\mathrm{He/H}) = 0.49$ between the two lowest solutions, the measurement would be reached near He/H ≈ 1.2, which is below every solution obtained and is an extrapolation, not a solution. The earlier crossing at 8.212 was a consequence of the base metals cooling the wind; they no longer do.

Neither branch is an independent prediction of the model. The number from the prescribed base is the reservoir that the observed line picks out over that base, and the escape rate, which is an output and not a fit, barely moves along the way. Making the base consistent with the elemental fluxes does not change that on the profile branch. That iteration returns the same He/H at the match from trial fluxes a factor of ten apart, single-valued to a few parts in 10^6 (Sect. 15.), but at $K_{zz} = 10^9$ the eddy term mixes the column uniformly all the way to the microbar matching level, so it hands back very nearly the reservoir it was given. What it adds is that a composition is consistent with a photochemical lower atmosphere carrying its own C, N and O, not that the composition is measured.

The same helium now absorbs almost equally over the two bases. On the new set of models, at essentially one composi-

tion, the metal-free scalar base at He/H = 2.13 and the flux-consistent solution on the photochemical column at 2.0924 give: $T_{\max}$ 5730 against 5415 K, the metastable peak 34.6 against 35.8 cm^{-3}, the metastable column over 2–20 R_p 45.2 against 45.7 cm^{-2} in units of R_p, and the equivalent width 1.3749 against 1.4765 % Å. The photochemical column now makes the *stronger* line, by 7.4 per cent, where the 2026-08-30 solutions made it weaker by a factor 2.6. Raising the reservoir still reheats the wind, and far more strongly than before: at He/H = 10.01 the mean mass per particle at 2 R_p is 3.17 $m_{\rm H}$ against 2.15, the same deposited energy is shared among fewer and heavier particles, and $T_{\max}$ rises to 7682 K (Sect. 15.6, Fig. 24).

The escape rate is flat on every axis, and the crossings span a factor 59. $\log_{10}(\dot{M}/\mathrm{g\,s^{-1}})$ = 7.84–7.87 on the GJ 1132 proxy across composition, diffusion and eddy coefficient alike, 7.90–7.91 across the whole flux-consistent sequence, and 8.42–8.57 on the GJ 699 proxy. Against that, the composition the line implies runs from 0.0526 to 3.10 over choices this measurement cannot arbitrate. It is strongly conditional on the assumed spectrum and on what is placed at the lower boundary, and weakly and monotonically on the assumed mixing. Any composition read off this line (the paper's H : He $\lesssim 10^{-3}$ as much as any value here) inherits those conditions.

(ii) The width is not reached on any axis, and the deficit is the same everywhere. At the equivalent-width crossing the profile is 3.3 times too deep and 3.3 times too narrow (red depth 4.085 against the measured 1.254 per cent, FWHM 0.2533 against 0.841 Å), and no composition fixes it. Composition changes how much helium absorbs, not how the absorption is spread in velocity. Measured directly, by convolving the solved profile with a Gaussian and asking which Gaussian reproduces the observed line, the missing spread is 22.22 km s^{-1} FWHM (σ = 9.44 km s^{-1}) in the well-mixed limit, 22.22 km s^{-1} with diffusion and the adopted K_{zz}, 22.15 km s^{-1} for the flux-consistent solution on the solved photochemical column, and 21.97 km s^{-1} on the 5.1× brighter spectrum. That is a spread of 1.1 per cent over four solutions that differ by a factor 40 in composition (Sects. 12.2, 12.4, 15.7 and 14.).

Supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, with nothing tuned to them. What produces it is unidentified. It is not thermal, since the solved gas cools below 1000 K over the region where the line forms, and it is not bulk outflow, which reaches 1.1 km s^{-1} at 20 R_p on the fainter spectrum and 3.6 on the brighter. Read as turbulence it is $\mathcal{M}$ = 2.2–3.9 against the sound speed these solutions themselves carry, supersonic on every convention and well above what the literature is willing to assume. One mechanism would supply width without a free parameter, the metastable bump produced by adiabatic cooling of Schulik & Owen (2025). It is present in these solutions and was measured, but it sits at 0.4–2.5 km s^{-1} (Sect. 12.6). The shortfall is in the energy and momentum budget, not in the composition, and it is a published problem rather than a defect of this setup (Sect. 12.).

The paper's own p-winds fit does not reproduce the line either. It is broader than the wind solved here, because its temperature is imposed at 5000–6000 K where the line forms, but at FWHM 0.575 Å with its turbulence term on (0.458 without it) it is still 1.5 times narrower than the measured 0.841, and it overshoots the blue component threefold. Adding broadening does not rescue it. Its absorbing power is 25% below the measurement, so no convolution can reach the observed depth and width at once. The width is a shortfall the two codes share. That is also what the audit of the measurement side leaves (Sect. 12.3): the measurement is not at fault, and neither one-dimensional steady-state wind reproduces it.

Two caveats attach to the outer wind rather than to the line. The first is that the domain stops inside the critical point.

These solutions are the subsonic part of a transonic escape whose critical point sits at $40.06\,R_p$, outside the $30\,R_p$ domain they are solved on, so what closes them is the rule applied at the outer boundary and not a critical point. It costs the mass flux almost nothing: $\log_{10}\dot{M} = 7.8737$, 7.8740, 7.8735 at 30, 45 and $60\,R_p$, a spread of 5×10^{-4} dex over a factor two in domain size, and the He I 10830 equivalent width moves by at most 1.7 per cent. What it does cost is the outer profile: at $28\,R_p$ the $30\,R_p$ solution is 25 per cent too hot, 29 per cent too dense and 22 per cent too slow against the converged answer, so nothing above about $10\,R_p$ in these profiles should be quoted (Sect. 16.).
The second is that the gas is not collisional out there either: $\mathrm{Kn}_{\mathrm{bulk}}$ passes 0.1 at 7.30–$7.44\,R_p$ and reaches 0.68–0.71 at $29\,R_p$, while the He 10830 layer at 1.1–$3\,R_p$ sits at $\mathrm{Kn} \leq 0.012$ and the line analysis is untouched.
Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

One further caveat attaches to the composition. Above about $4\,R_p$ the ionization state of this wind is frozen into the flow. The recombination and photoionization Damköhler numbers fall to 10^{-2} and below, so the photoionization equilibrium the wind is *solved* with is not the composition that gas actually carries. The advection-corrected ionization profiles say so. These are computed after the wind is solved: they integrate the ionization along the flow instead of assuming it balances locally, and they are what the transit tool consumes. The difference is not small. The neutral hydrogen fraction differs by a factor 4.5 at the outer boundary, and the He I 10830 equivalent width read from the solution's own composition is 15.51 % Å against 1.17 from the corrected one, a factor 13.3. Every line number in this memo is read from the advection-corrected profiles. The solution's own composition is not usable above $4\,R_p$ and is not quoted. Carrying the three ionization fractions as transported variables is designed but not implemented (Sect. 5.).

Three systematic effects ride on the absolute numbers, and no convergence test catches any of them. They are stated here because they bound how far any single crossing should be read.

1. The outer region is inherited from the starting state and is not resolved by the domain. A converged wind takes its 20–$30\,R_p$ state from the solution it was started from, and the size of the residual $\|R\|$ cannot tell two such states apart. Sect. 16. bounds what that costs the line: moving the outer boundary to 45 and $60\,R_p$ moves the equivalent width by at most 1.7 per cent and the mass flux by 5×10^{-4} dex, while the profiles above $10\,R_p$ move by 22–29 per cent.
2. The tolerance of the base-flux iteration is visible in the answer. It stops when the undamped elemental imbalance falls to 0.05, and the solutions end at 0.045 for helium and 0.018 for hydrogen. Tightening it further is refused, because the radial spread of the flux over the measurement window, 3.7×10^{-3} in hydrogen and 4.9×10^{-3} in helium, would then approach the imbalance being decided on it.
3. Which chemistry produced the column. Every column here is Photochem 0.9.0 with the local corrections listed in Appendix A.

Everything below uses inputs verified against the authors' own released material (Zenodo 20723095): their p-winds

driver script, their Mega-MUSCLES v23 spectra, their reduced transmission spectrum and their likelihood grid. Every number and figure here can be regenerated from the files listed in Appendix A.

5. What changed since 2026-08-30

Note added 2026-10-04. The history of the 2026-09 re-solves. The corrections described here are in the code; the numbers they produced are superseded by Sect. 1.. The frozen-in composition of Sect. 5.6 is now carried by ionization transport, and the solution and advection-corrected equivalent widths agree to 1 per cent.

Every EXHALE number in this memo was re-solved between 2026-09-13 and 2026-09-16. This section says why the older answers were discarded, what was corrected, and how far the answers moved. The case-by-case record and the working notes behind each item are listed in Appendix A.

Throughout this section and the rest of the memo, a state is called *converged* when it satisfies the discretized conservation equations for mass, momentum and energy, and the equations that set the composition, to a stated tolerance. Because the flow is a wind only above about $1.2\,R_p$, that check is applied above $1.2\,R_p$. The solution proceeds in cycles: the wind is solved at fixed composition, then the composition is allowed to adjust at fixed wind, and so on.

The set of models was solved twice. The campaign of 2026-09-14 was superseded three days later by the correction to the lower boundary described in Sect. 5.8, and every case was solved again. The set of record is the one solved on 2026-09-16, on a single build. Of the 86 cases in it that have no molecules in the wind, 79 converge, each in one cycle started from its own previous state. The seven that did not are the lowest-XUV cases; five of them have since certified with the Roe numerical flux, solved after this campaign, and the two left are cases at 0.01 of the fiducial XUV (Table 7, Sect. 15.9). The results of 2026-09-14 are kept unchanged, and the two generations are compared case by case in the working notes of 2026-09-15 on the re-solve. A number labeled *preL21* below is from 2026-09-14; everything else is from the final pass.

5.1 Why the 2026-08-30 solutions were discarded

They were not stationary. Measured equation by equation the way the code now measures them, the archived state carries an imbalance of order unity everywhere between the base and about $1.1\,R_p$. In every cell below $1.049\,R_p$ the energy equation is dominated by the divergence of its own flux rather than by heating and cooling ($|\mathrm{d}F_3|/|\Gamma - \Lambda| = 7 \times 10^5$ at cell 1, 4.9×10^1 at cell 16), and the momentum equation is out by a constant 2.8 per cent of the local weight over the whole column. The imbalance belongs to the state and not to the way it is measured. The velocity sawtooth in the base layer contributes nothing to the energy equation, and putting the archived heating back into the same hydrodynamic state lowers the worst cell only from 1.979 to 1.930. The old runs reported convergence because they looked only above $1.2\,R_p$ and never below it.

Interpolating such a state onto the current grid and re-solving does not repair it. All four routes tried stall: the Newton solve at $||R|| = 1.6$–2.0 for 56 iterations; a time-marching restart whose fractional radial spread of the mass flux never falls below the 10^{-2} handoff in 5600 steps; a cold two-stage march still at a spread of 9.5 after 954 steps; and the stationary route, which does not converge and whose composition step finds no admissible advance.

Separately, the code has moved in ways that reach this wind directly: the quadrature over the photon grid (a 9.5 per

cent overcount of the H I photoionization rate removed), the cell width $\mathrm{d}r_j$, the Jupiter radius (+2.26 per cent in R_0), the helium mass, the CODATA constants, the spectrum below 13.6 eV on which the He($2\,^3S$) population depends, and a fix to an extra row the transit reader was picking up. The new Jupiter radius also changes the grid, so no archived state can be reloaded without interpolation.

5.2 The well-balanced reconstruction

No case in this campaign converges unless the discretization is the one that keeps a hydrostatic atmosphere exactly hydrostatic. At the base of this planet the Mach number of the flow is 2.7×10^{-7}. The contact speed that the numerical flux puts through a base face is then set by the third-order truncation error of the reconstruction in the face pressure rather than by the velocity, which that error exceeds by one to three decades. The energy equation of every base cell therefore measures its own largest term against itself, and no starting state can lower it. A relative density jump d at a face enters the interface flux through its acoustic part as $d\,\rho c_s$, which is $d/\mathcal{M}$ times ρv. At $\mathcal{M} \sim 10^{-6}$ a jump of one per cent is a six-decade error in the flux.

The remedy is the reconstruction of Käppeli & Mishra (2016). Every cell carries its own hydrostatic equilibrium, the reconstruction and the Riemann solver act on the *departure* from it, and the momentum source is the face-pressure difference of that same equilibrium. At a discrete hydrostatic equilibrium the flux difference and the source then cancel in the algebra instead of in floating point. The term-by-term statement, the mapping onto the code and the reason this is not the default are in the working notes of 2026-09-13 on hydrostatic discretization. The table below shows what it does here, from the same starting state, the same input and the same build.

Table 5: The three hydrodynamic conservation equations after the first cycle (largest cellwise imbalance, each measured against the largest term in its own equation) and the outcome of the stationary solve on the reference case, $K_{zz} = 10^9$, He/H = 1.6261.

option	hydro	mass	momentum	energy	outcome
none	not converged	1.08	3.2×10^{-2}	0.63	cycle refused
low-Mach damping at 2.0×10^{-2}	not converged	1.32	1.4×10^{-2}	1.24	cycle refused
hydrostatic-exact reconstruction	converged	2.6×10^{-10}	1.8×10^{-14}	4.6×10^{-9}	converged at cycle 8

Low-Mach damping does nothing to the energy equation, because it is a fourth difference of the *velocity* and does not touch the truncation error in the pressure. Three starting states whose base velocities differ by a factor 10^4 reach one solution (ρ to 3.4×10^{-10}, v to 2.5×10^{-8}, $\dot{M}$ to seven digits), so the answers of this campaign do not depend on which solved state they were started from.

5.3 The secondary-ionization coupling, applied from the first evaluation

By default the code switches on the Shull & van Steenberg (1985) channel, in which a fast photoelectron ionizes further atoms, only once a time-marching run has first converged. The stationary route takes no time step, so a state solved under the default is written with that channel off. The advection-corrected profiles apply it unconditionally, and

every number quoted here comes from those, so they report the same wind with the coupling on. On the $K_{zz} = 10^9$, He/H = 1.6261 case we measure $\log_{10}\dot{M} = 8.29$ under the default against 7.87 with the channel on from the first evaluation, and $n(\mathrm{He}\,2\,^3S)$ differs by 74 per cent beyond $1.5\,R_p$, where the line forms. Every case of the campaign turns the channel on from the start.

5.4 The C, N and O line cooling at the local electron density

Six coolants that the photochemical column brings into the base (C I, C II, N I, N II, O I and O II) were evaluated in the coronal limit, that is, above the critical density of the levels they excite. Against the CHIANTI statistical equilibrium of the same ion, the coefficient used stood at 0.02 to 5.4×10^4 times the right value over the range of conditions the wind spans. The replacement is a density-resolved table read at the electron density of each cell, and it stands at 0.969 to 1.009 at every point of a 7×6 grid spanning 1000 to 20000 K and 10^4 to 10^9 cm^{-3}.

A second error sat in the closed-form fits those six terms used, and is removed along with them. The fits were built from CHIANTI *theoretical* transition energies, while the level populations that decide the emission use the *observed* level energies. The two differ by up to 14 per cent in ΔE for the O II $^4S^\circ$–$^2D^\circ$ excitation, which at 2000 K is a factor 14 in $\exp(-\Delta E/kT)$.

This is the change that reshaped the branch of this memo with the solved column. The metal content of the base is no longer a thermostat. At He/H = 2.13 over the scalar base, adding the column's C, N and O reservoirs moves $T_{\max}$ from 5730 to 5566 K and the red-pair equivalent width from 1.3749 to 1.3721 % Å.

5.5 The outer boundary

The LHS 1140 b wind is subsonic at $30\,R_p$, so continuing the flow freely across the outer boundary is not a passive extrapolation: one characteristic points inward, and the rule applied there is an external condition. Solving the same case on 45 and $60\,R_p$ domains settles what that costs. The wind is transonic, with its critical point at 40.06 and $40.08\,R_p$ in the two extended runs, and once the boundary is outside that point the two solutions agree with each other to better than 0.1 per cent everywhere.

The $30\,R_p$ answer holds for the quantities taken from it and not for the rest: $\log_{10}\dot{M} = 7.8737$, 7.8740, 7.8735 at 30, 45 and $60\,R_p$ and the He I 10830 equivalent width within 1.7 per cent, against 25 per cent in temperature, 29 in density and 22 in velocity at $28\,R_p$. No profile above about $10\,R_p$ is quoted from these runs. The exobase of the extended solutions sits at $29.0\,R_p$, below the critical point, so the continuum model fixes its own topology in gas it cannot describe. The insensitivity of the mass flux is a statement about the discrete problem and not a physical validation.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $\mathrm{H}^+ + \mathrm{H}$ and $\mathrm{He}^+ + \mathrm{He}$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(\mathrm{H}_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

5.6 The composition above 4 R_p

The wind is solved with a composition in photoionization equilibrium, and above about $4\,R_p$ that composition is not the one the gas carries. Over a flow time r/v the hydrogen recombines 1.7 times at $2\,R_p$ and 5×10^{-3} times at $29\,R_p$, and is photoionized 7×10^{-2} and 10^{-2} times at the same radii. The ionization state is frozen into the parcel and carried outward. The equilibrium neutral fraction falls from 0.963 at $2\,R_p$ to 0.346 at $29\,R_p$ while the advection-corrected one goes from 0.974 to 0.941, and the photoheating that follows the neutral density stands a factor 5.9 apart at $44\,R_p$.

The consequence for the line is a choice of which profile to read rather than a correction. The He I 10830 equivalent width read from the solution's own composition is 15.5120 % Å against 1.1702 from the advection-corrected one, a factor 13.3. Feeding the transit tool the solution's *temperature* together with the corrected composition instead moves it by 0.15 per cent. The line depends on the composition and hardly at all on anything else in those two files, and every line number here is read from the advection-corrected profiles. Carrying x(H II), x(He II) and x(He III) as transported fractions of each element's nuclei is designed but not implemented.

One defect was found and fixed on the way. The advection correction discarded roots returned by the nonlinear solver with a warning flag even where the iterate satisfied the energy equation to 10^{-17} of the largest term the equation holds, and because the correction marches outward cell by cell, a discard restarted the temperature profile at every cell above it. With the roots accepted on their residual, the 45 and $60\,R_p$ corrected temperatures agree to 0.19 per cent, which is the 0.28 per cent at which the two *solutions* agree.

5.7 The convergence procedure at low XUV

Five cases of the XUV grid did not converge in the first campaign, and the reason lies in the cycle rather than in the wind. Each cycle solves the wind at fixed composition and then relaxes the composition at fixed wind. The error of that alternation is the undamped distance the composition step returns, which the under-relaxation contracts by $1-\omega$ per cycle. The control that decides whether the cycles are making progress reads a different quantity, the distance of the state from being fully converged. Where that distance fails to fall it halves ω, taking the contraction from 2.00 to 1.14 per cycle, and after three such cycles it ends the loop. On these cases the two quantities disagree from the first cycle: the composition distance falls every cycle while the imbalance in the equation for the helium-to-hydrogen ratio rises. The case at 0.30 of the same spectrum, which does converge, is the control: it entered at the same composition distance, its helium-to-hydrogen imbalance also failed to fall over its first cycle, and it converged at cycle 13 with ω at 0.500 throughout. What refused the five was not the energy equation in the first cell. At the last cycle of the worst of them the three hydrodynamic equations read mass 2.30×10^{-8}, momentum 3.39×10^{-14} and energy 8.49×10^{-7} against a tolerance of 10^{-6}, while the helium-to-hydrogen equation read 2.12×10^{-2} against 10^{-5}.

On the corrected lower boundary these cases start further still from their fixed point, and the remedy is to start the Newton solver's artificial time step high, at 10^{8}, instead of letting it collapse from 1. MEASURED on five cases with a scalar base, that turns hours of refusing into minutes of converging: the case at 0.20 of the fiducial spectrum and He/H = 2.13 went from four cycles in 6 h 22, still refusing, to converged in 17 minutes. Nine of the thirteen cases treated this way converged within the hour. *Four did not*: the run over the scalar base at 0.10 of the fiducial with He/H = 9.7, and the photochemical cases at 0.15, 0.20 and 0.25, all keep the same behavior at $\mathrm{d}\tau_0 = 10^{8}$, the step

falling back to order ten while the mass equation in the first cell holds $\|R\|$ near 1.5. Starting the step higher is not what they need. Separately, the three cases at 0.01 of the fiducial did not solve on either boundary in this campaign. That is attributed to the Newton model of a wind subsonic to Mach 10^{-8} over its whole lower atmosphere; the finite-difference Jacobian and the convergence criterion were both measured and both cleared. All four of the first group have since certified with the Roe numerical flux in place of HLLC, and so has the scalar case at 0.01 with $\mathrm{He}/\mathrm{H} = 2.13$, reached by a continuation in XUV on 2026-09-21; the other two cases at 0.01 are not certified (Sect. 15.9, Table 22).

5.8 The base boundary, which chose its branch on the wrong quantity

This is the change that forced the second campaign, and it is the largest single movement in the set of models. The base face of a converged LHS 1140 b state is an *inflow* face carrying the wind's own mass flux. The boundary condition decided whether to treat it as inflow or as reversal using the cell-centered product $\rho_1 v_1 r_1^2$ of the first interior cell. At this base that product is not a flux: it is a two-cell oscillation in the velocity, sitting on the cell centers. MEASURED on the fiducial case, the face mass flux from the Riemann solver is $+1.00000\,F_{\mathrm{wind}}$ at every one of the 500 faces including the base face (largest departure 1.6×10^{-9}), while the cell-centered product reads $-2.00\,F_{\mathrm{wind}}$ at cell 1 and -2.27 at cell 2. The boundary was therefore running 97 per cent in the reversal branch, where the face density is the interior's isentrope rather than the reservoir's, and the two differ by $\rho_{\mathrm{rev}}/\rho_{\mathrm{res}} = 0.531$ at the common face pressure. The 226 K reservoir prescribed for the lower atmosphere was entering the state with a weight of 0.026, and the first cell of a "226 K base" run sat at 417.6 K.

No choice of blending width repairs that, because the discriminant has the wrong sign: a narrower window takes the reversal branch more completely. Deciding the branch on the mass flux the state actually carries is the fix. A second error sat beside it. The blending window was 10^{-6} in face Mach number, chosen against a much hotter planet, while LHS 1140 b's own base runs at Mach 4.6×10^{-7}, *below* the window, so even with the sign repaired a fifth of the reservoir's entropy would be blended away on a clean inflow. The window is now 10^{-8}. On the same state the branch reads $M_{\mathrm{branch}} = +4.02 \times 10^{-7}$ and a reversal weight of zero: the reservoir sets the entropy of the gas that enters, which is what the boundary exists to do.

What it did to the set of models, over the 79 cases that converge both ways: the first physical cell falls by a median 46 per cent in temperature and rises by a median factor 1.82 in density, $\dot{M}$ falls by a median 2.0 per cent with no case rising, and the red-pair equivalent width falls by a median 2.0 per cent. On the fiducial case the first cell goes from 417.6 to 238.0 K. The profiles cross near cell 21 and agree to a fraction of a per cent in the outer wind. Four solutions of the GJ 699 sequence are identical across the correction, because on that weak spectrum the interior isentrope at the base already *was* the reservoir, so the branch the old boundary took wrongly happened to give the right state. The correction changes what was wrong and nothing else.

5.9 What a written state means when it is read back

Two further items tightened the relation between a state and the files that hold it, and neither changes physics.

A converged state did not reproduce its own convergence when it was read back. The pair of output files states the density twice, once directly and once as the sum over species, and the two agree only while the composition closes its

own mass. It does not close exactly, and the loader took the species as the authority, so the state handed to the check stood some 2000 units in the last place from the one that had been solved. At this base the hydrodynamic equations cancel their largest term by 6.2×10^5, and at that cancellation 2000 units in the last place of the density is the whole imbalance. The loader now takes the density from its own column and scales the loaded species onto it, one factor per cell, which leaves every element ratio and every ionization split where the file put them. One unit in the last place of the density still moves the largest cellwise imbalance of the energy equation by about 11 per cent, so identical reproduction of a convergence check is not available, and the stated contract says so.

The composition a sweep returns is now put on the density the sweep was given, and the mass closure $\max_j |\sum_i f_i A_i - 1|$ is reported with the convergence check. A companion question, whether the written cooling column is the cooling of the written state, was raised as a defect and is answered in the negative: on every state the current code produces, that column and the total of the cooling breakdown agree to between 3.4×10^{-15} and 3.0×10^{-10}. The factor 1.8 that raised the question was measured on a regression case pinned from an earlier generation of the code. What was missing was the check itself, which is now there.

5.10 The molecular rate network, and the artificial time step

One item asked which chemistry is right about the base layer, where the state handed over from the lower atmosphere stands 7 to 21 times above the root of the wind network's own equation for the transported H_2. It retired one reaction, added the channel of Black (1978), and rewrote another. Those repairs are in the build every number here stands on. They move the molecular test cases and leave an atomic wind alone, so nothing in Sects. 11. to 16. depends on them.

A second item took both ends of the schedule for the Newton solver's artificial time step: a shrink on damped steps that could pin the step at its floor, and a growth that could climb to 10^{14} while the residual stood at 1.9. The rule now in place clips the cut of an accepted step at ten and floors it a fixed span below the step the solve has already earned at a whole step, with a cap on how many damped steps may run in a row. Turning the guard off reproduces the earlier arithmetic exactly. What it does to the XUV grid of this planet is measured below.

5.11 How far the answers moved

Two things follow from the table. Against 2026-08-30 the branch with a prescribed base moved by 2 to 19 per cent in composition and by about 0.1 dex in $\dot{M}$: its structure is unchanged and its numbers are revised. In the branch with a solved column the sign of the effect reversed. With the base metals no longer cooling the wind, the photochemical column makes a *stronger* line than the scalar base at the same composition, and the sequence of solutions that used to straddle the measurement now stands entirely above it. Against preL21 the movement is smaller and it is all one thing: the base cell is the reservoir that was prescribed instead of the interior isentrope, so every case is a little cooler and denser at its first cell, a little smaller in $\dot{M}$, and a little weaker in the line. The crossings of the groups with diffusion walk *up* by 1.5 to 14 per cent, most of it at $K_{zz} \leq 10^5$, where the equivalent width responds most weakly to composition and a 2 per cent shift in it buys a long step in He/H. The two well-mixed crossings move less: by 0.8 per cent on the GJ 1132 proxy and by 0.002 per cent on the GJ 699 one, whose solutions the correction left as they were.

Table 6: Three generations of the same quantities. The archived column is read from the record the 2026-08-30 runs left; *preL21* is the campaign of 2026-09-14; *final* is the set of record solved on 2026-09-16. The intermediate generation is not tabulated: it agrees with the final one to seven decimal digits or better in every entry.

quantity	archived 2026-08-30	preL21	final
well-mixed crossing He/H, GJ 1132	0.4132	0.4012	0.4044
well-mixed crossing He/H, GJ 699	0.0483	0.0526	0.0526
crossing He/H, $K_{zz} \to 0$	3.81	2.72	3.10
crossing He/H, $K_{zz} = 10^9$	1.6108	1.4922	1.5161
crossing He/H, $K_{zz} = 10^{11}$	0.837	0.878	0.893
crossing He/H, flux-consistent sequence	8.212	none	none
EW of the flux-consistent solution at He/H $\simeq 2.09$ [% Å]	0.545	1.5006	1.4765
T of the first cell, fiducial case [K]	–	417.6	238.0
$\log_{10}\dot{M}$, GJ 1132	7.7–7.8	7.84–7.88	7.84–7.87
$\log_{10}\dot{M}$, flux-consistent sequence	7.40–7.41	7.91–7.92	7.90–7.91
$\log_{10}\dot{M}$, GJ 699	8.58	8.43–8.57	8.42–8.57
matched Gaussian FWHM [km s^{-1}]	21.95–22.23	21.97–22.22	21.97–22.22

5.12 The campaign, group by group

Table 7 collects the whole campaign on one page: twenty-one groups of models, the reservoirs solved in each, where that group's own solutions cross the measured equivalent width, and the range of the escape rate over the group. It indexes the sections that follow, and it also records what is not solved: two cases at 0.01 of the fiducial XUV, which are what is left of the seven lowest-XUV cases this campaign did not solve, and the groups with molecules in the wind, which are still being worked and whose solved cases are recorded as they stand.

Two of the molecular cases do now converge, and they are the first LHS 1140 b winds solved with H_2, H_2^+, H_3^+ and HeH^+ carried through the wind rather than left at the handoff: He/H = 2.13 at $\log_{10}\dot{M} = 7.91$, red-pair $EW = 1.5758$ % Å and depth 5.63 per cent, and 9.7 at 7.95, 2.4212 % Å and 7.68 per cent, both with element diffusion at $K_{zz} = 10^9$. Both stand above the measured 1.108, as the flux-consistent solutions do. These numbers are from the spectra of 2026-09-16; both cases have been run again since, and the spectra now in their directories (written 2026-09-20) give $EW = 1.5604$ and 2.4060 % Å through the operator of Sect. 11.. The rest of their record is not re-quoted here.

[2026-09-24] Five more cases of the same group now carry a certificate, each confirmed by four consecutive re-entries that change the state by at most 1.2×10^{-12}: He/H = 0.55 ($\log_{10}\dot{M} = 7.59$, $EW = 0.1015$ % Å), 1.45 (7.85, 1.0964), 1.52 (7.86, 1.1532), 1.6 (7.86, 1.2155) and 1.8 (7.88, 1.3584), all through the operator of Sect. 11.. The two adjacent certified rungs 1.45 and 1.52 straddle the measured 1.108, and the log-log chord between them crosses it at He/H = 1.4643; from 1.45 to 2.13 the profiles, $\dot{M}$, the outer Mach number and the H_2 front move smoothly with He/H, while the 0.55 state is a different wind (an H_2 layer to $1.13\,R_p$, $x_{H_2} \simeq 10^{-2}$ in the far wind, a peak temperature of 3400 K against 5200 K), and the interval between 0.55 and 1.45 has not been sampled. The crossing is of the equivalent width alone: at He/H = 1.45 the model red line is 4.07 per cent deep, 3.3 times the fitted depth of C26, and correspondingly narrower,

Table 7: Every group of models and what it returns, on one build apart from the XUV-grid cases named below. The crossing is where the group's own solved solutions reach the measured red-pair equivalent width, 1.108 ± 0.030 % Å, and $\dot{M}$ is the range over the group. The rows of the XUV grid also hold cases solved after this campaign, READ from `LHS1140b/MODELS.md` (sections 3.1, 7 and 8, as of 2026-09-23): the photochemical cases at 0.15, 0.20, 0.25 and the scalar He/H = 9.7 case at 0.10, which stopped solving on the corrected lower boundary and certify with the Roe numerical flux, and the scalar He/H = 2.13 cases from 0.07 down to 0.01, reached by a continuation in XUV on 2026-09-21, and the scalar He/H = 9.7 rungs at 0.0675, 0.0662 and 0.065, continued from 0.07 on 2026-09-23. The scalar He/H = 9.7 cases at 0.06, 0.0624 and 0.01 and the photochemical case at 0.01 are not certified (Sect. 15.9). In the groups with molecules the cases without a certificate are marked so; why each stops is recorded in the working notes on this campaign.

group	reservoirs solved	crossing He/H	$\log_{10}\dot{M}$
prescribed scalar base, GJ 1132 proxy			
well mixed	0.083–1000 (9)	0.4044	7.84–7.86
$K_{zz} = 0$	0.55–3.9 (6)	3.1010	7.84–7.87
$K_{zz} = 10^5$	2.6–3.93 (5)	3.0706	7.87
$K_{zz} = 10^6$	0.55–3.74 (6)	2.8930	7.84–7.87
$K_{zz} = 10^7$	0.55–3.18 (4)	2.5365	7.85–7.87
$K_{zz} = 10^8$	0.55–2.41 (4)	1.9729	7.85–7.87
$K_{zz} = 10^9$	0.55–9.7 (7)	1.5161	7.85–7.87
$K_{zz} = 10^{10}$	0.55–1.29 (4)	1.1580	7.86
$K_{zz} = 10^{11}$	0.55–0.93 (4)	0.8927	7.86
$K_{zz} = 10^9$, C/N/O base	2.13 (1)	-	7.87
prescribed scalar base, GJ 699 proxy			
well mixed	0.042–1000 (6)	0.0526	8.42–8.57
photochemical column, base consistent with the elemental fluxes			
$K_{zz}(p)$ from the profile	2.09–12 (9)	none: every solution above	7.90–7.91
XUV grid, lower atmosphere held fixed			
scalar, $K_{zz} = 10^9$	0.01–0.33× at He/H 2.13; 0.065–0.33× at 9.7	depth limit at 0.28×	5.79–7.36
	0.01, 0.06, 0.0624× at 9.7: not certified		
photochemical column	0.10–0.33× at He/H 9.7	limit below 0.10×	6.84–7.39
	0.01×: not certified		
molecular chemistry in the wind			
scalar base, well mixed	0.083, 0.55 not certified; 2.13 not solved	-	-
scalar base, $K_{zz} = 10^9$	0.55–9.7 (7) certified;	1.4643	7.59–7.94
	0.083 not certified		
photochemical column	9 not certified; 2.09 not solved	-	-

so the molecular solutions share the line-shape disagreement of the other unbroadened EXHALE spectra of this memo. Convolved as in Sect. 12.2, the 1.45 solution reaches the measured width at $f = 22.21\,\mathrm{km\,s^{-1}}$, with depth 1.224 per cent, blue 0.160, ratio 7.64 and $EW = 1.060\,\%$ Å, and the broadened EW crosses 1.108 at He/H = 1.511 between the certified 1.45 and 1.52 (`docs/lhs1140b_catalog_state.tex`, section 5). The remaining molecular cases are not certified and are not quoted.

6. Inputs, and how each was pinned

6.1 System

Planet and star from Cadieux et al. (2024) as quoted by C26: $R_p = 1.730 \pm 0.025\,R_\oplus$, $M_p = 5.60 \pm 0.19\,M_\oplus$, $T_{\rm eq} = 226\,\mathrm{K}$, $a = 0.0946\,\mathrm{AU}$, $M_\star = 0.1844\,M_\odot$, $R_\star = 0.2159\,R_\odot$. One trap: EXHALE converts R_p with its own internal constants (mean radii, $R_J = 6.9911 \times 10^9$ cm, $R_\oplus = 6.3725 \times 10^8$ cm), so the input must be built with those rather than with IAU equatorial values. The wrong pair costs 2.2% in radius and 4.4% in transit depth.

6.2 Lower boundary condition

EXHALE anchors the lower boundary at the planetary radius R_p by fixing the pressure there, which is the standard CETIMB condition, at $P_0 = 1.0\,\mu\mathrm{bar} = 1.0\,\mathrm{dyn\,cm^{-2}} = 1.0\,\mathrm{erg\,cm^{-3}}$ (0.1 Pa). The corresponding total number density n_0 at the base follows from the ideal gas law at the equilibrium temperature $T_0 = T_{\rm eq} = 226.0\,\mathrm{K}$:

$$n_0 = \frac{P_0}{k_B T_0 n_{\rm tot,bc}} = \frac{1.0\,\mathrm{dyn\,cm^{-2}}}{(1.380649 \times 10^{-16}\,\mathrm{erg\,K^{-1}}) \times 226.0\,\mathrm{K}} = 3.20485 \times 10^{13}\,\mathrm{cm^{-3}}, \tag{2}$$

where $n_{\rm tot,bc} = 1.0$ for neutral atomic gas. In logarithmic form, $\log_{10}(n_0/\mathrm{cm^{-3}}) = 13.506$, which is what the input file carries. When the boundary is anchored on pressure, EXHALE re-evaluates and enforces this relation internally, so the density entered in the input is consistent with the pressure by construction.

6.3 Stellar spectrum

The paper describes its normalization two ways that cannot both hold. The Supplement's prose says the proxy spectrum is scaled so that its integrated X-ray flux matches the one measured for LHS 1140, while the quoted constants ($A_{1132} = 0.59$, $A_{699} = 0.056$) are the catalog X-ray flux *ratios*, a factor 1.8 apart. The authors' script settles it:

```
Lx_ratio_1132 = 0.59
norm_constant_1132 = Lx_ratio_1132*(d_1140/(smax_1140*au2m))**2
```

applied to the plain constant-resolution v23 product. That combination also reproduces the fiducial XUV flux the main text quotes, $0.033\,\mathrm{W\,m^{-2}}$ (we measure $33.6\,\mathrm{erg\,s^{-1}\,cm^{-2}}$ over 10–1300 Å). All results here use it. The proxy spectrum and the paper's own APEC fit disagree in the X-ray band (4.4 against $2.9\,\mathrm{erg\,s^{-1}\,cm^{-2}}$ at the orbit). That inconsistency comes from the paper's procedure and is not introduced here.

6.4 Measurement

The comparison target is the authors' released Fig. 3B spectrum (129 samples, vacuum, planetary rest frame, corrected for correlated noise), not a reading of a figure. Fitting it with our own three-Gaussian extractor, the same one applied to every model spectrum here, reproduces the paper's MCMC to within its published uncertainties:

	this extractor	paper
blended red depth	1.254%	$1.24^{+0.22}_{-0.23}$%
blue depth	0.198%	$0.25^{+0.14}_{-0.12}$%
red/blue	6.35	$6.7^{+12.7}_{-3.1}$
FWHM	0.841 Å	$0.86^{+0.15}_{-0.27}$ Å

so models and measurement can be compared through one instrument.

6.5 The atomic input that sets the line: He($2\,^3S$) destruction on atomic hydrogen

Every composition this memo quotes is read off the metastable helium population, and in this wind that population is drained almost entirely by one reaction, He($2\,^3S$) + H. Its rate coefficient is therefore an input of the same standing as the spectrum, and it has been changed since the earlier draft. The grounds are given here because every number below moved with it.

The code carried the fit of Taylor et al. (2025), Table 2, transcribed correctly: $1.9\times10^{-9}(300/T)^{0.07}$ below 4000 K and $9.1\times10^{-9}(300/T)^{0.50}$ above it. Those two branches do not meet: at 4000 K they give 1.5849 and 2.4921×10^{-9}, a jump of 1.5724. A rate coefficient is a continuous function of temperature, and "it is the published value" is not a defense of an expression that is discontinuous in a state variable. In these solutions the seam falls at 1.11–$1.18\,R_p$, inside the region where the line forms. In the figures of the earlier draft it showed up as a one-cell step of 33 per cent in $n(2\,^3S)$.

The fit also disagrees with its own paper. Taylor et al.'s Fig. 19 (bottom right) plots the quantity Table 2 is meant to represent, their Maxwell–Boltzmann average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections, and that curve has a maximum near 1400 K. A monotonically decreasing power law cannot produce a maximum, so the tabulated low branch cannot be the same quantity; digitized, the two differ by 1.13–1.62 over 500–4000 K. Above the seam the high branch is worse: 2.23×10^{-9} at 5000 K, above every cross-section determination collected in García Muñoz (2025), Fig. 5, which lie between about 1.3 and 1.7×10^{-9} there. Their appendix records how this happened: an equation already implemented in their own code, fitted in two pieces by least squares, with nothing tying the pieces together at the join.

What is used now. García Muñoz (2025), A&A 698, A199, publishes in the caption of its Fig. 5 a closed form for the Maxwell–Boltzmann average of the Movre & Meyer (1997) total ionization cross sections,

$$k = 10^{-9}\exp\left(\frac{c}{T} + d_1\ln T + d_2(\ln T)^2 + d_3(\ln T)^3\right)\ \mathrm{cm^3\,s^{-1}}, \tag{3}$$

with $c=-8.64804\times10^{1}$, $d_1=-2.86766\times10^{-1}$, $d_2=+8.68445\times10^{-2}$ and $d_3=-5.73001\times10^{-3}$, stated there as accurate to better than 2% over 200–10^4 K. Measured here, it reproduces that paper's own Table A.5 to 0.4%, and it is continuous at 4000 K to the eighth digit (1.35630953×10^{-9} at 3999.9 K against 1.35630905 at 4000.1).

The change is not one paper's fit traded for another's. Taylor's own Fig. 19 and the García Muñoz curve agree to 10–20%. The outlier is the tabulated fit, and that is what the code was using.

The 0.9:0.1 **branching.** Equation (3) is the *total* ionization rate: Penning ionization, $\mathrm{He}(2\,^3S) + \mathrm{H} \rightarrow \mathrm{He}(1\,^1S) + \mathrm{H}^+ + e^-$, and the associative channel that leaves HeH^+, sharing one cross section. García Muñoz splits them "an average 0.9:0.1", and the machine-readable network file deposited with that paper (Zenodo 15453858) applies that split through two amplitudes, one per channel. EXHALE now does the same. The split is more than a scale factor. The total rate removes the metastable atom, so every sink must keep the full sum; using 0.9 there would make the observable, $n(2\,^3S)$, 10 per cent too high. The 0.9 scales only the terms that leave a lasting proton behind and deposit the Penning exothermicity, and the remaining 0.1 becomes a source of HeH^+ in the molecular system.

Effect on the numbers. The new coefficient is lower than the old one everywhere in this wind (new over old: 0.69 at 1000 K, 0.86 just below the seam, 0.54 just above, 0.81 at 10^4 K), so the metastable level lives longer and the line is stronger at fixed composition. Re-solved, the case at $\mathrm{He/H} = 0.55$ goes from $EW = 1.109$ to 1.310 % Å and the one at 2.13 with diffusion at $K_{zz} = 10^9$ from 1.129 to 1.338, with $\log_{10}\dot{M}$ unchanged in both. Every crossing therefore moves down, and the composition sequences were re-solved rather than scaled by a local slope, because a +18 per cent shift is well outside the range over which such a slope is valid. A detector that looks for a one-cell step in the metastable population flags no cell at the seam in any of the runs quoted here.

7. Reproducing the paper's p-winds model

We ran p-winds at the authors' settings: their spectrum normalization, $R_p = 0.154\,R_J$, $b = 0.23$, five-phase transit averaging, their turbulent broadening term on, and their fitted line-of-sight bulk velocity $v_{\mathrm{wind}} = 2.26\,\mathrm{km\,s^{-1}}$. The composition and outflow are the MCMC medians ($\dot{M} = 2.03 \times 10^8\,\mathrm{g\,s^{-1}}$, $T = 5160$ K, $\mathrm{H:He} = 1.01 \times 10^{-3}$), the set of values their Fig. 4 legend states for the plotted curve. Our reproduction gives a red depth of 1.376% and FWHM 0.575 Å, against the 1.32% and ~ 0.71 Å read off their Fig. 4. The depth matches to 3%, but the line is about 20% too narrow.

The width residual does not depend on which point in parameter space is drawn. Running the same reproduction at the grid maximum-likelihood point ($1.83 \times 10^8\,\mathrm{g\,s^{-1}}$, 5131.6 K, 8.338×10^{-4}) gives red 1.366% and FWHM 0.581 Å, so the two choices differ by less than 0.01 Å in width. What produces the extra width in the published curve has not been identified.

Two things here cost time to find.

First, the curves drawn over the spectrum in the paper's Fig. 3 are the three-Gaussian empirical fits, not p-winds; the physical model is overlaid only in Fig. 4. The close agreement in Fig. 3 is a fit to the data, and says nothing about the model's line shape.

Second, the paper's plotted model is visibly broader than a plain p-winds run. We first read that as an undocumented instrumental kernel, and it is not one. Their script turns the turbulent broadening term on and averages over five transit phases. Both widen the line, and together they account for the difference.

8. Turbulent broadening: what the literature does

Since that switch turned out to matter, all 43 papers held with this study were checked: every PDF converted to text, searched for "turbulen", "microturbulen" and "non-thermal broadening", and each hit read in context. Fifteen matched; most use "turbulent" for eddy diffusion K_{zz} or for flow structure in 3D simulations, not for line broadening. Section 8.1 tabulates the ones that bear on broadening.

The convention, stated most explicitly by Yan et al. (2024), replaces the thermal Doppler width by

$$\Delta\nu_D = \frac{\nu_0}{c}\left(\frac{2kT}{m} + v_{\rm turb}^2\right)^{1/2}, \qquad v_{\rm turb} = \sqrt{\frac{5kT}{3m}} = c_s, \tag{4}$$

the monatomic sound speed, as assumed by Salz et al. (2018), Lampón et al. (2020) and Czesla et al. (2022). The p-winds implementation uses $\sqrt{5kT/6m}$ against a Doppler width that lacks the factor 2, which is the same physics; in the convention EXHALE uses for the Doppler parameter the same term is $b \to \sqrt{11/6}\,v_{\rm th}$, a factor 1.354. That is what the optional turbulence term in our transit tool applies. It is opt-in and off by default, since the field does not agree that it belongs.

8.1 Paper by paper

Adopt it as an assumed value. Salz et al. (2016), Eq. (5): "b is the Doppler parameter, $v_{\rm th}$ the thermal Doppler velocity, and for the microturbulence $v_{\rm micro}$ we use the sound speed", applied when synthesizing the transit profile. Their atmosphere solver separately carries a constant $1\,{\rm km\,s^{-1}}$ microturbulence at the lower boundary, a different quantity, which they note "has little impact on the overall atmospheric structure". Lampón et al. (2020, 2021) is the reference implementation, with turbulent broadening a free or assumed parameter of 1D isothermal Parker fits, and Czesla et al. (2022) follows it. Cherubim et al. (2026) inherits the choice by leaving the p-winds turbulence term on, so the LHS 1140 b retrieval carries it. The paper text does not say so anywhere.

Test it, then leave it out. Yan et al. (2024) give the cleanest statement of the convention and of the doubt. They record that "these authors assumed that the turbulence velocity $v_{\rm turb} = \sqrt{5kT/3m}$, which corresponds to the speed of sound of a monatomic gas and, thus, to the transonic case with a Mach number of 1", then keep it out of their nominal models and test $v_{\rm turb} = 0.5$, 1.0 and $1.5\,c_s$ as a sensitivity. Their finding for He 10830 is that turbulence "decreases the absorption depth slightly at the line center, but doesn't significantly alter the absorption at the line wings". They argue Mach 1 is likely an overestimate, since in the diffuse ISM turbulent velocity anti-correlates with temperature and the Mach number is below unity at a few thousand K. They conclude that "the turbulence effect is likely less significant than the one explored in this paper". Yan et al. (2019) computes the Doppler width explicitly "assuming no turbulence".

Argue against it. Murray-Clay et al. (2009), on García Muñoz's (2007) proposal to broaden Lyα by wind turbulence: "to generate the large velocities observed, the energy in such turbulence would need to exceed the thermal and bulk kinetic energies in the mean flow by a factor of ~ 100. Such energy requirements seem insurmountable." Koskinen et al. (2013b) find that "non-thermal broadening such as that proposed by Ben-Jaffel and Hosseini (2010) does not appear to be necessary to explain the current observations", while allowing that stellar-wind interaction may produce broadening

turbulence. Schulik & Owen (2025) offer a physical alternative, in which adiabatic cooling produces a metastable-helium population bump at high velocity and widens the line with no free parameter, and they are direct about the isothermal case: that model "can only explain the broadness of the data under physically inconsistent assumptions, i.e. those of even larger mass-loss rates, not supported by the energy input into this planet, and an empirical free parameter, turbulent broadening (Lampón et al. 2021)". They add that $T = 10^4$ K already maximizes what thermal broadening alone can give.

Need it even so. Taylor et al. (2026) is the closest analogue to the present work: a self-consistent model (energy balance, multispecies diffusion, non-LTE H($n = 2$)) that reproduces the HD 189733 b line core but not its width. They write that "several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework", and quantify it as $\sim 12\,\mathrm{km\,s^{-1}}$ of additional non-thermal broadening. They suggest, calling the suggestion speculative, that it is plasma turbulence from ions trapped on low-latitude planetary field lines, strengthening with stellar activity. Rumenskikh et al. (2022), as reported there, instead imposed reduced hydrogen line cooling and prescribed zonal jets of 10–20 $\mathrm{km\,s^{-1}}$ in a 3D multifluid model.

Matches that are not about broadening. Recorded so the sweep need not be repeated: Wogan (2025), Lavvas (2014, 2019), Robeling (2026) and the second half of Koskinen et al. (2013b) use "turbulent" for eddy diffusion K_{zz}; the 2026 preprint arXiv:2607.18193 uses it for flow-flow interaction in a 3D stellar-wind simulation; Jensen (2012) for photospheric turbulence driving stellar activity; Taylor et al. (2025) for gravity-wave breaking. The Vidal-Madjar et al. (2003) *Nature* letter matches only on its first page, which carries the tail of the preceding letter, on Crab pulsar radio emission; the authors themselves do not discuss turbulence.

8.2 Empirical baseline: turbulence in Solar System giant planets

Whether sonic or supersonic ($v_{\mathrm{turb}} \sim c_s$) turbulence is physically plausible in a planetary atmosphere can be checked against the measured turbulence and non-thermal velocity fields in the stratified layers of Jupiter and Saturn:

- *Troposphere and cloud deck ($\sim$ 0.1–1 bar).* Cloud-tracking measurements from *Voyager*, *Galileo*, *Cassini*, and *Juno* (e.g., Salyk et al. 2006; Del Genio et al. 2007; Ingersoll et al.) isolate the fluctuating eddy velocities $v' = \sqrt{\overline{u'^2 + v'^2}}$ from the mean zonal jets (100–500 $\mathrm{m\,s^{-1}}$). The resulting RMS turbulent eddy velocity is $v' \approx 3$–$10\,\mathrm{m\,s^{-1}}$ on Jupiter (locally 20–30 $\mathrm{m\,s^{-1}}$ in the Great Red Spot shear zone) and 5–15 $\mathrm{m\,s^{-1}}$ on Saturn. Compared to the tropospheric sound speed ($c_s \approx 800$–$900\,\mathrm{m\,s^{-1}}$), this is deeply subsonic turbulence, with Mach numbers $\mathcal{M} \sim 0.005$–0.02.
- *Stratosphere ($\sim$ 10 μbar–10 mbar).* High-resolution submillimeter spectroscopy of impact-delivered HCN/CO molecules with ALMA (e.g., Cavalié et al. 2021) and Herschel reveals stratospheric jet velocities up to 300–400 $\mathrm{m\,s^{-1}}$ in the auroral zone, but the associated non-thermal velocity dispersion (turbulence and wave motions) remains limited to tens of $\mathrm{m\,s^{-1}}$ up to $\sim$ 100–200 $\mathrm{m\,s^{-1}}$ ($\mathcal{M} \lesssim 0.2$).

- *Thermosphere and auroral ionosphere ($\lesssim 1\,\mu$bar).* Ground-based high-resolution ($R \sim 10^5$) infrared spectroscopy of H_3^+ line profiles (e.g., with VLT/CRIRES, Keck, and CFHT) disentangles thermal Doppler broadening from non-thermal velocity fields. In non-auroral mid-latitudes, the non-thermal velocity is $v_{\rm nth} \approx 0.2$–$0.5\,{\rm km\,s^{-1}}$. In the auroral oval, where intense magnetosphere–ionosphere coupling, ion drag, and Joule heating inject substantial kinetic energy, non-thermal broadening reaches $v_{\rm nth} \approx 0.8$–$2.0\,{\rm km\,s^{-1}}$.

These measurements show that pure hydrodynamic planetary turbulence is strictly subsonic ($\mathcal{M} \ll 1$), and that even strong magnetospheric auroral driving in upper thermospheres caps non-thermal broadening at $\lesssim 2\,{\rm km\,s^{-1}}$.

8.3 Turbulence and 3D hydrodynamic simulations of exoplanets

Beyond 1D parameterizations, 3D simulations show how turbulent mixing, non-thermal velocity structure and line broadening arise in several classes of exoplanet:

1. 3D hot Jupiter escape and circulation.

- *3D atmospheric escape and kinematic line broadening.* Hydrodynamic 3D escape simulations (e.g., McCann et al. 2019; Debrecht et al. 2020; Carolan et al. 2020, 2021; Wang & Dai 2021; MacLeod & Oklopčić 2022) show that apparent transmission line broadening comes mainly from large-scale 3D kinematic velocity gradients (10–50 km s^{-1}) across the day-to-night outflow, Coriolis-deflected tails, and tidal distortion, rather than from unresolved isotropic microscale turbulence. The collision between the expanding planetary wind and the incoming stellar wind creates a shear layer prone to Kelvin–Helmholtz instabilities and turbulent mixing at the contact discontinuity. In 3D multifluid models of HD 189733 b, Rumenskikh et al. (2022, 2023) show that upper-atmospheric zonal jets of 10–20 km s^{-1} combined with 3D outflow geometry reproduce the observed broad transmission profiles without invoking ad hoc microturbulence.

- *3D general circulation models (GCMs) and macroturbulent mixing.* In the observable lower-to-middle atmosphere (1 bar to $\sim 1\,\mu$bar), 3D GCMs (e.g., Showman & Polvani 2011; Showman et al. 2013, 2015; Parmentier et al. 2013; Charnay et al. 2015; Zhang & Showman 2018) show that equatorial superrotation (1–5 km s^{-1}) and vertical mixing ($K_{zz} \sim 10^7$–10^{11} cm^2 s^{-1}) are driven by planetary-scale macroturbulence (non-linear interactions between planetary-scale Kelvin and Rossby waves and the mean flow) and overturning circulation. The sub-grid parameterization of small-scale turbulent dissipation (hyperdiffusion, Smagorinsky closures, or Shapiro filters) remains a known source of numerical sensitivity in GCM wind speeds and jet stability (Heng et al. 2011; Polichtchouk et al. 2014; Skinner & Cho 2021).

- *Magnetohydrodynamic (MHD) turbulence in ultra-hot atmospheres.* For planets with $T \gtrsim 1500$ K, thermal ionization of alkali metals (Na, K) couples the gas to the planetary magnetic field. 3D MHD simulations (e.g., Perna et al. 2010, 2012; Rauscher & Menou 2013; Rogers & Showman 2014; Rogers & Komacek 2014; Beltz et al. 2022) show that magnetic winding of the field by supersonic zonal winds generates Lorentz forces (magnetic drag) that brake the flow, producing 3D MHD turbulence, Ohmic dissipation, and time-dependent variability in wind patterns and hotspot offsets.

2. 3D simulations across other planetary classes. 3D fluid simulations reach well beyond hot Jupiters, into the other main classes of exoplanet and substellar object:

- *Warm Neptunes and sub-Neptunes.* For warm Neptunes such as GJ 436 b and GJ 3470 b, 3D particle-mesh and hydrodynamic escape simulations (Bourrier et al. 2015, 2016; Villarreal D'Angelo et al. 2021; Carolan et al. 2021) give large asymmetric cometary hydrogen tails spanning tens of planetary radii, which account for the 56% Lyα transit obscuration and the He 10830 Å absorption through 3D stellar radiation pressure and charge-exchange interactions. In metal- and water-rich sub-Neptune atmospheres (e.g., GJ 1214 b, K2-18 b), 3D GCMs model cloud/haze circulation, photochemical transport, and vertical eddy mixing ($K_{zz} \propto P^{-0.4}$; Charnay et al. 2015; Innes & Pierrehumbert 2022).

- *M-dwarf habitable zone terrestrial planets.* Tidally locked rocky worlds (such as TRAPPIST-1 e/f/g, Proxima b, and LHS 1140 b) have been modeled extensively with 3D climate GCMs (Turbet et al. 2016, 2018; Wolf et al. 2017; Sergeev et al. 2020, 2022; the THAI intercomparison, Fauchez et al. 2020, 2022). These simulations show how thick dayside convective cloud decks raise the planetary albedo and prevent runaway greenhouse states, while stationary Rossby–Kelvin wave patterns redistribute heat to the nightside and keep the atmosphere from freezing out. Complementary 3D multi-species MHD simulations (Dong et al. 2017, 2018; Cohen et al. 2014) assess O^+ and H^+ ion escape driven by extreme stellar coronal mass ejections and flaring winds.

- *Lava worlds and ultra-short-period rocky planets.* For scorched rocky planets with dayside magma oceans (e.g., 55 Cancri e, K2-141 b, CoRoT-7 b), 3D hydrodynamic models of mineral-vapor atmospheres (Castan & Menou 2011; Nguyen et al. 2020; Zilinskas et al. 2022; Kang et al. 2021) show supersonic dayside-to-nightside winds ($\mathcal{M} > 1$–3, $v \sim 1$–3 km s^{-1}), standing hydraulic jumps/shocks, and mineral vapor condensation leading to rock rain on the nightside.

- *Brown dwarfs and directly imaged giant planets.* In isolated brown dwarfs and young, directly imaged giants (e.g., HR 8799 b/c/d/e, VHS 1256 b), 3D convective-radiative GCMs (Tan & Showman 2019, 2021a,b; Showman & Kaspi 2013; Zhang & Showman 2014) couple rapid planetary rotation (strong Coriolis forces) with deep geothermal convection and silicate/iron cloud radiative feedbacks. The resulting turbulent vortices, patchy cloud bands, and storm dynamics explain the observed rotational photometric variability.

8.4 Where that leaves this comparison

There is no settled treatment. Turbulence at $v_{\rm turb} = c_s$ is a common convention in 1D isothermal Parker-wind fits, and hence in the paper being reproduced, while Solar System observations and self-consistent or 3D models show that isotropic microscale turbulence cannot reach the required supersonic levels. 3D models instead supply line width through kinematic flow gradients, zonal jets, and magnetospheric or plasma interactions. Turning the $v_{\rm turb} = c_s$ term on in EXHALE is therefore right for a like-for-like comparison with C26 and wrong as a physical default, which is how it is implemented.

9. Searching for a match: the composition scan

Note added 2026-10-04. The crossings of this section (prescribed metal-free base, 2026-09 physics without heat conduction in the model) were not recomputed with the present equations. On the present equations with the solved photochemical column there is no crossing (Sect. 1.2).

Once the star, the planet and the spectrum are fixed, composition is the free parameter left to search over. This section searches it in the well-mixed limit, with element diffusion off, so that the ratio in the reservoir is also the ratio everywhere in the wind. That is the simpler of the two mixing assumptions available. Sect. 12.4 repeats the same scan with binary H/He diffusion active, where the answer depends on the eddy coefficient as well.

Eleven compositions were solved, from solar (He/H = 0.083) to He/H = 10^4, each a converged steady state with a residual $\|R\| < 10^{-3}$. One case, He/H = 100, stalls at 1.18×10^{-3} and is recorded as such. Nothing is imposed: mass, momentum and energy are solved with photoionization heating and radiative cooling, so $\dot{M}$ and $T(r)$ are outputs.

He/H	$\log_{10}\dot{M}$ [g s^{-1}]	red [%]	red (turb) [%]	FWHM [Å]	EW [% Å]
0.083	7.77	1.13	1.02	0.233	0.279
0.25	7.75	2.49	2.21	0.244	0.645
0.50	7.76	3.83	3.35	0.255	1.040
0.55	7.76	4.05	3.54	0.257	1.109
0.70	7.78	4.71	4.09	0.261	1.311
1	7.80	5.59	4.83	0.268	1.598
10	7.76	7.12	5.97	0.298	2.270
10^3	7.74	5.46	4.55	0.315	1.844
10^4	7.73	5.40	4.49	0.316	1.825
observed	-	1.254	-	0.841	1.108 ± 0.030

Nine of the eleven runs are listed; He/H = 0.6 falls between two tabulated entries and He/H = 100 is the case whose residual stalls. The three helium-dominated rows are corrected values. Their depths and equivalent widths were an artifact of the older advection correction, which pinned the metastable population to its equilibrium value across the hydrogen ionization front. They fell by 89–93% when the correction was rebuilt around the slowest species it solves for (at He/H = 10, EW 21.53 $\rightarrow$ 2.270 % Å). The crossing of Sect. 11. is interpolated inside He/H ≤ 1 and is untouched by that correction. The escape rate itself is nearly composition-independent, $\dot{M} \simeq 6 \times 10^7$ g s^{-1}, about a third of the 2.03×10^8 the retrieval imposes.

This table is the 2026-08-30 scan, kept as the record of what was run then and not as the current value of any entry. It predates both the replacement of the He($2\,^3S$)+H rate coefficient of Sect. 6.5 and every change of Sect. 5.. The current well-mixed sequence is the nine solved compositions of the well-mixed group, He/H = 0.083 to 1000, and the crossing of Sect. 11. is read from it. For one point of comparison, He/H = 0.55 re-solved on the current code gives EW = 1.3118 % Å and FWHM 0.2598 Å with $\log_{10}\dot{M}$ = 7.85, against the 1.109 and 0.2573 of this table.

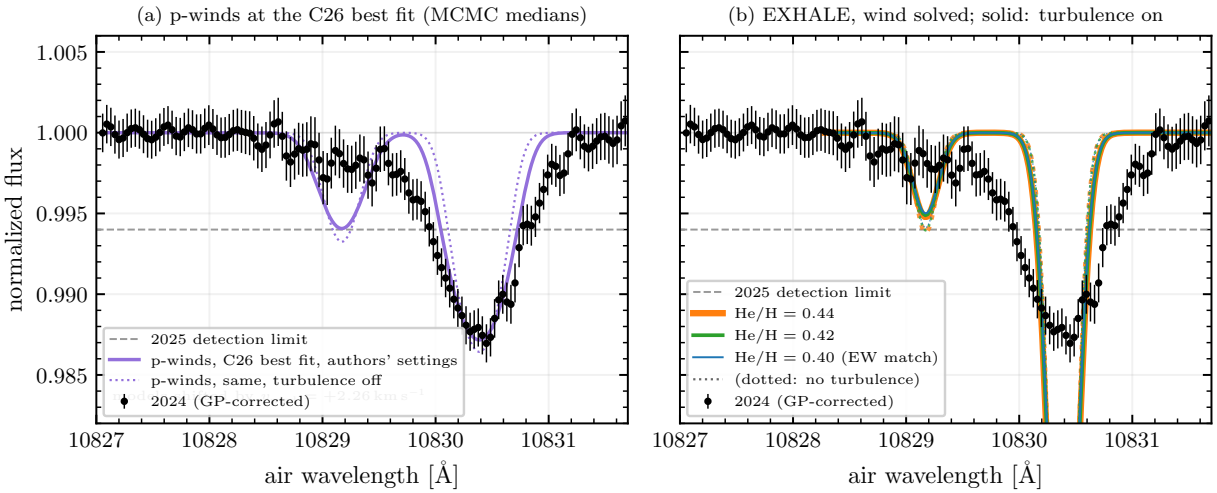


Figure 8: The measurement is in black. On the left is the paper's own p-winds model at its best fit, the MCMC medians its Fig. 4 legend states, computed with the authors' released settings; that model is the solid curve and the dotted one is the same run with turbulence off. On the right is EXHALE at the three well-mixed compositions of record that bracket the equivalent-width crossing 0.4044: He/H = 0.40, the solved case nearest the crossing, with 0.42 and 0.44. For each of them a second spectrum was synthesized on the same solution with the turbulence term on, so the solid EXHALE curves carry the same turbulence term p-winds uses and the dotted ones are without it. Over that range of composition the profile barely moves, so the three nearly coincide; the composition dependence of the absorbing power is Fig. 11. Neither code applies a line-of-sight bulk velocity, so both sets of model curves are shifted by the $v_{\mathrm{wind}} = +2.26\,\mathrm{km\,s^{-1}}$ that C26 fit. The shift is applied to the plotted curves only: every equivalent width in this memo is computed on the unshifted spectra, the observation in the planetary rest frame of its released file. The figure is drawn in the presentation of the paper's Fig. 4.

10. Why the two models disagree

Note added 2026-10-04. The EXHALE profiles of this section are 2026-09 states; the p-winds side is unaffected. The present profiles are cooler at the line-forming radii and carry their ionization by transport (Sects. 1. and 2.).

At the same composition the two models are not small perturbations of each other. The *total* metastable column is comparable (at H : He = 10^{-3} EXHALE's radial column over 1–20 R_p is 1.04 of p-winds'), but its distribution is not, and transit depth is absorbing *area*:

	p-winds	EXHALE
50% of the 2^3S column inside	$1.31\,R_p$	$5.7\,R_p$
90% inside	$1.58\,R_p$	$11.7\,R_p$

Both radii are read from the metastable profile in local equilibrium, which is what Fig. 9 draws. The advection-corrected profile that the transit integral consumes is more concentrated, and for this helium-dominated composition it collapses onto the base once the correction is built around the metastable level's own timescale rather than hydrogen's. This section is therefore a statement about the two wind solutions; every transit number quoted elsewhere in the memo is measured on the advection-corrected profiles.

The star is $13.6\,R_p$ across, so metastable helium spread to $10\,R_p$ covers most of the disk while a column confined to $1.6\,R_p$ can darken only a narrow ring. The difference in distribution comes from the wind structure, which p-winds imposes and EXHALE solves. Take $r = 10\,R_p$ at the same composition and spectrum. p-winds is isothermal at 6100 K by construction, while EXHALE's energy balance gives 1499 K. p-winds flows at 4.3 km s^{-1} and EXHALE at 0.51. The cold, slow, denser gas keeps helium neutral and the metastable level populated far out, $n(2^3S) = 48$ cm^{-3} against 2×10^{-4} (Fig. 9). That radius is at the edge of what these $30\,R_p$ solutions resolve (Sect. 16.). The same three comparisons at $5\,R_p$, well inside it, read 6100 against 3102 K, 1.9 against 0.20 km s^{-1} and 1.3×10^{-2} against 47 cm^{-3}.

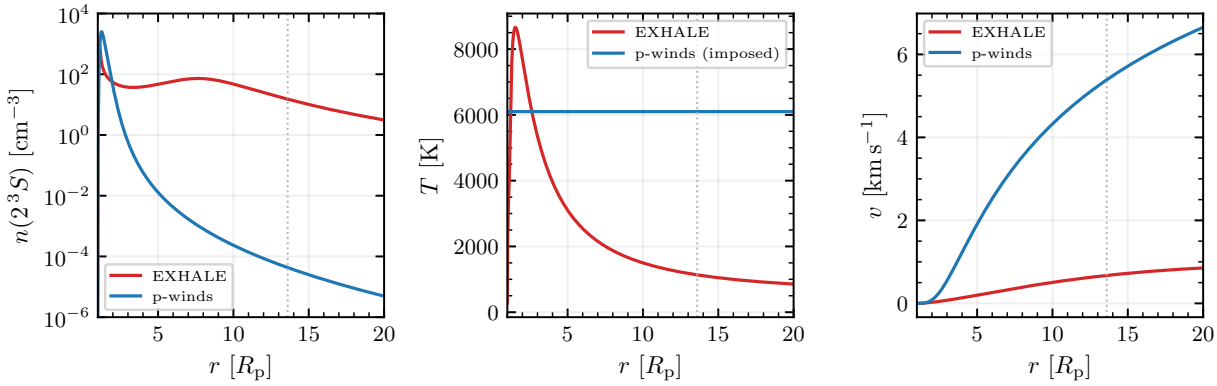


Figure 9: Same composition (H : He = 10^{-3}), same spectrum, two wind structures. Dotted vertical line: the stellar radius, $13.6\,R_p$.

10.1 Density and ionization, the same pair

Figure 9 draws three of the five profiles the comparison was set up to make; the other two, mass density and hydrogen ionization, are in Fig. 10, from the same pair of solutions.

The densities are the part that agrees. Outside the base the ratio EXHALE / p-winds runs from 0.15 at $1.2\,R_p$ through unity near $2\,R_p$ to 2.7 at $5\,R_p$, and stays inside a factor 2.3 from $3\,R_p$ out. At the base cell itself EXHALE sits 176× higher, which is the two lower boundary conditions and not a profile difference: p-winds sets its base density from the imposed $\dot{M}$ through the Parker solution, EXHALE from its own base state.

The ionization does not agree at all. EXHALE has hydrogen completely ionized above $1.03\,R_p$, with a neutral hydrogen density of exactly zero in the output, while p-winds keeps it 91–94% neutral throughout, $x(\mathrm{H\,II})$ rising only from 0.056 at $1.2\,R_p$ to 0.086 at $20\,R_p$. Neither end of that is the other's error, and both need an explanation.

p-winds' hydrogen is frozen far below its own photoionization equilibrium. Integrating the same spectrum against the Verner et al. (1996) ground-state cross section gives an unattenuated rate $\phi = 5.17 \times 10^{-8}\ \mathrm{s^{-1}}$ (p-winds' internal value for this file agrees to 0.1%), an ionization time of 224 d, while the flow time r/v through the imposed $\dot{M} = 2.03 \times 10^{8}\ \mathrm{g\,s^{-1}}$ falls to 2.6×10^{5} s at $2\,R_p$ and 2.6×10^{4} s at $10\,R_p$. Equilibrium on p-winds' own density and temperature would be $x(\mathrm{H\,II}) = 0.76$ at $2\,R_p$ and 0.997 at $5\,R_p$ (dotted in Fig. 10b); the wind carries the gas out before it gets there. Shielding is not what holds it down: hydrogen is a trace species here and the neutral hydrogen column above $1.2\,R_p$ is optically thin at every energy.

EXHALE's hydrogen goes the other way: it is more ionized than any photoionization balance on its own structure allows. With the same ϕ, its own electron density at $3\,R_p$ ($2.0 \times 10^{6}\ \mathrm{cm^{-3}}$, set by helium) and its own temperature there (5247 K), primary photoionization alone gives $x(\mathrm{H\,II}) = 0.059$: close to what p-winds carries, and nowhere near 1. An exact zero can only come from a hydrogen ionization source that does not vanish with the neutral hydrogen density, and the code has one. The Shull & van Steenberg (1985) secondary channel adds to the hydrogen photoionization rate a term $R_{\mathrm{sec,HI}}/n(\mathrm{H\,I})$, where $R_{\mathrm{sec,HI}}$ is accumulated over the photoelectrons of every absorber. In a helium-dominated gas that numerator is helium's. Evaluating it at $3\,R_p$ on this solution's own radiation field (the He I photoelectrons above the code's 40 eV secondary threshold, with $f_{\mathrm{ion,HI}} = 0.091$ at the local ionized fraction $x = 0.246$) gives $R_{\mathrm{sec,HI}} = 4.9 \times 10^{-2}\ \mathrm{cm^{-3}\,s^{-1}}$ against a hydrogen recombination rate of 6.6×10^{-3}, a factor 7.5. Complete ionization follows.

That is an approximation used outside the composition it was made at. The Shull & van Steenberg fits are functions of the ionized fraction alone; the helium abundance of the gas their Monte Carlo followed is fixed inside them and is not the $\mathrm{He/H} = 10^{3}$ of this run. Their two branches hand hydrogen 7× helium's share of the secondary ionizations ($f_{\mathrm{ion,HI}}/f_{\mathrm{ion,HeI}} = 0.3908/0.0554$ at $x \to 0$), where the abundances say the opposite by three orders of magnitude. It is recorded here rather than changed, because the reach was measured and is bounded. The neutral hydrogen density reaches an exact zero only at $\mathrm{He/H} \geq 10$ over a prescribed metal-free base: above $6.6\,R_p$ at $\mathrm{He/H} = 10$, above 1.06 at 100 and above 1.03 at 1000. It does not happen in any of the solutions this memo quotes science from: $\mathrm{He/H} = 0.55$ well mixed, 2.13 with diffusion at $K_{zz} = 10^{9}$, and every flux-consistent solution out to $\mathrm{He/H} = 12$, where $x(\mathrm{H\,II})$ at $2\,R_p$ runs 0.060 to 0.178 and no cell is fully ionized. The $\mathrm{H:He} = 10^{-3}$ comparison of this section is the only place in the memo that sits inside it, and the quantity it distorts is hydrogen rather than the metastable helium the line is made of.

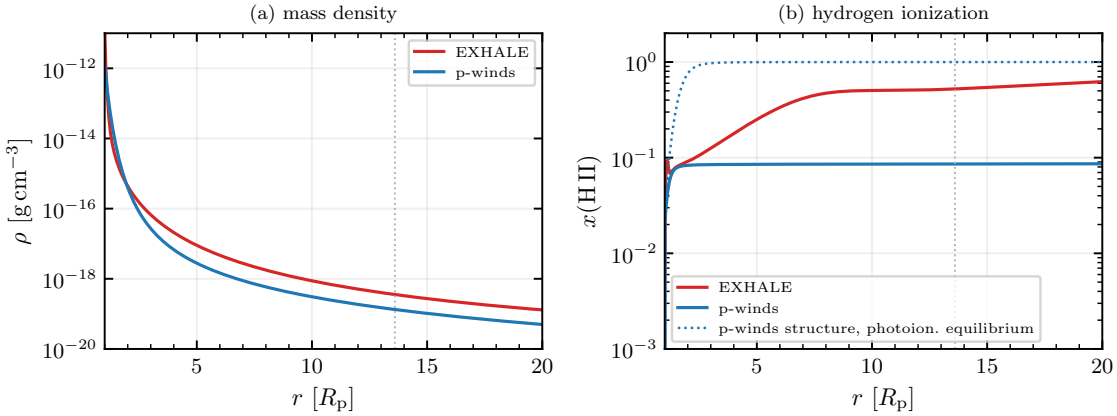


Figure 10: The other two profiles of the same comparison (H : He = 10^{-3}, same spectrum): (a) mass density, (b) hydrogen ionization. The dotted curve in (b) is photoionization equilibrium evaluated on p-winds' own density and temperature, so its distance from the solid p-winds curve is how far the imposed flow holds hydrogen out of equilibrium. Dotted vertical line: the stellar radius.

11. What can be matched: the equivalent width

Note added 2026-10-04. The equivalent widths here are of the 2026-09 states. On the present equations the amount of absorption is not matched at any He/H (Table 1).

Depth and width trade against each other under any broadening, but the equivalent width does not: it is unchanged by the instrument profile, by the turbulence term and by the phase averaging. It is therefore the quantity on which a self-consistent model can be matched without first settling the broadening question.

Over the blended red pair (vacuum 10832.6–10834.2 Å), the released spectrum gives $EW = 1.108 \pm 0.030$ % Å: the trapezoid over its own 44 samples in the window (10832.622 to 10834.175 Å). Since 2026-09-23 every model equivalent width in this memo is measured through that same operator: the model curve (instrument-convolved until 2026-10-06, since then the planet-rotation and instrument-convolved column 4; the two differ by 10^{-6} %Å for this planet) is interpolated linearly onto those 44 samples and integrated with their trapezoid weights. Before that date each model was integrated with a trapezoid on its own, finer grid inside the nominal window. The two differ by at most 1.5×10^{-4} % Å on the unbroadened EXHALE spectra of the set of models and 1.8×10^{-4} on the unbroadened p-winds curves, and by up to 0.6 per cent on the curves broadened by the added Gaussians of Sect. 12.2, whose wings reach the window edges. EXHALE crosses the measured value at

$$\mathrm{He/H} = 0.4044 \quad (0.3877\text{–}0.4237 \text{ on the } EW\ 1\sigma), \qquad \mathrm{H:He} = 2.5, \tag{5}$$

that is 4.9× solar, a helium mass fraction of 0.62. The crossing is read between two solved compositions, 0.40 at $EW = 1.1011$ and 0.42 at 1.1324 % Å against the observed 1.1084 ± 0.0295, in the well-mixed group; the case at 0.40 is 0.25σ from the measurement. Turning the turbulence term on moves the crossing to 0.3985, 1.5 per cent away, so the estimate does not depend on the broadening treatment. The value sits three orders of magnitude away from the paper's H : He $\lesssim 10^{-3}$, and the reason is the structural difference of the previous section: a wind that spreads metastable helium

over most of the stellar disk needs far less helium to absorb as much.

Two assumptions are carried inside that number and are taken up later: the gas is well mixed, and the spectrum is the GJ 1132 proxy. Sect. 12.4 rescans it with element diffusion active at every decade of the eddy coefficient, and Sect. 14. rescans it on the brighter proxy. Both move it, the spectrum by much the larger factor.

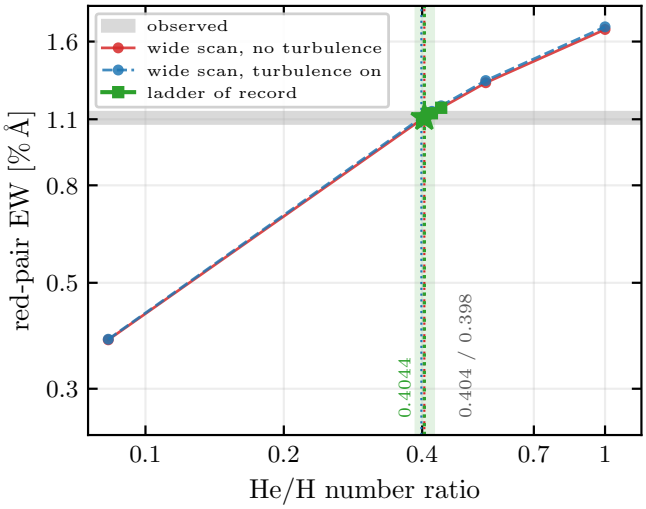


Figure 11: Red-pair equivalent width against composition, with and without the turbulence term, compared with the measured band. The crossing is insensitive to the broadening treatment: each solved composition of the well-mixed group up to He/H = 1 carries both syntheses of the same solution, so the two curves differ only in the broadening, and they cross the measured band at He/H = 0.4044 without the turbulence term and 0.3985 with it, 1.5 per cent apart. The three compositions 0.40, 0.42 and 0.44 drawn on top are where the spacing brackets the measurement directly.

12. What cannot be matched: the profile

Note added 2026-10-04. The conclusion of this section stands: the width demand on every 2026-10 state is f = 21.92–22.28 km s^{-1} (Sect. 1.3). Its EXHALE numbers are of the 2026-09 states. One statement no longer holds: the broadened line now has the right shape but is 21 per cent too shallow, because the equivalent width is no longer matched.

Matching the equivalent width is not matching the line. At the crossing He/H = 0.4044 the profile is 3.3 times too deep (4.09% against 1.254%) and 3.3 times too narrow (0.253 against 0.841 Å). The equivalent width states how much absorbing power the model has. The profile states how that power is spread in velocity, and composition does not act on the second: every solved composition gives 0.23–0.32 Å, rising only to 0.26–0.39 Å with the turbulence term.

12.1 How much is missing, in velocity

The measured FWHM is 0.841 Å, that is 23.3 km s^{-1} at 10833 Å. Three ways to supply it, and what each would require:

- *Thermal alone.* A Doppler width of 23.3 km s^{-1} FWHM in helium needs $T = 4.7 \times 10^4$ K. Helium is fully ionized far below that, so there would be no metastable population left to absorb. This is excluded.

- *Bulk outflow.* A line-of-sight velocity spread of order $\pm 12\,\mathrm{km\,s^{-1}}$ across the line-forming region would do it. The EXHALE solution matched on equivalent width reaches $1.1\,\mathrm{km\,s^{-1}}$ at $20\,R_p$; the paper's p-winds model, with its imposed temperature, reaches 4–7. Both are an order of magnitude short.
- *Non-thermal (turbulent) broadening.* Added in quadrature to what EXHALE already has, closing the gap takes $22.2\,\mathrm{km\,s^{-1}}$ FWHM: $9.4\,\mathrm{km\,s^{-1}}$ of turbulent velocity (σ), against the $2.9\,\mathrm{km\,s^{-1}}$ that the sound-speed convention of Sect. 8.1 supplies for helium at 5000 K. Of the three, this is the one that can be measured directly, and Sect. 12.2 measures it.

The model is short by an amount of velocity comparable to the escape speed, and no choice of composition supplies it.

12.2 The broadening the data demand, measured

Note added 2026-10-04. The convention of this subsection is now the adopted rule (user decision 2026-10-04): every model line compared with an observed transit profile is convolved with a Gaussian whose FWHM f is solved so that the extractor returns the observed width, and its equivalent width is the window equivalent width of that broadened curve (update log section 87). The numbers in the table below are of the 2026-09 solution at $\mathrm{He/H} = 0.40$.

The three routes above can be replaced by one measurement. Take the EXHALE profile matched on equivalent width as solved, convolve it with a Gaussian of FWHM f in velocity, and ask which f reproduces the measured line. The convolution is applied to the excess absorption $(\max T - T)/\max T$ built from the planet-rotation and instrument-convolved synthetic spectrum (column 4 of the curve file; the instrument-only column 3 up to 2026-10-05, the two differ by 10^{-6} %Å for this planet). Depth, width and the red/blue ratio are then read off with the same three-Gaussian extractor used on the data, and the equivalent width through the same operator as the observed one (Sect. 11.): the broadened curve interpolated onto the observation's samples in the vacuum window 10832.60–10834.20 Å and integrated with their trapezoid weights. Only f is varied, and every number in Table 8 is measured the way the corresponding observed number was.

Table 8: The EXHALE profile matched on equivalent width (the solved composition $\mathrm{He/H} = 0.40$ of the well-mixed group, nearest the crossing at 0.4044; no turbulence term) convolved with an added Gaussian of FWHM f. Observed values on the last row: red and blue depths from the C26 MCMC posterior, width and equivalent width measured here on their spectrum. The convolution, the extractor and the window are the ones used for every figure in this memo.

f [$\mathrm{km\,s^{-1}}$]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.085	0.586	6.97	0.253	1.101
10	2.346	0.313	7.49	0.440	1.101
15	1.730	0.228	7.59	0.597	1.097
20	1.351	0.177	7.64	0.765	1.079
22.2	**1.229**	**0.161**	**7.65**	**0.841**	**1.065**
25	1.103	0.144	7.66	0.938	1.042
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

The width is reached at $f = 22.22\,\mathrm{km\,s^{-1}}$ FWHM, that is $\sigma = 9.44\,\mathrm{km\,s^{-1}}$. At that value the depth is 1.229% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.161% against $0.25^{+0.14}_{-0.12}$, and the ratio 7.65 against $6.7^{+12.7}_{-3.1}$, so all three fall inside the observed 1σ interval at the value fixed by the width alone. None of the three was tuned. Figure 12 shows the profile and the two crossings.

The number does not depend on which solution is broadened. The neighboring compositions of the same group, He/H = 0.42 and 0.44, ask for 22.21 and $22.20\,\mathrm{km\,s^{-1}}$. The $K_{zz} = 0$ run at He/H = 3.0 asks for 22.23 and the $K_{zz} = 10^9$ run at 1.50 for 22.22. Solar composition asks for 22.40, although that run carries a third of the observed absorption (1.533% red, EW 0.380) and so matches the shape while missing the amount. All six are solved cases, and the spread over them is 0.9 per cent. [2026-09-24] The molecular $K_{zz} = 10^9$ solutions that straddle the measurement, He/H = 1.45 and 1.52, ask for 22.21 and $22.20\,\mathrm{km\,s^{-1}}$, and the whole certified molecular ladder from 0.55 to 2.13 for 22.12 to 22.32. The required broadening is therefore set by the velocity field rather than by composition or by the broadening treatment already in place, which is what Sect. 12. inferred. The equivalent width falls by 3% between $f = 0$ and $f = 22.2$ because the Gaussian wings carry absorption out of the red-pair integration window, not because absorbing power is lost.

This is a description, not an explanation. One free parameter applied to a solved profile reproduces the measured line, and nothing in that says what produces $22.2\,\mathrm{km\,s^{-1}}$. The candidates are listed below, and the measurement sharpens them by fixing the number each has to deliver.

One conversion narrows that list. If the missing $22.2\,\mathrm{km\,s^{-1}}$ is called turbulence, it has to be turbulence in gas whose sound speed the solution itself fixes. Evaluating $c_s = (\gamma kT/\mu m_H)^{1/2}$ with $\gamma = 5/3$ on the advection-corrected profiles of the well-mixed He/H = 0.55 solution, taking μ from the solved H I/H II/He I/He II/He III and electron densities, gives $c_s = 5.3\,\mathrm{km\,s^{-1}}$ at $2R_p$ ($T = 4108\,\mathrm{K}$, $\mu = 2.00$), 5.6 at the metastable peak $1.53R_p$ (4607 K, 2.03) and 3.2 at $10R_p$ (1437 K, 1.90); weighted over the line-forming region by $n(2\,^3S)\,r$ (integrated over r), $T = 2498\,\mathrm{K}$, $\mu = 1.93$ and $c_s = 4.2\,\mathrm{km\,s^{-1}}$ [corrected 2026-09-24: the value printed before, 4.1, did not follow from its own T and μ, and the Mach numbers below moved with it]. This is the sound speed of the bulk gas, the quantity a Mach number is defined against, not the helium thermal speed that enters the line profile.

Against that, the Gaussian is supersonic on any convention. Read as a line-of-sight velocity dispersion, $\sigma = 9.44\,\mathrm{km\,s^{-1}}$ is $\mathcal{M} = 2.2$. Read in the convention of Lampón et al. and p-winds, where v_{turb} is the quantity added to $b^2 = 2kT/m + v_{\mathrm{turb}}^2$, it is $v_{\mathrm{turb}} = \sqrt{2}\,\sigma = 13.4\,\mathrm{km\,s^{-1}}$, $\mathcal{M} = 3.2$; as an isotropic three-dimensional turbulent speed, $\sqrt{3}\,\sigma = 16.3\,\mathrm{km\,s^{-1}}$, $\mathcal{M} = 3.9$. The $v_{\mathrm{turb}} = c_s$ convention of Sect. 8.1 supplies only 3–5 $\mathrm{km\,s^{-1}}$ in this gas, about a third of what the data demand. The literature reviewed there points the other way: Yan et al. (2024) argue $\mathcal{M} = 1$ is already an overestimate at a few thousand K, and Murray-Clay et al. (2009) call the energy requirement of supersonic turbulence "insurmountable". Supplying the shortfall under the name of turbulence therefore looks physically hard to sustain, and the measurement points to the candidates of the next section (an ordered velocity field, three-dimensional flow, or heating the 1D energy equation does not carry) rather than to a missing thermal width.

12.3 Is the demand an artifact of the measurement?

The number above is a difference between two widths, so it is worth no more than the two definitions behind it. Before the missing velocity is treated as physics, the observing and fitting side was audited on its own. In that audit every number was recomputed from the released spectrum, the released MCMC chain and the code in this repository.

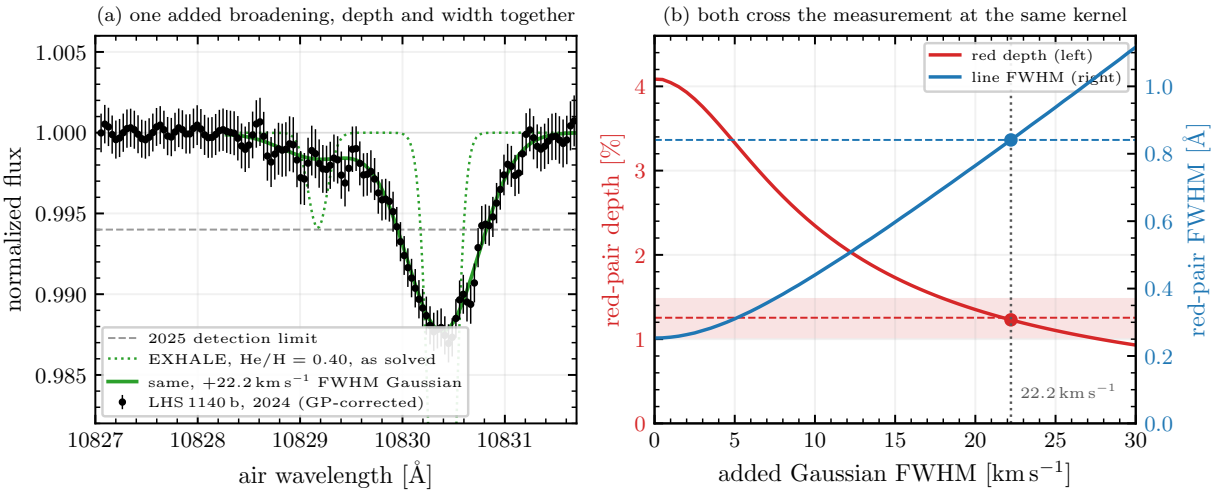


Figure 12: Left: the EXHALE profile matched on equivalent width as solved (dotted, its core running below the plotted range) and convolved with a 22.22 km s^{-1} FWHM Gaussian (solid), against the measurement. Both are shifted by the same $v_{\rm wind} = +2.26$ km s^{-1} used in the earlier figures. Right: red-pair depth (red, left axis) and red-pair FWHM (blue, right axis) as the added Gaussian is scanned. Dashed lines are the observed values, the shaded band the C26 1σ interval on the depth; the dotted vertical marks where the width is matched. Both curves meet the measurement at the same Gaussian.

Nothing on the measurement side supplies the width. Term by term, in the units the requirement is stated in:

- *The instrument profile.* The WINERED HIRES-Y kernel the Supplement states, and validates against nearby telluric features, is 10833 Å/68,000 = 0.1593 Å = 4.41 km s^{-1} FWHM. The observed line is 5.4 times wider, so it is resolved, and removing the kernel in quadrature moves the intrinsic width from 23.9 to 23.4 km s^{-1}.
- *The triplet splitting.* The 0.0897 Å separation of the red pair is present on both sides of the comparison. It is 0.7 per cent of the observed blend width but 8 per cent of the model's, because at $\sigma = 0.35$ Å the pair is a single hump; the asymmetry nearly cancels, moving the requirement by less than 0.1 km s^{-1} and in the direction that makes the gap larger, since the extractor already credits the model with width it does not get from gas motion.
- *Stellar rotation.* With $P_{\rm rot} = 131$ d and $R_\star = 0.216 R_\odot$, $v_{\rm eq} = 0.083$ km s^{-1} and the rotational broadening is 0.14 km s^{-1}. The Rossiter–McLaughlin distortion and the center-to-limb residual scale with the same $v \sin i$ and are correspondingly irrelevant. Neither is mentioned in the paper or its Supplement, and on this star there is nothing to mention.
- *Planetary rotation.* Tidally locked: 0.032 km s^{-1} at R_p and 0.097 at $3R_p$. It is already inside the model curves, which carry the solid-body spin.
- *Orbital motion.* 0.93 km s^{-1} across the full transit if left uncorrected, but the spectra are shifted into the planetary rest frame before averaging, which leaves the 0.022 km s^{-1} of the ephemeris uncertainty; a 300 s exposure smears 0.037 km s^{-1}.

Together these come to at most about $0.4\,\mathrm{km\,s^{-1}}$, and most of that is removed as soon as each side is deconvolved from its own kernel. Done that way, component width against component width and both intrinsic, the requirement is $22.00\,\mathrm{km\,s^{-1}}$ FWHM ($\sigma = 9.34\,\mathrm{km\,s^{-1}}$), against the 22.22 the convolution scan above returns without deconvolving either side. The audit moves the number by $0.2\,\mathrm{km\,s^{-1}}$, not by a factor. The audit itself was run on the earlier crossing; what is carried here is the difference it measured, $-0.22\,\mathrm{km\,s^{-1}}$, applied to the present scan.

One defect was on our side, and it is corrected here. The transit synthesis convolves the He 10830 line with a Gaussian of FWHM λ/R, and until this revision every LHS 1140 b run in this memo took R from the transit tool's built-in default, the CARMENES 8×10^4, whereas this observation was made at WINERED's 68,000. Every model curve quoted here has been re-synthesized at the WINERED value, which the runs now read from their own setting, and the built-in default is left alone because the other planets in this repository were observed through other instruments. The correction widens the modeled blend by about 0.013 Å (at $\mathrm{He/H} = 0.55$, $0.244 \rightarrow 0.257$ Å), lowers the modeled peak depth by about 5 per cent, and leaves every equivalent width in this memo unchanged to 10^{-4} % Å, because a convolution conserves absorbing power. So it moves no crossing and no composition, only the widths and the broadening they demand, the latter by $0.12\,\mathrm{km\,s^{-1}}$. The p-winds curves were already computed at $R = 68{,}000$, so the comparison of Sect. 12.5 is now through one instrument on both sides.

The paper's own model does not reach the width either, and that changes what the shortfall means. Reproducing the authors' p-winds setting at their retrieved medians and measuring it with the same extractor gives FWHM 0.458 Å with turbulence off and 0.575 Å with it on, against the measured 0.841. The retrieval that produced the published parameters therefore lands 1.5 times short of the width printed in the same paper. Taken with the accounting above, the reading this supports is that the observation is not described by a one-dimensional steady-state wind of either code, rather than that the wind solved here carries a defect the comparison model does not. That is an interpretation. What is measured is that both models are too narrow and that the measurement side does not account for the difference.

A second difference runs the same way. The p-winds curve reaches part of its width through saturation: its red-to-blue amplitude ratio is 2.36 with turbulence off and 2.57 with it on, against the optically thin limit of 8, whereas the measurement sits at 6.35 and the solutions here at 6.9–7.0. The premise that this line is optically thin, so that column density cannot be traded against width, holds for the data and for these solutions but not for the comparison model.

Two remarks on the published width, offered as observations rather than conclusions. First, the three widths the Supplement's stellar-variability test reports as FWHM (0.22176 Å for the stellar helium line, 0.323 for the in-transit feature and 0.392 for the pre-ingress feature) appear to be Gaussian standard deviations rather than full widths. Fitting the same three-Gaussian model to the same spectrum before its Gaussian-process treatment, with our extractor, returns $\sigma = 0.3241$ Å against their 0.323, a 0.3 per cent match, where the corresponding FWHM is 0.763 Å and matches nothing on that list. The $\sim 9\sigma$ separation the paragraph concludes with is a difference between two widths and is unaffected; the absolute values are affected. Second, the asymmetric uncertainties on the headline FWHM appear to be transposed. The deposited MCMC chain (identified by its shift column reproducing the published $0.072^{+0.080}_{-0.073}$ Å exactly) gives $2.3548\,\sigma = 0.865^{+0.261}_{-0.155}$ Å ($23.93^{+7.24}_{-4.30}\,\mathrm{km\,s^{-1}}$), where the paper prints $0.86^{+0.15}_{-0.27}$ Å and $23.9^{+4.2}_{-7.5}\,\mathrm{km\,s^{-1}}$: the magnitudes agree once the two are swapped. Neither remark can be settled from outside the collaboration, since the printed bars may come from a chain other than the deposited one, and neither changes the comparison except in one

direction: on the deposited posterior the 1σ lower bound on the measured width is 0.709 Å rather than 0.59, so narrow models are excluded somewhat more firmly than the printed interval suggests.

What the audit could not exclude. The star's own metastable helium line is broad: taking the Supplement's 0.22176 Å as a standard deviation, it is 0.522 Å = 14.5 km s^{-1} FWHM as observed and 13.8 km s^{-1} with the instrument kernel removed. An imperfectly divided stellar line would therefore inject a residual much wider than the modeled planetary line, though still short of the observed feature on its own. The released data do not include the stellar template, so a mixture of a narrow planetary line with a displaced stellar residual could not be tested here and is not excluded. Whether the published σ is itself deconvolved from the instrument profile was also left unsettled; it is worth 1.7 per cent in width either way.

12.4 The same demand, under diffusion and the adopted K_{zz}

Everything above was solved in the well-mixed limit. When helium and hydrogen are allowed to separate by diffusion, helium settles out of the base unless an eddy term lifts it, and how much of it reaches the line-forming region is set by the eddy coefficient K_{zz}, which is not measured for this system. The range the literature leaves open at the 1 μbar base of these runs, the set of cases solved, and the adoption of $K_{zz} = 1.0 \times 10^9$ cm^2 s^{-1} are recorded in the notes listed in Appendix A.

The composition, and how much of it K_{zz} carries

Letting helium and hydrogen separate moves the equivalent-width crossing of Fig. 11 away from He/H = 0.4044, and by how much depends on K_{zz}. The composition was therefore rescanned at every decade of the eddy coefficient from $K_{zz} = 0$ to 10^{11} cm^2 s^{-1}, by the same route in every case: the same stationary solve, the same advection-corrected profiles, the same transit synthesis, and the same red-pair equivalent width on the vacuum window 10832.60–10834.20 Å, through the operator of Sect. 11.. Table 9 is the result, with four to seven solved compositions in each group. Every entry is interpolated between two solved compositions, none extrapolated, but the brackets are not all equally tight: at 10^7, 10^8 and 10^9 the case below the crossing is the He/H = 0.55 one that each group carries for the sensitivity scan, so the reading spans a factor 2.7–4.9 in composition rather than the 12–17 per cent the other decades give.

The full curve is three regimes rather than one power law (Fig. 13). Above 10^7 the degeneracy is shallow and very nearly a power law, $\mathrm{d}\log(\mathrm{He/H})/\mathrm{d}\log K_{zz} = -0.115$ (a least-squares fit over 10^8–10^{11}; –0.109, –0.114, –0.117 and –0.113 decade by decade), so $\mathrm{He/H} \propto K_{zz}^{-0.115}$. Below 10^7 the curve rolls over (–0.057 across 10^6–10^7, –0.026 across 10^5–10^6) onto a plateau at He/H = 3.10, the value the line implies when the eddy term does nothing.

The shape has survived every revision; the level has not. Decade by decade the exponent was –0.021, –0.071, –0.130, –0.150, –0.151 and –0.133 on the 2026-08-30 solutions, against –0.026, –0.057, –0.109, –0.114, –0.117 and –0.113 here: the three regimes and their ordering are the same, the shallow branch is 24 per cent shallower and the roll-off below 10^7 is steeper. What moved is the level, and not by a uniform factor: the plateau falls to 0.81 of its archived value while the 10^{11} end rises to 1.07 (last column of Table 9). That is the shallow branch flattening, not a rescaling.

The plateau sits where it does because the eddy term is added to the molecular diffusion coefficient. The base coefficient of the $K_{zz} = 0$ run is $D_{\mathrm{eff}} = 9.276 \times 10^5$ cm^2 s^{-1} (measured at the $r = 1.00029\,R_p$ face, dominant pair He I–H I), so $K_{zz} = 10^6$

Table 9: Where the red-pair equivalent width crosses the measured 1.108 ± 0.030 % Å, at every decade of the eddy coefficient, with helium and hydrogen free to separate by diffusion throughout. The crossing is read along the log–log chord through the group's solved compositions, and the 1σ range propagates the measurement error alone, within one K_{zz}. At 10^{11} the upper edge of that range lies above the group's highest solved composition (EW = 1.1364 at He/H = 0.93) and is not bracketed. The decades 10^1–10^4 were not scanned in composition on the current code: on the 2026-08-30 solutions their equivalent width at a fixed composition differed from the $K_{zz} = 0$ one by at most 6×10^{-8} of the measurement error, which is the entry they carry. The last column is the ratio to the 2026-08-30 value (Table 6).

K_{zz} [cm^2 s^{-1}]	He/H at the crossing	1σ range	bracketing runs [EW in % Å]	new/old
0 (molecular only)	3.1010	2.9942–3.2093	3.0 [1.0798], 3.5 [1.2175]	0.814
10^1–10^4	not distinguishable from $K_{zz} = 0$ by this line (archived 2026-08-30)			
10^5	3.0706	2.9677–3.1771	3.0 [1.0880], 3.35 [1.1862]	0.809
10^6	2.8930	2.7928–2.9949	2.8 [1.0804], 3.19 [1.1948]	0.800
10^7	2.5365	2.4824–2.5904	0.55 [0.1586], 2.70 [1.1996]	0.826
10^8	1.9729	1.9195–2.0263	0.55 [0.3089], 2.05 [1.1513]	0.867
10^9 (adopted)	**1.5161**	1.4644–1.5744	0.55 [0.4794], 1.50 [1.0996]	0.941
10^{10}	1.1580	1.1104–1.2077	1.15 [1.1031], 1.29 [1.1867]	1.019
10^{11}	0.8927	0.8542–(not bracketed)	0.865 [1.0866], 0.93 [1.1364]	1.067
none (well mixed)	0.4044	0.3877–0.4237	Fig. 11	0.979

raises it by 108%, 10^5 by 11% and 10^4 by 1.1%. The crossings in the table say the same from the other side: 10^5 moves the inferred reservoir by 1.0 per cent against the 6.9 per cent the measurement error alone allows, and 10^6 is the first decade that moves it by more than that error. At the other end the eddy term never reaches the well-mixed limit either: even at 10^{11} the inferred reservoir, 0.8927, is 2.2 times the well-mixed 0.4044.

Over the span the literature leaves open at the base pressure of these runs the inferred reservoir ratio moves only from 1.16 to 1.97, a factor 1.7, against a factor 100 in K_{zz} that moves the equivalent width at fixed composition by a factor 2.1 (He/H = 0.55, from 0.3089 to 0.6463 % Å): the line responds to composition much faster than it responds to K_{zz}, so a large change in the eddy coefficient is bought back by a small change in the reservoir. The composition inferred from this line under diffusion is therefore only weakly conditional on K_{zz}, as $K_{zz}^{-0.115}$, and strongly conditional on the assumed spectrum. The spectrum moves the well-mixed crossing by a factor 7.7, from 0.4044 to 0.0526 (Sect. 14.), which is more than twice the entire K_{zz} lever.

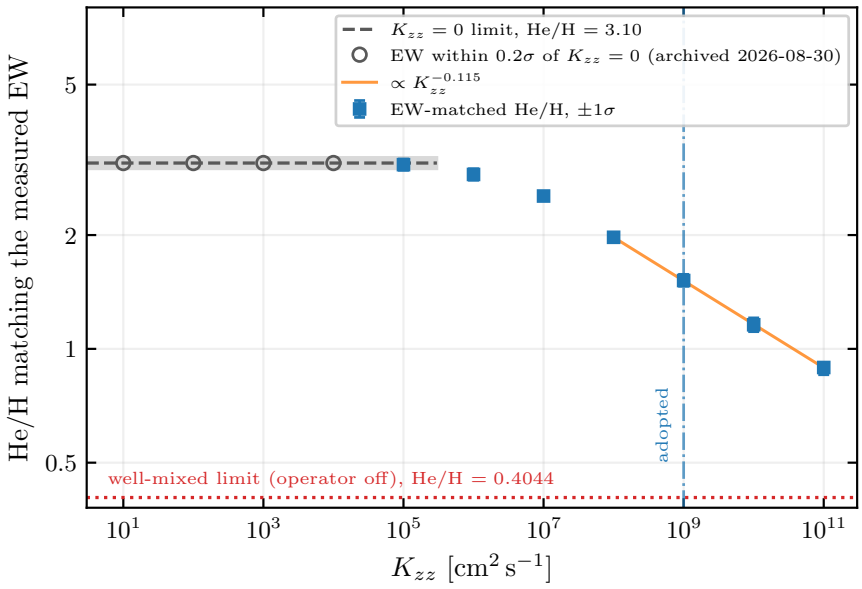


Figure 13: The composition–mixing degeneracy in full. Each square is the He/H reservoir ratio whose red-pair equivalent width equals the measured one, solved at one decade of K_{zz} with helium and hydrogen free to separate by diffusion (Table 9); the error bars propagate the measurement error alone and are smaller than the symbols over most of the range. The landscape has three parts. Below $K_{zz} \simeq 10^5$ the eddy term is a small correction to the molecular coefficient the base already carries, the inference sits on a plateau at He/H = 3.10 (dashed, with its 1σ band), and the open circles mark the decades whose equivalent width, on the 2026-08-30 solutions, differed from the $K_{zz} = 0$ one by less than a fifth of the measurement error. Between 10^6 and 10^7 the curve rolls off. From 10^8 up it is the shallow power law $K_{zz}^{-0.115}$ (orange, fitted to those four points). Neither end of the axis reaches the well-mixed crossing of 0.4044 (dotted): with diffusion active, helium settles at every eddy coefficient tried, and He/H inferred from this line is a statement about the assumed mixing as much as about the planet. The adopted $K_{zz} = 10^9$ cm^2 s^{-1} is marked.

What K_{zz} does act on strongly is the helium profile itself, and Fig. 14 separates that from the wind. At a fixed reservoir He/H = 0.55, raising K_{zz} from 0 to 10^{10} lifts the homopause from below the base to 1.154 R_p and the metastable peak

by a factor 24; between 10^8 and 10^{10} the factor is 2.3, while the velocity at $10\,R_\mathrm{p}$ moves by 8 per cent, the density at $2\,R_\mathrm{p}$ by 28 per cent and $\log_{10}\dot{M}$ by 0.02 dex. Along the sequence of Table 9, where each K_{zz} carries its own crossing composition, $\log_{10}\dot{M} = 7.86$–7.87 in every run: the escape rate does not carry the K_{zz} dependence at all.

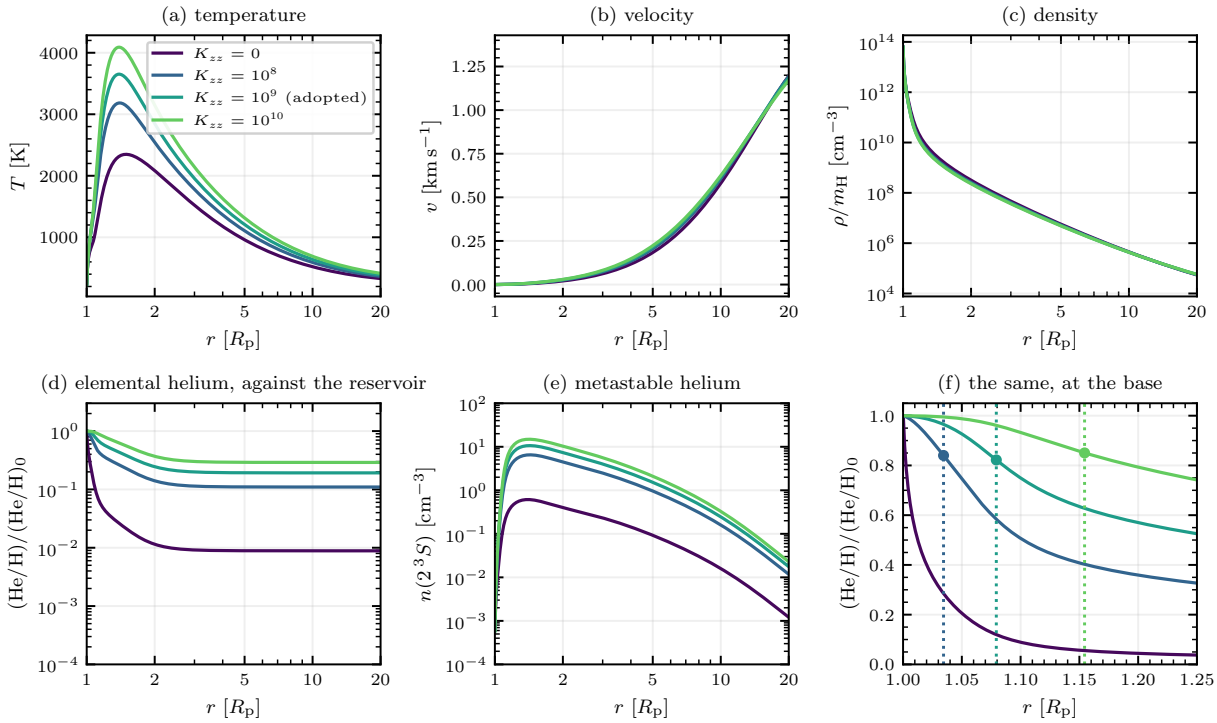


Figure 14: Four solutions differing only in the eddy coefficient, at a fixed reservoir He/H = 0.55 with helium and hydrogen free to separate by diffusion: $K_{zz} = 0$, 10^8, 10^9 (adopted) and 10^{10} cm^2 s^{-1}, dark to light. The eddy coefficient acts on the composition, not on the wind. Panels (d)–(f), the elemental helium against its base reservoir value and the metastable population: at $1.05\,R_\mathrm{p}$ the elemental ratio is 0.206, 0.745, 0.927 and 0.988 of the reservoir and at $5\,R_\mathrm{p}$ it is 0.009, 0.110, 0.193 and 0.291, while the peak $n(2^3S)$ is 0.61, 6.5, 10.6 and 14.9 cm^{-3}. In panels (b) and (c) the velocity at $10\,R_\mathrm{p}$ moves by 8% and the density at $2\,R_\mathrm{p}$ by 28%, and $\log_{10}\dot{M}$ by 0.02 dex. Panel (a) is the exception: T at $1.5\,R_\mathrm{p}$ rises from 2349 to 4006 K, because the helium the eddy term lifts is itself absorbing. Panel (f) repeats (d) on a linear axis over the base, where the dots and dotted lines mark each run's homopause, the radius at which its own molecular coefficient equals its K_{zz}, at 1.034, 1.079 and $1.154\,R_\mathrm{p}$, with the $K_{zz} = 0$ homopause below the base. The table of fluxes through the cell faces that the homopause is read from is written only when the run is asked for it, and comes from these same four solutions post-processed a second time with that output turned on.

The width the data demand, again

The remaining question is whether the width the data demand moves with the composition. It does not. Repeating the measurement of Sect. 12.2 on the solved composition He/H = 1.50, the one nearest the $K_{zz} = 10^9$ crossing at 1.5161 (red-pair EW 1.100 % Å, red depth 4.091%, FWHM 0.252 Å), with the same convolution, the same three-Gaussian extractor and the same integration window, gives Table 10.

Table 10: The diffusive solution matched on equivalent width (the solved composition He/H = 1.50, nearest the crossing at 1.5161 for $K_{zz} = 10^9$ cm^2 s^{-1}) convolved with an added Gaussian of FWHM f, measured as in Table 8.

f [km s^{-1}]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.091	0.588	6.96	0.252	1.100
10	2.344	0.313	7.48	0.440	1.099
15	1.728	0.228	7.58	0.597	1.095
20	1.349	0.177	7.63	0.765	1.077
22.2	**1.227**	**0.161**	**7.64**	**0.841**	**1.063**
25	1.101	0.144	7.65	0.938	1.041
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

The width is reached at $f = 22.22$ km s^{-1} FWHM, the same value to four digits as the well-mixed solution of Sect. 12.2 solved without diffusion. At that Gaussian the depth is 1.227% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.161% against $0.25^{+0.14}_{-0.12}$ and the ratio 7.64 against $6.7^{+12.7}_{-3.1}$, all three inside the observed 1σ interval at the value fixed by the width alone, as in the well-mixed case. The velocity the profile demands is therefore unchanged by turning diffusion on and by the adopted eddy coefficient. Diffusion changes how much helium is there, not how fast it moves. The equivalent width at that Gaussian, 1.063 against 1.108 ± 0.030, falls 1.5σ below the measurement, for the reason given in Sect. 12.2: the wings carry absorption out of the integration window. Figure 15(a) shows the radial number density profiles of the hydrogen and helium species in this diffusive steady state, and Figure 15(b) the broadened transit spectrum against the measurement.

12.5 Broadening cannot repair a deficit in equivalent width

The same scan was applied to the p-winds curves, in their own vacuum frame and with the same three-Gaussian extractor and the same integration window (Table 11). The convolution conserves absorbing power: it redistributes the excess in velocity and leaves the equivalent width unchanged, apart from the few percent that the wings carry out of the integration window. Whether one Gaussian can reproduce depth and width together is therefore decided before the scan starts, by the equivalent width the curve already has at $f = 0$.

The p-winds best fit carries 0.836 % Å against the measured 1.108 ± 0.030, 25% short. At the Gaussian that reproduces the width, $f = 17.8$ km s^{-1}, the red depth is 0.926% against $1.24^{+0.22}_{-0.23}$, about 1.4σ low. The broadening moves the disagreement rather than removing it. Unbroadened, the curve is too deep in the blue component (0.534% against $0.25^{+0.14}_{-0.12}$, some 2σ high) and too narrow. Broadened to the measured width, the blue component falls to 0.339%, inside 1σ, and the red component drops below the measurement. The blue side improves, the red side does not, and the total is fixed.

The intensity ratio barely moves, 2.57 at $f = 0$ against 2.73 at the matched Gaussian. It is the amplitude ratio of a saturated pair, set by how the two components share the optical depth, and a common convolution acts on both alike. It stays well below the measured $6.7^{+12.7}_{-3.1}$, though the observed interval is wide enough that this alone is not a

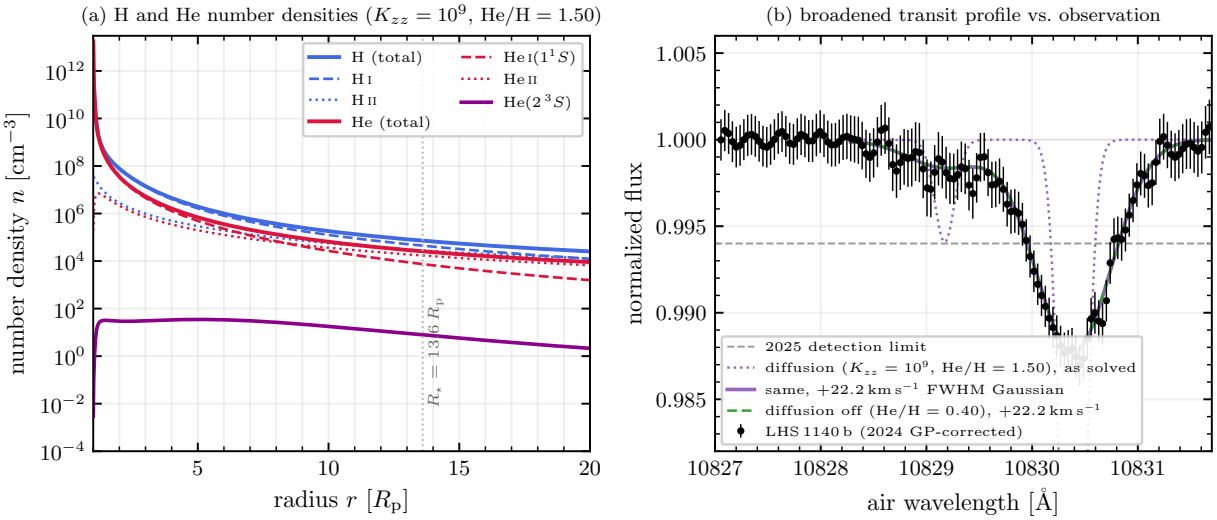


Figure 15: The diffusive solution matched on equivalent width (He/H = 1.50 at $K_{zz} = 10^9$ cm^2 s^{-1}). **(a)** Absolute number densities of hydrogen species (total H, H I, H II) and helium species (total He, He I(1^1S), He II, and metastable He(2^3S)) as a function of planetary radius r [R_p]. At the lower boundary ($r = 1.0\,R_p$), He/H = 1.50; diffusion lets helium settle outward relative to hydrogen, with the local abundance falling to He/H = 0.456 at $2\,R_p$ and 0.355 at $10\,R_p$. Dotted vertical line marks the stellar disk radius ($13.6\,R_p$). **(b)** The transmission spectrum as solved (dotted purple, its core running below the plotted range) and convolved with the 22.22 km s^{-1} FWHM Gaussian that reproduces the measured width (solid purple), against the measurement; the well-mixed curve solved without diffusion, at its own matched Gaussian (dashed green), is repeated from Fig. 12. The two broadened curves are indistinguishable: their Gaussians agree to four digits. All model curves are shifted by the fitted $v_{wind} = +2.26$ km s^{-1}.

Table 11: The same added-Gaussian scan applied to the p-winds curves, with the EXHALE row of Table 8 repeated for comparison. The Gaussian is the one that reproduces the measured width, 0.841 Å; the depths and the ratio are read at that Gaussian. Equivalent widths are for the unbroadened curve, over the vacuum window 10832.60–10834.20 Å and through the operator of Sect. 11..

curve	EW(f=0) [% Å]	f [km s^{-1}]	red [%]	blue [%]	red/blue
p-winds best fit, turbulence on	0.836	17.8	0.926	0.339	2.73
p-winds best fit, turbulence off	0.701	20.1	0.777	0.301	2.58
p-winds depth-matched, 6100 K	0.480	20.2	0.535	0.140	3.82
EXHALE, He/H = 0.40	1.101	22.2	1.229	0.161	7.65
observed	1.108 ± 0.030	-	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$

discriminator.

Matching the depth as well would require 32% more absorbing power, and that is not something a convolution supplies: it is a change in $\dot{M}$, temperature or composition. Fitting the broadening as a free parameter alongside those, which is what the measurement asks for, would presumably shift the C26 parameter estimates, since the composition is constrained through the same absorbing power that would have to rise.

The asymmetry with EXHALE is not a statement about one code outperforming the other. The EXHALE curve entered the scan with its equivalent width already set to the observed value by the choice of $\mathrm{He/H} = 0.40$ (Sect. 12.2), so only the shape remained free, and one Gaussian settled it. The p-winds curve entered at the composition C26 report, whose equivalent width falls short, so no Gaussian could settle both. The two scans test the same physical statement, that broadening redistributes and does not create, from two different starting points.

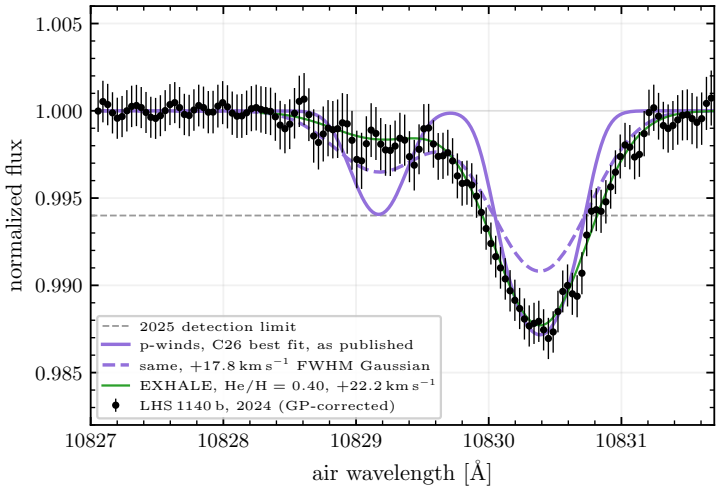


Figure 16: The C26 p-winds best fit as published (solid purple) and convolved with the 17.8 km s^{-1} FWHM Gaussian that reproduces the measured width (dashed purple), against the measurement. The EXHALE $\mathrm{He/H} = 0.40$ curve at its own matched Gaussian (22.2 km s^{-1}, thin green) is repeated from Fig. 12. Broadened to the observed width, the p-winds curve passes above the red core, so the width is matched and the depth is not. All curves are shifted by the same $v_{\mathrm{wind}} = +2.26$ km s^{-1}.

12.6 The adiabatic-cooling population bump, looked for

Schulik & Owen (2025) propose that the missing width is already contained in the metastable population itself. Writing the steady state of their Eq. (1) as

$$n_{2^3S}(T_e) = \frac{\alpha(T_e)}{q(T_e) + t_{\mathrm{ion}}/n_e}\, n_{\mathrm{He}^+}, \tag{6}$$

they note that as adiabatic expansion drops the temperature "from $\simeq 10^4$ K to $\simeq 10^3$ K", "a change in slope in the population occurs as the dominant triplet loss rate $q(T_e)$ drops significantly at lower temperatures": the metastable fraction climbs above the curve computed with the rates frozen at the peak temperature. The rise is "quenched" where the advection timescale overtakes the excitation to 2^1S, after which $\dot{m}_{2^3S} = 4\pi r^2 v\, n_{2^3S} m_{\mathrm{He}}$ falls through ionization and the $2^3S \rightarrow 1^1S$ decay. They credit Yan et al. (2022) and Biassoni et al. (2024) with first identifying the effect, and their

conclusion states the condition it must satisfy to be observable: "The helium triplet population bump must be situated at high velocities to help explain anomalous line widths."

EXHALE carries the temperature dependence the mechanism needs, and it enters in the balance written above. The equation the code solves for the metastable population is

$$n_e\left(n_{\mathrm{He}^+}\alpha_{2^3S}+n_{\mathrm{He\,I}(1^1\mathrm{S})}\,q_{13}\right)=n_{2^3S}\left[\Gamma_{2^3S}+A_{31}+n_{\mathrm{H\,I}}Q_{31}\right.\\ \left.+n_e(q_{31a}+q_{31b}+b_{2^3S})\right], \tag{7}$$

with every coefficient a function of T: the recombination $\alpha_{2^3S}=2.10\times10^{-13}(T/10^4)^{-0.778}$, the collisional losses q_{31a} ($2^3S\to2^1S$, 0.80 eV) and q_{31b} ($2^3S\to2^1P$, 1.40 eV) as Bray (2000) collision strengths folded with $\exp(-\Delta E/kT)$, the destruction term Q_{31} on atomic hydrogen as the García Muñoz (2025) closed form of Eq. (3), collisional ionization of the metastable, and the radiative decay $A_{31}=1.272\times10^{-4}\ \mathrm{s}^{-1}$. Between 5000 and 1000 K the loss $q_{31a}+q_{31b}$ falls by a factor 1.3×10^3 while α_{2^3S} rises by a factor 3.5, so the ingredients of the bump are present and are not an imposed parameterization.

The result was measured by reconstructing that balance in Python from the solved profiles, recovering from the balance itself the one term not written to file, the 2^3S photoionization rate Γ_{2^3S}, and then recomputing the reference curve with α, q_{13}, q_{31a}, q_{31b} and Q_{31} frozen at the peak temperature. Each transcribed coefficient was checked against the compiled Fortran over 300–10^4 K: Q_{31} agrees to 8×10^{-16} and the other five to 1.2×10^{-5} or better, the residue being the single-precision literals the Fortran carries.

Table 12: The metastable bump in the solved winds. Columns 2–4 locate the n_{2^3S} maximum in the equilibrium profile and give the temperature and outflow speed there; the last three characterize the absorber-weighted ($w=n_{2^3S}r^2\,dr$) distribution of $|v_r|$ over the advection-corrected 2^3S column that the transit tool integrates. The He/H = 1000 solutions have no outward density maximum (their n_{2^3S} peaks at the base), and are marked †. The six runs are the two well-mixed groups, on the two proxy spectra, and all six are converged ($\|R\|\le1.7\times10^{-7}$).

case	r_{peak}	T	v	med. $\lvert v_r\rvert$	>2	>5
	$[R_p]$	[K]	[km s^{-1}]	[km s^{-1}]	[%]	[%]
GJ 1132 proxy spectrum						
He/H = 0.55	5.57	1616	0.28	0.41	0.0	0.0
solar He/H	4.17	1367	0.15	0.44	0.0	0.0
He/H = 1000	1.08†	3258	0.00	0.33	0.0	0.0
GJ 699 proxy spectrum						
He/H = 0.050	5.82	917	1.56	2.49	63.8	0.0
solar He/H	5.74	982	1.50	2.41	62.1	0.0
He/H = 1000	1.05†	3166	0.00	1.37	9.9	0.0

The bump is there. In every case the metastable fraction $f_3=n_{2^3S}/n_{\mathrm{He}}$ rises monotonically outward across the whole cooling leg, from the temperature maximum to a maximum of f_3 at 9–15 R_p (a rise of 1.2–4.4×10^3 in the equilibrium profiles and 20–140 in the advection-corrected ones), and the rise follows from the temperature dependence in the way

Schulik & Owen describe. On the GJ 1132 proxy at the well-mixed He/H = 0.55 case the equilibrium maximum sits at 5.57 R_p where T = 1616 K, against $T_{\max}$ = 4789 K at the inner peak: the temperature has fallen to 0.34 of its peak, which is the regime the mechanism asks for.

The quench is visible too: in the advection-corrected profiles the density maximum has moved inward to 1.05–1.8 R_p and n_{2^3S} then falls monotonically, although $\dot{m}_{2^3S}$ itself does not drop past the peak on the GJ 1132 cases and falls by 33 per cent only on the helium-dominated He/H = 1000 run of the brighter spectrum.

The velocity condition, however, is not met, and the shortfall is large. At the bump the outflow is moving at 0.28 km s^{-1} on the GJ 1132 proxy and 1.6 km s^{-1} on the GJ 699 proxy. Weighting by absorbers over the whole 2^3S column, the median $|v_r|$ is at most 0.4 km s^{-1} and 2.5 km s^{-1} on the two spectra, and the largest $|v_r|$ anywhere in the column is 1.2 and 3.9 km s^{-1}. No part of the column on either spectrum reaches 5 km s^{-1}, against the σ = 9.4 km s^{-1} of extra broadening the measurement demands (Sect. 12.2). Radial velocity is the generous choice here: the line-of-sight projection of a radial field over the transit chord is smaller still. Figure 17 draws the three statements together.

So the bump exists in these solutions and contributes nothing measurable to the width. That is consistent with what Schulik & Owen themselves report for the coolest hosts: for their M2 case they find that "the transit signal probes the same velocity region for both treatments of thermodynamics, making the line insensitive to velocity", and they conclude that K stars, not M stars, are the favorable testbeds: G-type hosts suppress the bump by photoionizing the triplet, and M-type hosts have weaker tidal fields and so rarely drive fast enough outflows. LHS 1140 is an M4.5 dwarf, cooler than the M2 they model, and its planet is a temperate super-Earth rather than a hot giant. The mechanism appears to be present and correctly captured; the host is the wrong one for it to help.

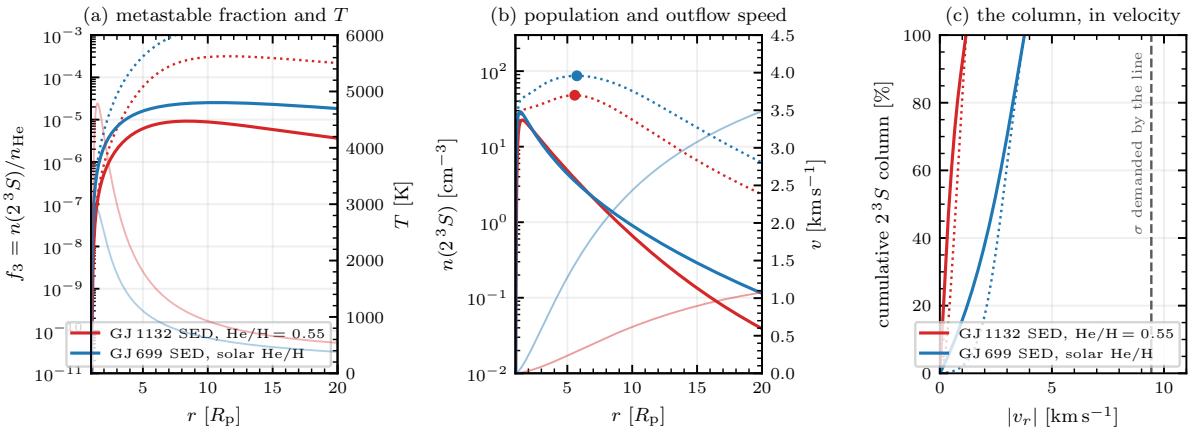


Figure 17: The adiabatic-cooling metastable bump in the EXHALE solutions (solid: advection-corrected, the profiles the transit integral uses; dotted: local equilibrium; faint curves on the right-hand axes). Two cases of Table 12 are drawn: He/H = 0.55 on the GJ 1132 proxy and solar He/H on the GJ 699 one. (a) The metastable fraction rises outward over the cooling leg, from the temperature maximum out to a maximum of f_3 at 11.6 and 11.8 R_p in equilibrium, which is the bump in the sense of Schulik & Owen (2025). (b) The equilibrium n_{2^3S} peaks at 5.6–5.7 R_p (markers); advection moves the maximum inward, to 1.4–1.5 R_p, and quenches the outer rise. (c) The absorber-weighted distribution of $|v_r|$ over the 2^3S column: the whole column lies below 4 km s^{-1}, far short of the σ = 9.4 km s^{-1} the measured width demands.

12.7 The width in the wider He 10830 literature

The comparison so far has been between this model and this measurement. Against the published population, the measured width is ordinary and it is the model that is anomalous.

The measured width is typical. Few He 10830 analyses fit the width freely rather than predicting it; where they do, the answer is 19–33 km s^{-1} FWHM. Table 13 collects them. LHS 1140 b's $23.9^{+7.2}_{-4.3}$ km s^{-1} is an unremarkable member of that set. Guilluy et al. (2024) is the only survey that tabulates a free FWHM for several planets at once; their GJ 3470 b entry should not be taken at face value, since they write that "The larger FWHM for GJ 3470 b is believed to be attributed to residuals from OH line contamination". The entries are quoted as published (different instruments, different extractors, no common deconvolution of the instrumental profile), so the clustering is a statement about published numbers rather than a uniformly reduced sample.

Table 13: Published He 10830 transit widths where the width was fitted freely. Conversions use 10833 Å. The last two rows are velocity dispersions of a different definition, given for scale, not as FWHM.

system	instrument (R)	FWHM [km s^{-1}]	source
LHS 1140 b	WINERED (68,000)	$\mathbf{23.9^{+7.2}_{-4.3}}$	this memo, on the released spectrum
WASP-107 b	GIANO-B ($\sim$ 50,000)	$32.6^{+4.2}_{-3.6}$	Guilluy et al. (2024), Table A.3
HAT-P-11 b (3 nights)	GIANO-B	29.3/19.0/22.9	Guilluy et al. (2024), Table A.3
WASP-69 b	GIANO-B	26.6	Guilluy et al. (2024), Table A.3
GJ 3470 b	GIANO-B	60.3	Guilluy et al. (2024); OH residuals
4 young sub-Neptunes	NIRSPEC (32,000)	27–33	Zhang et al. (2023)
SPIRou detections	SPIRou	$\sim$ 22	Masson et al. (2024)
HD 189733 b	CARMENES	$b = 15.1 \pm 0.6$	Salz et al. (2018), Table 3
HAT-P-32 b	CARMENES	$v_{t,a} = 15.2 \pm 3.1$	Czesla et al. (2022)

Zhang et al. (2023) treat that width as *evidence for* photoevaporation: "The width is the most consistent metric and is within 10% of 1.1 A for all planets", because "For a Parker outflow, the sound, bulk outflow, and thermal speeds are all close to sqrt(2kB T/mu mH) = 9.4 km s-1 for T = 7000 K and mu = 1.3, leading to an FWHM of 2.355 sigma $\sim$ 22 km s-1." That argument needs a 7000 K hydrogen-dominated transonic outflow. This solution has 5200 K at $2R_p$, a mean mass of 2.6–3.5 amu because of the helium enrichment, and an outflow of 0.024 km s^{-1} there; the same construction returns the model's 2.4 km s^{-1}, not 9.4.

The turbulence term the field offers. There is a single turbulence formula in this literature, $v_{\rm turb} = \sqrt{5kT/3m}$ with m the mass of a helium atom (Lampón et al. 2020, Eq. 16), adopted unchanged by p-winds as its Eq. (21) (Dos Santos et al. 2022). It is the adiabatic sound speed of a monatomic gas, that is, a Mach number of one, and it is a fixed function of T, not a free parameter. Evaluated on this solution it gives 4.2, 4.2, 3.6, 3.0 and 2.7 km s^{-1} at 1.2, 2, 3, 5 and $8R_p$, with a metastable-weighted mean of 3.8–4.0 km s^{-1} over 1–$8R_p$. Against the required $\sigma = 9.34$ km s^{-1} that is short by a factor 2.1–3.7 across the absorbing region and 2.3–2.5 on the absorber-weighted average. The measurement asks for

Mach 2–4 and the term on offer is Mach 1. It cannot absorb the requirement, whatever one concludes about its physical justification. Three published analyses report the same failure in their own systems: Pallé et al. (2020) on GJ 3470 b ("The inclusion of the broadening of the lines due to turbulence (vturb = 5kT/3m, where m is the mass of a helium atom), in addition to the standard Doppler broadening, was not enough to explain the measured broadening in the observations"), Lampón et al. (2021) on HD 189733 b ("Including a turbulence broadening component (see Sect. 3.3), the profile broadens, but it is still narrower than the measured absorption. ... Hence, our hydrodynamic model alone is not able to explain the width of the absorption profile."), and Czesla et al. (2022) on HAT-P-32 b ("Thermal broadening alone remains insufficient to explain the observed line width, even if the turbulence term is included.").
Where a width is instead fitted freely, it comes back two to three times larger: Salz et al. (2018) obtain $b = 15.1 \pm 0.6\,\mathrm{km\,s^{-1}}$ (extended model, $f = 20\%$) and 11.3 ± 0.5 (compact) at mid-transit for HD 189733 b, and Czesla et al. (2022) obtain $v_{t,a} = 15.2 \pm 3.1$, 11.2 ± 3.1 and $11.7 \pm 1.0\,\mathrm{km\,s^{-1}}$ for HAT-P-32 b. Our own $9.34\,\mathrm{km\,s^{-1}}$ sits at the low end of that set.

An independently written code on another planet. Taylor et al. (2026) apply a 1D self-consistent model of the same class (energy balance, lower-atmosphere coupling, multispecies diffusion, non-LTE H($n = 2$)) to HD 189733 b and report the same shortfall: "several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework", at "a factor of $\sim$ 2–3" in width. The axes they varied are the axes varied here: "We tested variations in the free parameters (heating efficiency, K_{zz}, stellar activity level/XUV input, base composition, and lower-boundary placement): configurations that increase the width either deepen the core transit depth too much or violate the Hα constraints, and none simultaneously reproduce both core depth and width." Their proposed mechanism is stated by them as speculative, "Given that excited He is mostly produced by recombination of helium ions, this broadening can be driven by plasma turbulence ... While these ideas are speculative, we encourage further exploration of plasma effects and magnetic field interactions", and its first clause is the same physics Sect. 12.8 derives independently below. Rumenskikh et al. (2022), as reported there, reached the observed width in 3D only by prescribing zonal jets of 10–$20\,\mathrm{km\,s^{-1}}$ together with reduced hydrogen line cooling; Schulik & Owen (2025) offer the adiabatic-cooling population bump as the one route with no free parameter, which Sect. 12.6 shows is present here and worth 0.25–$1.5\,\mathrm{km\,s^{-1}}$.

What follows. The width shortfall is not specific to EXHALE and not specific to LHS 1140 b. It is an unresolved problem of the 1D steady-state model class. The evidence for that statement is threefold: the measured width is a typical member of the published population (Table 13); the deficit is reproduced at the same factor of 2–3 by an independently written code, on a different planet, around a star at the opposite end of the activity range, after testing the same axes (Taylor et al. 2026); and the field's own remedy is a term that is Mach 1 by construction and that its users report in print to be insufficient. Every published route to the observed width adds velocity by hand (a sonic turbulence term, a free Gaussian, or prescribed jets), and none of them follows from the 1D energy equation.
One citation needs a note here, as an observation rather than a correction to anyone's result. Lampón et al. (2023) and Yan et al. (2024) both cite Salz et al. (2018) as a source of the $v_{\mathrm{turb}} = \sqrt{5kT/3m}$ convention, but "turbulen" does not occur anywhere in the published Salz et al. (2018), which fits a free Doppler parameter instead. The convention itself

is stated in Salz et al. (2016), Eq. (5); the 2018 paper appears to be intended as the source of the measured *width*. As written, the attribution suggests that a Mach-1 turbulence term reproduced the HD 189733 b width, when in that paper it was a free 15.1 km s^{-1} Doppler parameter.

12.8 The size of the non-thermal candidates

If the missing velocity is not thermal and not in the 1D energy equation, the next question is which non-thermal mechanism could supply $\sigma = 9.34\,\mathrm{km\,s^{-1}}$. The estimates below are order-of-magnitude requirements computed on the reference solution, one of the solutions made consistent with the elemental fluxes. The full set, including the stellar-wind standoff, the turbulent and kinetic power budgets, charge exchange and radiation pressure, is in the working notes on the non-thermal candidates.

Magnetic. Setting the Alfvén speed equal to the required dispersion, $v_A = B/\sqrt{4\pi\rho} = 9.34\,\mathrm{km\,s^{-1}}$, and reading ρ off the advection-corrected profile, the field needed in the line-forming region is $B = 0.423$, 0.118, 0.043 and 0.016 G at 1.2, 1.5, 2 and $3\,R_p$. Continued inward as a dipole, $B_s = B(r/R_p)^3$, those are surface fields of 0.73, 0.40, 0.34 and 0.43 G: a shallow minimum near $2\,R_p$, so a dipole of a few tenths of a gauss would put v_A near the required value over the whole region at once. As a scale comparison that is unremarkable: it lies between Earth's surface dipole (≈ 0.3 G) and Jupiter's (≈ 4 G). The difficulty is what such a field would do rather than whether a planet can have it: the plasma β at that strength is 0.28–0.38 over 1.2–$3\,R_p$, so the field would be dynamically dominant and would restructure the outflow rather than decorate it, and this solution does not model that. The wave energy flux required to sustain $\sigma = 9.34\,\mathrm{km\,s^{-1}}$, $\rho(3\sigma^2)v_A$ through a sphere, is 12–870 times the solution's total XUV heating of 1.22×10^{20} erg s^{-1} over the same radii, which only a star-planet interaction driven by a wind of at least solar strength at this orbit could feed. The field strength does not exclude this route. The power the field would have to be given does constrain it, as does the fact that the comparison is being made for a 131 d rotator with $L_X/L_{\mathrm{bol}} = 6.5 \times 10^{-6}$.

Where a metastable atom can be made. This is the constraint that bears hardest on any mechanism that manufactures fast helium somewhere else (a stellar-wind interaction region, a charge-exchange or energetic-neutral population), and brings it into the beam. He 2^3S is fed by He^+ recombination and destroyed by Penning ionization with atomic hydrogen. The collisional channel from the singlet ground state supplies a fraction 1×10^{-13} of the source at $1.2\,R_p$ and 2×10^{-14} at $2\,R_p$, falling to 10^{-33} by $5\,R_p$: the metastable population is a recombination product without qualification. Table 14 evaluates the two timescales on the solution, with $\alpha_{2^3S} = 2.10 \times 10^{-13}(T/10^4)^{-0.778}$ and the García Muñoz (2025) destruction rate of Eq. (3). Both columns are recomputed here from the solved profiles. The replacement of the rate coefficient leaves alone the wind the table stands on: between the last solution of this case that predates the replacement and the re-solved one, T agrees to 1.2 per cent and t_{rec} to 1.8 at every radius, because the metastable is a trace population and not an energy-budget term. The last three columns are not preserved that way and should not be: they *are* the destruction rate, and it is the quantity Sect. 6.5 replaced, so they move by factors of 1.0–1.8 between the two fits. A reconstruction that reproduced them would be evidence that the replacement had not been applied.

Two requirements follow, and they are severe. First, recombination is slower than destruction by 7.5×10^3–4.8×10^4 everywhere, so only a tiny standing fraction of the helium is ever metastable and the population is set locally and

Table 14: Metastable helium source and sink timescales in the reference solution, and the distance an atom moving at the required $9.34\,\mathrm{km\,s^{-1}}$ covers within its total lifetime. The reference solution is re-solved on the current build. Renormalizing the composition onto the density, one of the changes of Sect. 5., moves T here by at most 0.3 per cent and t_{rec} by at most 1.4: this is a trace-population diagnostic on a wind that does not move.

r [R_p]	T [K]	t_{rec} [s]	t_{Penning} [s]	t_{life} [s]	$L = 9.34\,\mathrm{km\,s^{-1}}\,t_{\mathrm{life}}$ [R_p]
1.2	5111	8.4×10^4	1.72	1.72	0.0014
2.0	5283	8.9×10^5	59.6	59.1	0.049
3.0	3866	3.5×10^6	3.8×10^2	3.6×10^2	0.30
5.0	2662	1.6×10^7	2.8×10^3	2.1×10^3	1.7
8.0	2088	6.0×10^7	1.5×10^4	5.1×10^3	4.2
20	538	2.6×10^8	2.8×10^5	7.7×10^3	6.3

continuously. Second, and more restrictive, within its own lifetime an atom moving at the required speed travels $0.05\,R_p$ at $2\,R_p$ and $0.30\,R_p$ at $3\,R_p$ (Table 14, last column). The inner metastable helium, which carries roughly a third of the absorption, cannot have been manufactured anywhere but where it is seen. A stellar-wind or energetic-neutral population would therefore have to be re-ionized and recombine *inside* the absorbing region: the recombination time there is 10^6–10^7 s against a crossing time of $5\,R_p$ at $9.34\,\mathrm{km\,s^{-1}}$ of 5.9×10^3 s, short by three to four orders of magnitude. The candidate is not excluded as a source of *some* broad outer emission, but it cannot be the source of the bulk of the line.

Three-dimensional flow. With $P = 24.737$ d and tidal locking, $\Omega = 2.94 \times 10^{-6}\,\mathrm{s^{-1}}$ and the corotation speed Ωr is $0.032\,\mathrm{km\,s^{-1}}$ at the surface, 0.098 at $3\,R_p$ and 0.94 at the $29\,R_p$ domain edge: already measured in Sect. 12.3 and two orders below the requirement. A pressure-driven day-night flow is bounded above by the thermal energy available, that is, by a few times the 4–5 $\mathrm{km\,s^{-1}}$ sound speed in the complete-conversion limit, and independently by the measured line shift, whose posterior caps any coherent line-of-sight flow of the whole absorbing column at $4.7\,\mathrm{km\,s^{-1}}$. Both are small here for the same reason: the driver of 3D circulation is the day-night temperature contrast, and LHS 1140 b is a temperate planet at $T_{\mathrm{eq}} = 226$ K around an M4.5 dwarf, with an irradiation and a rotation rate one to two orders of magnitude below the hot Jupiters for which 10–20 $\mathrm{km\,s^{-1}}$ jets are prescribed. Taylor et al. (2026) reach the same conclusion for HD 189733 b, where the case is far more favorable: "typical GCM wind speeds (a few kilometers per second) are unlikely to supply the additional $\sim 12\,\mathrm{km\,s^{-1}}$ required to match the observed width."

What this does not settle. These are requirement estimates evaluated on *this* solution's density, temperature and ionization structure. They say what each mechanism would have to be given in order to produce $9.34\,\mathrm{km\,s^{-1}}$ here, and on that basis none of them is comfortable: the magnetic route needs a power it has no obvious source for around this star, the fast-neutral route cannot make metastables where the absorption is, and the 3D route is short by two orders of magnitude. They do not identify which mechanism is in fact responsible, and they do not exclude the possibility that the absorbing gas is not this solution's gas at all: the measured pre-ingress signal, at least as broad as the in-transit one

at a phase when the planet occults nothing, is direct evidence that some of the signal comes from a structure no 1D radial model contains. Which mechanism supplies the width is not determined by this work, and that is also the state of the literature.

12.9 What would have to change

The measured line wants gas that is fast enough to be broad but not so extended that it absorbs too much. EXHALE at this XUV flux delivers neither. The candidates, in the order they should be tested:

1. *More energy*: tested (Sect. 14.), and it does not do it. The GJ 699 proxy carries five times the XUV of the GJ 1132 proxy used here, and the paper itself runs both. Solving the scan again on it raises $\dot{M}$ fivefold and moves the composition estimate by an order of magnitude, but leaves the required extra broadening at 21.97 km s^{-1} (GJ 699 proxy, the solution at He/H = 0.050 nearest its crossing) against 22.22 (GJ 1132 proxy, the solution at He/H = 0.40): the width gap is not an energy-input problem.

2. *A different helium distribution*: tested (Sect. 12.4), and it does not do it either. Letting helium settle by diffusion and lifting it back with an eddy term redistributes the absorber by orders of magnitude and moves the inferred composition with it, but the required broadening stays at 22.22 km s^{-1} (the diffusive solution at He/H = 1.50, $K_{zz} = 10^9$ cm^2 s^{-1}), the value without diffusion: it changes how much helium is there, not how fast it moves.

3. *Heating the code does not carry.* Taylor et al.'s trapped-ion turbulence, or any magnetically mediated dissipation, is outside a 1D hydrodynamic energy equation by construction. Sect. 12.8 sizes it: the field needed to make $v_A = 9.34$ km s^{-1} is a few tenths of a gauss, which is ordinary, but the wave power that field must carry is 12–870 times the XUV heating.

4. *Geometry.* A line-of-sight velocity field from day-night or zonal flow is a 3D effect; Rumenskikh et al. reach the observed width with 10–20 km s^{-1} jets, which is the right order. For this planet, though, rotation gives 0.03–0.10 km s^{-1} and the measured line shift caps any coherent line-of-sight flow at 4.7 km s^{-1} (Sect. 12.8).

5. *Adiabatic-cooling population bump.* Looked for (Sect. 12.6): the bump is present in these solutions, but it sits at 0.28–1.6 km s^{-1}, so it does not supply width.

The paper's composition constraint should be read in the same light: it inherits the isothermal assumption. Impose a 6100 K structure and only helium-rich gas absorbs enough; solve the energy balance on this spectrum and, in the well-mixed limit, the same absorption is reached near He/H = 0.40, and near 0.053 on the other proxy spectrum, or near 1.5 once helium is allowed to settle against the adopted eddy coefficient. No answer on that list is secure while the width is unexplained, because the width is what tells us whether the line-forming gas is the gas either model thinks it is.

13. The best each model can do

Note added 2026-10-04. The EXHALE model of this section is a 2026-09 state; the best present model is He/H = 0.65 (Sect. 1.3, Fig. 2).

Figure 18 puts only the two best models on the measurement: the paper's p-winds model at its own best fit (the MCMC medians), and the EXHALE solution whose equivalent width matches. They fail in different ways, and neither reaches the observed width. p-winds is the broader of the two, because its temperature is imposed where the line forms, but at 0.575 Å it is still 1.5 times narrower than the measurement, and it overshoots the blue component by a factor of about three. EXHALE carries the right total absorption but concentrates it into a line a third as wide as observed, so its core is too deep. Only the second failure can be undone by broadening alone (Sects. 12.2 and 12.5).

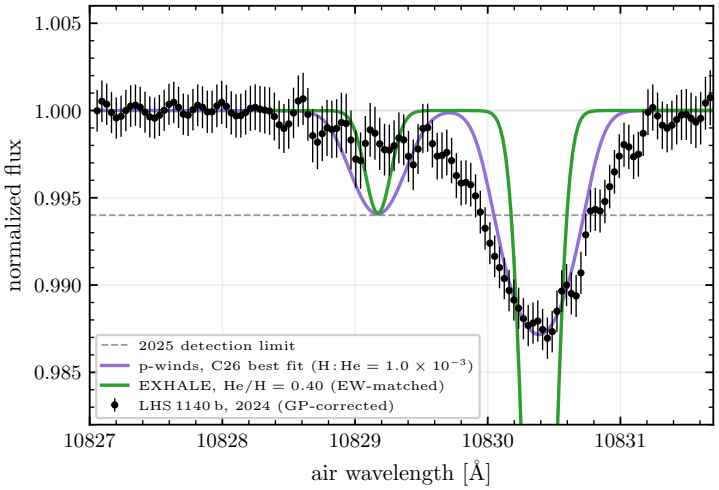


Figure 18: The two best models over the 2024 spectrum, in the presentation of the paper's Fig. 4: p-winds at the C26 best fit (the MCMC medians) with the authors' settings, and EXHALE at the composition matched on equivalent width, $\mathrm{He/H} = 0.40$. Both are shifted by the fitted $v_{\mathrm{wind}} = +2.26\,\mathrm{km\,s^{-1}}$.

Figure 19 shows what that EXHALE solution actually is. It is not a fast, hot outflow. The temperature peaks at 4292 K near $1.51\,R_p$, comparable to the isothermal value p-winds imposes, and then falls by adiabatic expansion to 855 K by $10\,R_p$ over the region that forms the line. The velocity reaches only $1.1\,\mathrm{km\,s^{-1}}$ at $20\,R_p$. Both elements stay mostly neutral through the line-forming region, and helium is the *more* ionized of the two: $x(\mathrm{H\,II}) = 0.03$, 0.07 and 0.16 at 2, 3 and $5\,R_p$ against $x(\mathrm{He\,II} + \mathrm{He\,III}) = 0.06$, 0.14 and 0.30. Helium carries the electrons that make the gas, and its ionization threshold is where this spectrum puts most of its photons. The metastable population peaks well out, at $5.2\,R_p$ with $n(2\,^3S) = 38\,\mathrm{cm^{-3}}$ in local equilibrium. A large, cold, slow, metastable-rich envelope of this kind gives the right equivalent width with the wrong profile: plenty of absorbers spread over most of the stellar disk, but almost no velocity spread to distribute the absorption in wavelength.

14. The other proxy: five times the XUV

Note added 2026-10-04. 2026-09 physics. The GJ 699 spectrum has since been run with the present equations (route B of the plan in Sect. 3., update log sections 95 and 98): a width-matched equivalent width of 3.3–3.7 at He/H 0.65–10, and the measurement matched by the GJ 1132 spectrum with 1.33–1.39 times its ionizing XUV at He/H 2.09.

The first item on the list above, more energy, can be tested without adding any physics, because the paper itself runs two proxy spectra. Both stand in for LHS 1140's unobserved XUV: GJ 1132 (used everywhere above) and GJ 699,

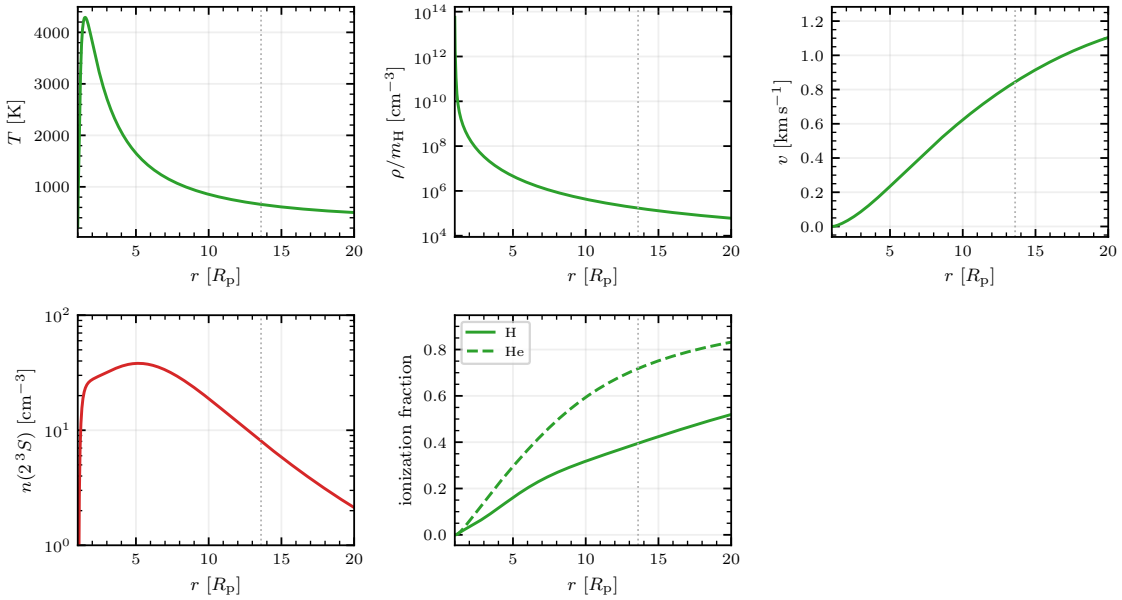


Figure 19: The EXHALE solution matched on equivalent width (the solved composition He/H = 0.40, nearest the well-mixed crossing at 0.4044): temperature, density, velocity, metastable helium density, and ionization fractions. Dotted vertical line: the stellar radius.

scaled to the orbit by the catalog X-ray flux ratios $A_{1132} = 0.59$ and $A_{699} = 0.056$ following the authors' script, the same convention adopted in Sect. 6.3. Integrated over 10–1300 Å at the orbit of the planet, the GJ 699 file carries 169.9 erg s^{-1} cm^{-2} against 33.5 for the GJ 1132 one: a factor 5.1 in XUV input.

On the EXHALE side only the spectrum changes. Each GJ 699 case is the matching GJ 1132 input with the spectrum file replaced. The luminosities stated in the input are irrelevant, because the code overwrites them with the integral of the spectrum it reads. Six reservoirs were solved, all of them converged.

14.1 Escape rate: the retrieval value, on this spectrum

Unlike a change of composition, the brighter spectrum moves $\dot{M}$:

	GJ 1132 proxy	GJ 699 proxy
$F_{\rm XUV}$ (10–1300 Å) [erg s^{-1} cm^{-2}]	33.5	169.9
$\log_{10}\dot{M}$ [g s^{-1}], He/H ≤ 1	7.84–7.86	8.55–8.57
$\dot{M}$ [g s^{-1}]	7.1×10^7	3.7×10^8
C26 retrieval on the same proxy	2.03×10^8	$4.22^{+1.14}_{-1.00} \times 10^8$

A factor 5.1 in XUV input gives a factor 5.1 in $\dot{M}$ here. On this spectrum EXHALE's escape rate agrees with the paper's retrieved value, $\log_{10}\dot{M} = 8.57$ at the crossing against 8.63, inside 0.06 dex and inside the retrieval's own ± 0.11 dex. It is also 1.3 times the energy-limited estimate the paper quotes for this proxy, 2.90×10^8 g s^{-1}. On the GJ 1132 spectrum the same code falls a factor 2.8 short of the retrieval. Whether a solved wind and an imposed one agree on $\dot{M}$ therefore depends on which proxy is assumed, and the disagreement reported in Sect. 12. is not a fixed property of the code.

14.2 Structure and line

Six compositions were solved on this spectrum, three of them (He/H = 0.042, 0.046 and 0.050) placed to bracket the equivalent-width crossing:

He/H	$\log_{10}\dot{M}$ [g s^{-1}]	$T_{\max}$ [K] (at r/R_p)	$v(20R_p)$ [km s^{-1}]	$n(2\,^3S)_{\max}$ [cm^{-3}] (at r/R_p)	red [%]	FWHM [Å]	EW [% Å]
0.042	8.57	2614 (1.47)	3.62	44.7 (5.8)	3.04	0.282	0.909
0.046	8.57	2645 (1.47)	3.60	48.9 (5.8)	3.29	0.282	0.986
0.050	8.57	2676 (1.47)	3.59	53.0 (5.8)	3.54	0.282	1.061
0.083	8.57	2919 (1.45)	3.49	86.7 (5.7)	5.40	0.284	1.627
1	8.55	6240 (1.34)	2.59	431 (6.8)	13.60	0.313	4.525
10^3	8.42	9518 (1.36)	2.03	2.6×10^4 (1.0)	7.51	0.359	2.872
observed	-	-	-	-	1.254	0.841	1.108 ± 0.030

The wind is hotter and faster than on the fainter spectrum: at He/H = 0.050 it reaches 3.6 km s^{-1} at $20R_p$ against 1.1 for the GJ 1132 solution matched on equivalent width, and the metastable population still peaks far out, near $6R_p$.

14.3 The composition estimate moves by an order of magnitude

Interpolating the equivalent width in log–log across the solved compositions and solving for the measured value gives

$$\mathrm{He/H} = 0.0526 \quad (0.0510\text{–}0.0543 \text{ on the } EW\ 1\sigma), \qquad \mathrm{H:He} = 19.0, \tag{8}$$

read between the solved cases at 0.050 (EW = 1.0609) and 0.083 (1.6271 % Å), the first of which is 1.6σ below the measurement and the second far above it, so the crossing is measured rather than extrapolated from the solar case.

The same code gives He/H = 0.4044 on the GJ 1132 proxy, an order of magnitude away, and the crossing passes the solar value on the way: the brighter spectrum wants a helium-*poor* atmosphere, 0.63× solar, where the fainter one wanted a helium-rich one, 4.9× solar. The measurement cannot arbitrate, because the two proxies are two guesses at the same unobserved spectrum and both reproduce the observed equivalent width once composition is free. The conclusion is therefore about the inference rather than about the planet: what the helium line constrains is the product of composition and XUV input, and any composition quoted from it (the paper's H:He $\lesssim 10^{-3}$ as much as the values here) is conditional on the assumed spectrum. On the present evidence the composition inference is limited by the assumed spectrum.

14.4 The width does not move

On this spectrum too the width does not move. Across the whole GJ 699 set the red-pair FWHM is 0.282–0.359 Å against the measured 0.841 Å. Measuring the deficit the way Sect. 12.2 does, by convolving with a Gaussian and asking which width reproduces the observation, the solution at the crossing needs 21.97 km s^{-1} FWHM of added velocity spread, against 22.22 km s^{-1} for the GJ 1132 solution matched on equivalent width and the same 22.22 for the

diffusive one of Sect. 12.4. Raising the XUV input by 5.1, $\dot{M}$ by 5 and the outflow velocity at $20\,R_p$ by 3 changes the required broadening by 1 per cent.

The reason is visible in the table: the outflow velocity that grows is radial and small compared with the missing spread, while the line-forming region stays cold. More energy is therefore tested and rejected as an explanation of the width. It moves $\dot{M}$ and the composition estimate, but not the profile. The remaining candidates, heating outside a 1D hydrodynamic energy equation, 3D velocity fields, and an adiabatic-cooling metastable bump, are unaffected by this test, and the last of them has since been looked for on both spectra (Sect. 12.6) and does not help either.

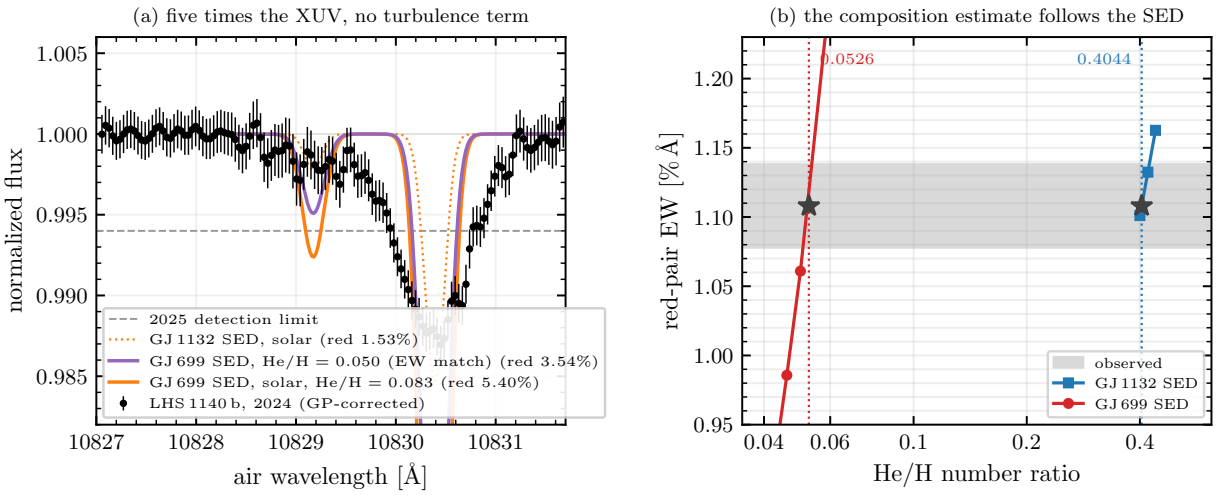


Figure 20: The GJ 699 proxy, five times the XUV of the GJ 1132 proxy used elsewhere in this memo. Left: the measurement with the EXHALE solution nearest the equivalent-width crossing 0.0526, the one at He/H = 0.050, and the solar case on this spectrum, with the GJ 1132 solar case dotted for contrast; all curves without the turbulence term and shifted by the fitted $v_{\rm wind} = +2.26\,{\rm km\,s^{-1}}$. The lines are deeper but no wider, the GJ 699 solar curve reaching 3.5× the depth of the GJ 1132 one (5.40% against 1.53%) while the added broadening the measurement demands of the two differs by $0.45\,{\rm km\,s^{-1}}$, or 2 per cent (21.95 against $22.40\,{\rm km\,s^{-1}}$). Right: red-pair equivalent width against composition for the two spectra, with the measured band and the two crossings; the composition the line implies moves from 0.4044 to 0.0526 when the assumed spectrum is changed.

15. Closing the loop: the composition as an output

Note added 2026-10-04. The closure procedure of this section is the one used in 2026-10, now with the energy coupling. Its flux-consistent sequence (equivalent width rising with He/H, every solution above the measurement) is superseded: on the present equations the sequence falls with He/H over 0.65–2.09 and every solution lies below the measurement (Sect. 1.2).

Every number above took the helium abundance as an input. The scans of Sects. 9. and 12.4 chose it, solved the wind, and asked what the line did. The reservoir at the base was a parameter set by hand, and the lower atmosphere below the 1 μbar matching level was a handful of single numbers. This section removes that freedom on one axis: the lower atmosphere is solved as a photochemical column, handed over as a profile, and the elemental fluxes are iterated until

they are continuous across the matching level. The composition at the match is then a solution of the coupled problem rather than a number chosen for it.

The three runs of this section have since been re-run end to end (chemistry, wind, measurement of the elemental fluxes and damped update) on Photochem 0.9.0, with the interface to it re-pinned, and on the current build, which is what Table 18 reports. What they establish is that the iteration is single-valued and that at $K_{zz} = 10^9$ it hands back the reservoir it was given, which is a statement about the iteration rather than about the line. The sequence of solutions of Sect. 15.7, which is what the composition is read from, was re-solved on both. The design, its tests and the runs are listed in Appendix A.

15.1 The setup

The column below the match is solved with Photochem (Wogan 2025) on the Zahnle reaction set restricted to H/He/N/O/C. The temperature is not prescribed: the accompanying climate solver is run with the tropospheric temperature solved for rather than imposed, so the stratospheric temperature is the skin temperature of the solution. The background gas is chosen by abundance rather than by name, and on this atmosphere that is helium rather than H_2. The climate solve is taken from a deep boundary at 20 bar, the shallowest one at which the bottom of the photochemical model fits inside the climate column. Chemistry and climate are run on the same set of elemental abundances, He/H starting at 2.09 (the $K_{zz} = 10^9$ crossing of Table 9) with Lodders C, N and O, on the GJ 1132 proxy spectrum at the planet's orbit and at the adopted constant $K_{zz} = 10^9$ cm^2 s^{-1}.

The result is handed to EXHALE as a profile of 102 levels from 16.73 bar to 1.10×10^{-8} bar, carrying p, r, T, n, ρ, K_{zz}, the molecular fractions and the elemental reservoirs He/H, C/H, N/H and O/H, matched to the wind at 1 μbar. Sect. 15.2 reads that solution column by column, and says which of the columns the wind is actually built from. The loop is then closed in the other direction. The elemental escape fluxes EXHALE computes are imposed back on the photochemical model at its top, $\phi_{\rm El} = \Phi_{\rm El}/(m_{\rm El}\, 4\pi r_{\rm top}^2)$, helium on He and hydrogen on H_2 at $\phi_{\rm H}/2$, which are the dominant carriers; the share carried by atomic hydrogen at that level is 10^{-4}. The pair is iterated by a damped fixed-point step ($\omega = 0.5$) on the mismatch $|F_{\rm measured} - \Phi_{\rm trial}|/|F_{\rm measured}|$ to a tolerance of 5%.

Two things about that measurement decide whether it means anything. The elemental flux is read in the *steady* window $r \geq r_{\rm esc}$ rather than in the interval the two models share: at a steady state $4\pi r^2(\rho X v + J)$ does not depend on radius, so a flux measured there is the flux through the matching level. The overlap window cannot carry it. From the match to the top of the profile the column spans 1.96 decades of pressure in $0.0098\,R_p$, entirely inside the region where the wind's mass flux is not yet its own, and the flux across it has a radial spread of 13 in $F_{\rm H}$ and 35 in $F_{\rm He}$. In the steady window the same solution holds 191 faces at spreads of 0.44% and 0.46%.

That flatness depends on how far the wind was converged, and we measured it: at a residual tolerance of 10^{-3} the same solution spreads by 7.4%, which would fail the validity test of the iteration itself, and continuing it to 10^{-4} brings it to 0.50% and moves $\log_{10}\dot{M}$ from 7.47 to 7.50. The iteration runs the wind at 10^{-4}.

The cross-check with a second chemistry code is not verified. The same handoff was set up with VULCAN (Tsai et al. 2017) on the production profile's own $T(p)$ and K_{zz}, and VULCAN did not converge on this cold helium-rich column: its convergence measure rose from 2.65 to 2.75 against a criterion of 0.01 with the timestep pinned, the limiting cell

being H_2O at the cold trap, and the run exited on its step limit. No VULCAN composition at the match was produced and no wind was run on one, so the spread between the two chemistry codes is not measured here.

15.2 The column the chemistry returns

The settings above produce one file, and everything EXHALE does on the profile branch rests on it, so the solution itself is read here column by column. Figs. 21 and 22 draw every column of the file except the two elemental-flux columns, which are constants carrying the trial fluxes the header states. The file measured here is the one-way handoff at iteration 0; the flux-consistent solutions of Sect. 15.7 carry their own copies and are compared at the end of this section.

Grid and thermal structure. 102 levels from 16.73 bar to 1.098×10^{-8} bar: the photochemical model's own domain, sitting inside the 20 bar column the climate step was taken from. Temperature runs from 520.3 K at the deep boundary to a floor of 185.05 K, which it reaches at 0.94 bar and holds at every level above. The stratosphere is isothermal at the skin temperature, and the tropopause the climate solve reports, 1.031 bar, is one level below the first isothermal level. K_{zz} is 10^9 cm^2 s^{-1} at all 102 levels, so the eddy coefficient the wind runs at and the one the column was mixed at are the same constant. The column spans 553.6 km in radius, 0.15608 to 0.16400 R_J, with the 1 μbar match at 0.16240 R_J = 1.782 $R_\oplus$, which fixes the base radius of the wind. Density falls from $n = 2.33 \times 10^{20}$ cm^{-3} ($\rho = 1.40 \times 10^{-3}$ g cm^{-3}) to 4.30×10^{11} cm^{-3} (2.58×10^{-12}).

The column is chemically almost frozen. Between the deep boundary and the match, 17 decades of pressure, q_{H_2} moves from 0.19258 to 0.19264, q_{CH_4} from 1.072×10^{-4} to 1.025×10^{-4} and q_{NH_3} from 3.158×10^{-5} to 2.919×10^{-5}. What changes at all is water, which condenses, and a set of trace photochemical products at the very top. In an eddy-mixed column at $K_{zz} = 10^9$ that is what one expects, and Sect. 15. makes the same point from the other side: the eddy term mixes the column uniformly all the way to the match.

Hydrogen stays molecular to the top. q_H rises from 5.9×10^{-21} at the deep boundary to 6.5×10^{-6} at the match and 4.0×10^{-5} at the model top, four to five decades below q_{H_2} throughout: there is no level in this column at which atomic hydrogen overtakes H_2. The base EXHALE is handed is an H_2 base in that sense. It does not enter the wind that way in these runs: EXHALE's molecular chemistry is off, and q_{H_2} is read but inert (see below).

The composition is cold and reducing. At 520 K and 17 bar CO is already negligible, $q_{CO} = 8.1 \times 10^{-18}$, and CO_2 smaller still. Carbon is CH_4 and nitrogen is NH_3, the opposite of the CO/N_2-dominated deep chemistry of a hot Jupiter. Counting carriers at the match against the hydrogen-nucleus fraction of the gas, $2q_{H_2} + q_H = 0.38532$:

- carbon: CH_4 gives 2.661×10^{-4} per H, 99.7% of the carbon the profile's molecular columns hold, with C_2H_2 and HCN supplying the rest;
- nitrogen: NH_3 gives 7.576×10^{-5} per H, 96.3%, with N_2 at 3.4%;
- oxygen: the water that survived the cold trap gives 4.831×10^{-7} per H, 99.9%.

Those sums come to 96.0, 96.0 and 97.6 per cent of the elemental columns $C/H = 2.778 \times 10^{-4}$, $N/H = 8.187 \times 10^{-5}$ and $O/H = 4.957 \times 10^{-7}$. The remainder sits in species the profile does not carry, so this accounting is a decomposition of the elemental ratios rather than an independent measurement of them.

This is the direct link to Sect. 15.6. The C and N that cooled the wind on the 2026-08-30 solutions and took its line down by 2.6× there (on the current C, N and O cooling they move it by 0.20 per cent, Sect. 5.4) are held in the column as CH_4 and NH_3, both of which the solution keeps in the gas phase across the trap, so they cross the tropopause intact and arrive at the match at essentially their deep values. Oxygen, whose carrier is water, does not.

Photochemistry acts only in the top decades. From the tropopause to the match, HCN goes $1.3 \times 10^{-13} \rightarrow 6.5 \times 10^{-8}$, N_2 $9.2 \times 10^{-9} \rightarrow 5.2 \times 10^{-7}$, C_2H_2 $2.5 \times 10^{-13} \rightarrow 1.1 \times 10^{-7}$, CO $5.2 \times 10^{-16} \rightarrow 1.8 \times 10^{-10}$ and OH $4.5 \times 10^{-27} \rightarrow 7.1 \times 10^{-11}$: several decades each, and all of them still trace. They do not move the elemental budget: the accounting above puts every one of them below 4% of its element.

What EXHALE consumes from this file was read in the code rather than inferred. The reader requires the pressure, the radius, the temperature, the total number density, the mass density, the eddy coefficient, the H_2 and atomic hydrogen fractions and the helium-to-hydrogen ratio to be present. It then takes, *at the matching level only*, the temperature as the base temperature, the radius as the base radius, the H_2 fraction as the base H_2 fraction, the helium-to-hydrogen ratio as the base reservoir, and the elemental ratios of C, N, O and of any other element the file carries as the metal abundances, through the same door a prescribed metal list uses. The matching pressure comes from the header. The eddy coefficient is the only quantity read over the whole profile, interpolated onto the radial grid through the file's own radius column; a single value of K_{zz} given in the input is then announced as inert.

Everything else goes unread. The number density and the mass density are required by the format but never used, since the base density follows from the input pressure and the base particle count from the base temperature, the helium-to-hydrogen ratio and the elemental ratios; the atomic hydrogen fraction is required and never used; the two elemental-flux columns are not read by the Fortran at all, since the iteration reads them on the Python side; and none of the molecular fractions beyond H_2 is read. The H_2 fraction itself acts only when a molecular base is switched on, in which case it enters the equation of state at the base. In the runs of this memo that option is off, and the configuration report the code prints at startup confirms it: the mass per hydrogen nucleus, 9.37393, and the particles per hydrogen nucleus at the boundary, 1.000116, are $1+4\,\mathrm{He/H}+\sum X_{\mathrm{El}}A_{\mathrm{El}}$ and $(1+\mathrm{He/H}+\sum X_{\mathrm{El}})/(1+\mathrm{He/H})$ with no H_2 term. So of the 23 columns, six numbers at one level plus one full column are what the wind is actually built from.

The flux-consistent solutions each carry their own solved column, and raising the reservoir changes that column in three measurable ways (Fig. 22d–f). The deep boundary cools, $520.3 \rightarrow 439.4 \rightarrow 387.9$ K at 16.73 bar for $\mathrm{He/H} = 2.09$, 5.01 and 10.31, while the stratosphere stays at 185 K. H_2 is diluted by number, $q_{H_2} = 0.1926 \rightarrow 0.0907 \rightarrow 0.0462$ at the match, with C/H and N/H unchanged to four digits because they are stated per hydrogen nucleus. The cold trap loosens: O/H at the match goes $4.96 \times 10^{-7} \rightarrow 6.43 \times 10^{-7} \rightarrow 9.09 \times 10^{-7}$, the trend the O/H column of Table 20 continues over 8.45–11.0. Oxygen is the only element the reservoir moves here, and it is the one whose carrier condenses.

15.3 The pressure form of K_{zz}

Every number in this subsection is the 2026-08-30 measurement. The mechanism it establishes does not depend on the physics that has changed since (Sect. 5.), but its magnitudes have not been re-measured on the current code.

The eddy coefficient of the last section is a stated number rather than a solved one, and it is stated in a particular way: constant, $1.0 \times 10^9\,\mathrm{cm^2\,s^{-1}}$ at every one of the 102 levels. The form is an assumption laid on top of the value. What

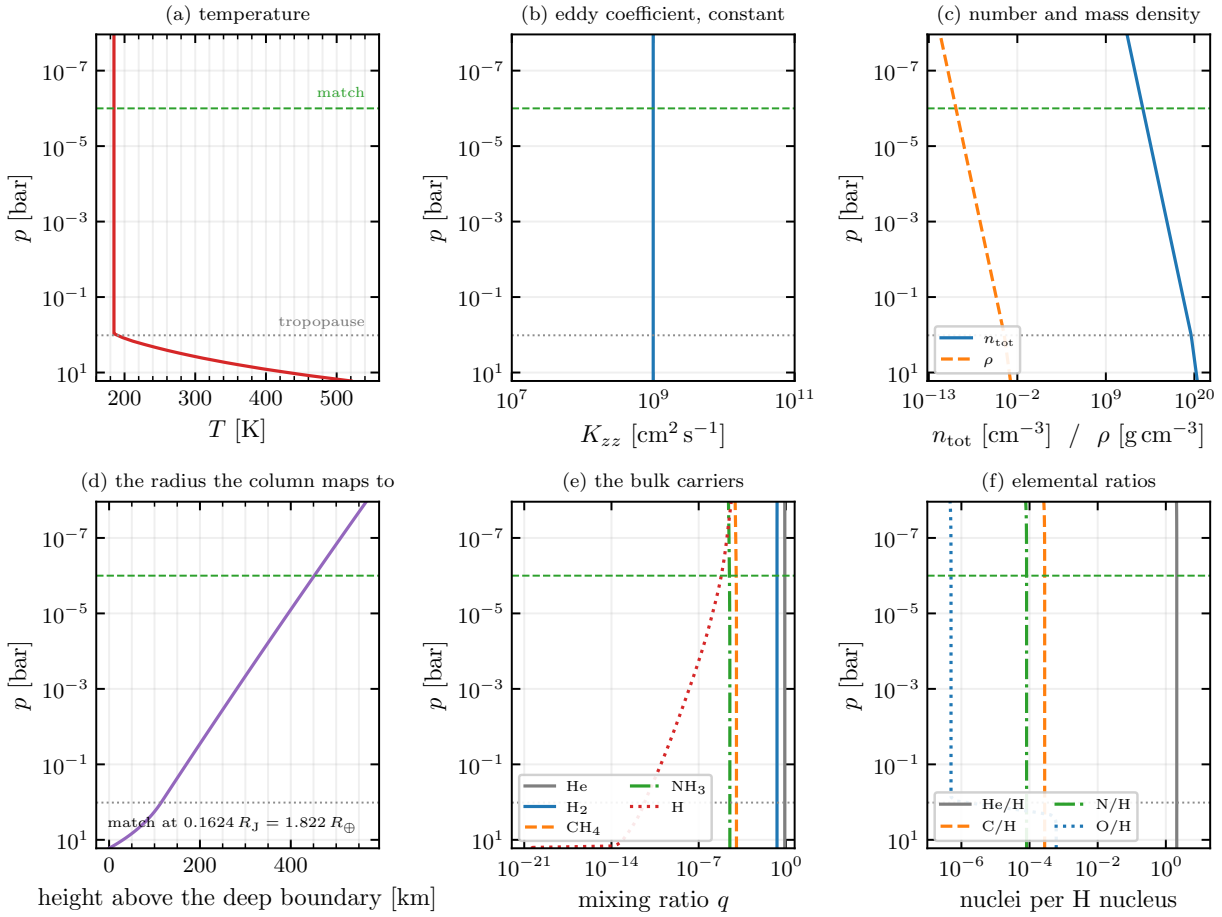


Figure 21: The photochemical column, its structure and the species that carry the bulk of each element. (a) the solved temperature; (b) the eddy coefficient, constant at 10^9 cm^2 s^{-1} over all 102 levels; (c) number and mass density; (d) the radius the pressure grid maps to, which is what fixes the base radius of the wind; (e) the species that hold the bulk: helium reconstructed as $\mathrm{He/H}(2q_{\mathrm{H_2}}+q_{\mathrm{H}})$, and atomic H four to five decades below H_2 everywhere; (f) the elemental ratios the profile hands over. The tropopause (1.031 bar, dotted) and the 1 μbar matching level (dashed) are marked on every panel. Fig. 26b,c shows the same climate and elemental ratios in the context of the iteration, while this figure shows the file itself, column by column.

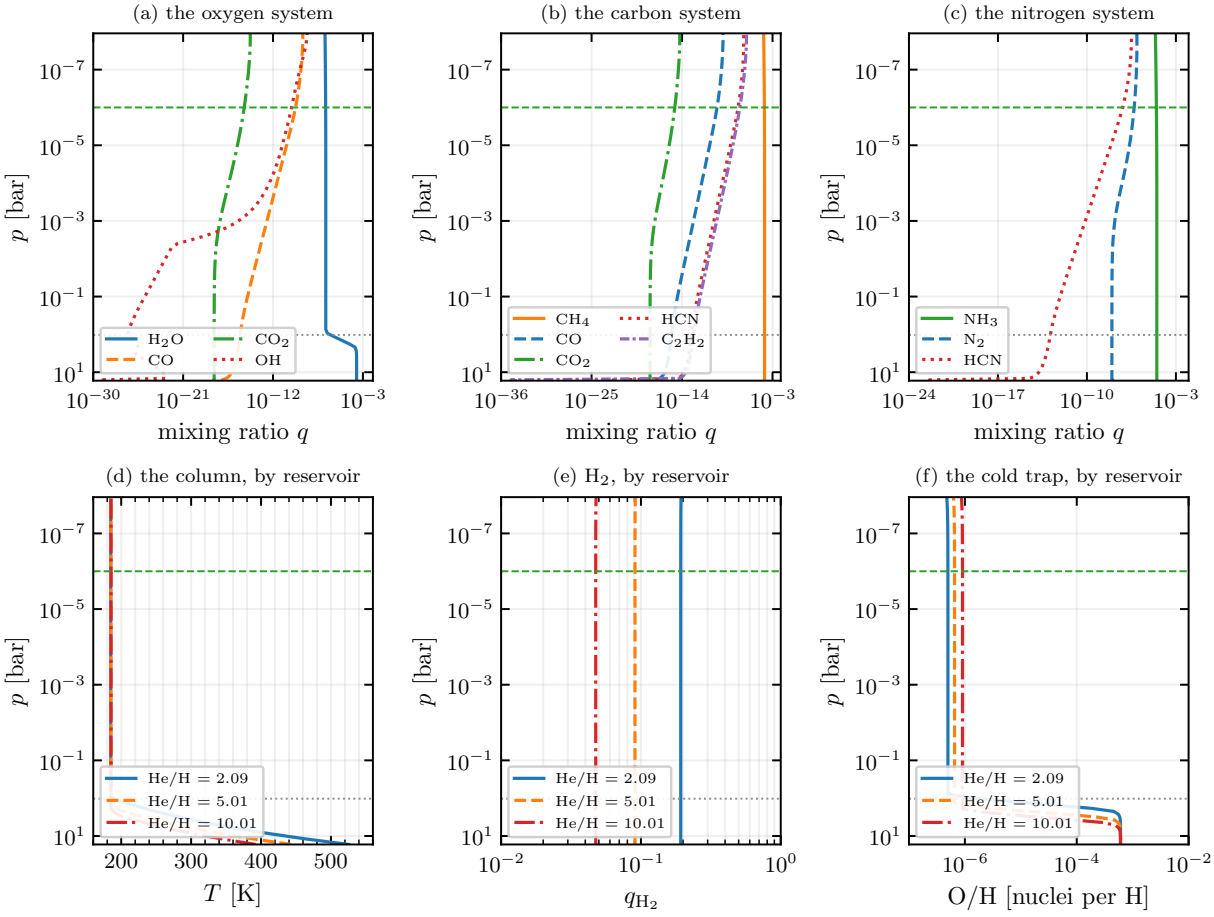


Figure 22: The molecular systems, and what the reservoir does to them. (a)–(c) the oxygen, carbon and nitrogen species of the same file: water condenses out at the tropopause while CH_4 and NH_3 cross it unchanged, and the photochemical products (HCN, C_2H_2, N_2, CO, OH) climb several decades in the top of the column while staying trace. (d)–(f) the same column solved at three reservoirs of the earlier sequence: the deep boundary cools as helium is added, H_2 is diluted by number, and the cold trap loosens, so oxygen is the one element the reservoir moves. Levels are marked as in Fig. 21.

the *value* carries is already measured: the equivalent width moves by a factor $\simeq 9$ between $K_{zz} = 10^8$ and 10^{10}, so the open question is whether the *shape* carries anything as well. We tested that against the constant, and the shape does not carry the conclusions of this memo. The reason is short: of the profile, EXHALE consumes elemental ratios and thermodynamics, and both are the same at the match to better than 1.2 per cent under either form.

What was tested. The saturated gravity-wave form,

$$K_{zz}(p) = 1.0 \times 10^9 \ \mathrm{cm^2\,s^{-1}} \left(\frac{p}{1.1 \times 10^{-8} \ \mathrm{bar}} \right)^{-1/2}, \tag{9}$$

anchored at the top of the profile and not at 1 bar. The anchor is the point of the test rather than a detail of it: EXHALE gives every cell above the shallowest tabulated level that level's value (Sect. 15.2), so anchoring at the top leaves the number the wind inherits above the file unchanged and confines the change to the interval the file describes. Equation (9) then implies 1.05×10^8 at the 1 μbar match, 1.05×10^5 at 1 bar and 2.56×10^4 at the 16.73 bar deep boundary: a factor 9.5 below the constant at the match, and four decades below it at the bottom.

Two cases separate the chemistry from the wind, because the profile hands over both:

- *A, chemistry only.* The column is solved on Eq. (9) and then its eddy coefficient alone is written back to the constant 10^9 before the handoff, so the wind sees the eddy coefficient it saw before and only the composition it is handed changes.
- *B, chemistry and wind.* The profile as produced; the wind reads $K_{zz}(p)$ as well, but only where the file speaks. The column reaches $1.0091\,R_\mathrm{p}$ above the match, and every cell above that takes the top level's value, so the change on the wind side is a shell nine thousandths of a radius deep in which the coefficient is 9.5 times lower at the base and equal at the top. The homopause under the constant sits at $1.056\,R_\mathrm{p}$, outside that shell. Case B is therefore a bounded test of the wind side rather than a test of lowering K_{zz} where helium settles; the unbounded version of that is the scan over magnitude, which is already measured.

The reference is the constant-K_{zz} profile re-solved from the same starting state and with the same build, so the comparison is not against a number recorded on an earlier occasion; the recorded number is carried in the tables anyway, because the difference between it and the reference is the size of the spread the procedure itself carries from one starting state to another. Two reservoirs of the earlier flux-consistent sequence were run, one at each end of it: He/H = 11.11 and 2.09, the latter on the branch started from a high trial flux. The test has since been repeated run by run on the current build, and the reasoning it was first reported under is confirmed: the coefficient is common to both sides of every entry, so a ratio between two chemistries at fixed everything else does not turn on it, and every ratio the test reports moves by less than the spread the procedure itself carries. The table below is the repeated version.

What did not move. The climate is identical: deep boundary 411.4 K, tropopause 2.636 bar at 185.5 K, cold-trap $f_{\mathrm{H_2O}} = 5.590 \times 10^{-8}$ and match temperature 185.478 K, the same digits under either form. That is structural rather than coincidental. The climate step solves the radiative-convective column with no eddy coefficient entering it, and hands the photochemistry a (P, T) pair with the eddy coefficient still unset, so the thermal structure, the tropopause and the trap the profile carries cannot depend on the eddy form.

The elemental ratios at the match, which with the temperature are what EXHALE consumes, move by at most 1.2 per cent:

Table 15: The pressure form of K_{zz} against the constant, at two reservoirs. *A* solves the chemistry on Eq. (9) and hands the wind the constant; *B* hands it Eq. (9) as well. The reference row is the constant-K_{zz} profile re-solved from the same starting state, and the row above it is the same profile solved on an earlier occasion from a different starting state: the spread of the procedure, against which the effect of the eddy form is to be read. All eight runs are on the build of 2026-08-30. *EW* is on this memo's red-pair window, comparable with the observed 1.108 ± 0.030 % Å.

run	red [%]	FWHM [Å]	*EW* [% Å]	*EW*/obs	He/H at $2R_p$	$\log_{10}\dot{M}$
reservoir He/H = 11.11						
recorded, other starting state	3.8857	0.2845	1.1798	1.065	1.5498	7.400
reference, $K_{zz} = 10^9$	3.9267	0.2845	1.1928	1.077	1.5664	7.410
A, chemistry only	3.9422	0.2845	1.1976	1.081	1.5743	7.410
B, chemistry and wind	3.9416	0.2845	1.1966	1.080	1.5582	7.410
reservoir He/H = 2.09						
recorded, other starting state	2.0683	0.2472	0.5445	0.491	0.2351	7.500
reference, $K_{zz} = 10^9$	2.0760	0.2472	0.5466	0.493	0.2359	7.500
A, chemistry only	2.0929	0.2472	0.5512	0.497	0.2377	7.510
B, chemistry and wind	2.0846	0.2467	0.5486	0.495	0.2355	7.510

Table 16: At the 1 μbar match, the ratio of the $K_{zz}(p)$ column to the constant-K_{zz} one. The top block is what EXHALE reads; the bottom block is what it does not.

quantity at the match	at He/H = 11.11	at He/H = 2.09
consumed by the wind		
X_{He}	0.9985	0.9985
X_C	0.9899	0.9903
X_N	0.9878	0.9875
X_O	0.9894	0.9821
T	1.0000	1.0000
carried, not consumed		
q_{CH_4}	0.443	0.665
q_{CO}	54.7	47.6
q_{CO_2}	9.3×10^3	4.3×10^3
q_{NH_3}	0.0038	0.0496
q_{N_2}	28.7	28.1
q_{HCN}	0.566	2.95
q_{H_2O}	0.714	0.804
q_H	4.99	5.55

What did move is the carrier species, by up to four decades. The lower part of Table 16 is the real effect of the eddy form, and it is large. Nitrogen changes carrier outright: at $\mathrm{He/H} = 11.11$, $q_{\mathrm{NH_3}}$ at the match falls from 6.32×10^{-6} to 2.43×10^{-8} while $q_{\mathrm{N_2}}$ rises from 1.21×10^{-7} to 3.46×10^{-6}, so a column that was NH_3-dominated becomes N_2-dominated. Carbon partly follows, CH_4 down by 2.3 and CO up by 55; CO_2 rises by nearly four decades from a negligible base. Deeper the redistribution is more extreme still (at 1 bar the same ratios are 1.3×10^5 for CO, 3.0×10^7 for CO_2 and 8.8×10^3 for N_2) because that is where the two forms differ most. *Elemental* nitrogen at the match nevertheless moves by −1.2 per cent, because a change of carrier conserves nuclei, and it is elemental nuclei that the handoff transmits.

The crossing. Converted with the local slope $\mathrm{d}\ln EW/\mathrm{d}\ln(\mathrm{He/H}) = 0.382$ measured over the 11.11–12.01 interval of the sequence these two belong to, the four equivalent widths move that sequence's crossing from 11.73 to 11.61 and 11.63 at $\mathrm{He/H} = 11.11$ and to 11.48 and 11.62 at 2.09: −0.25 to −0.10 in He/H, under a tenth of the 10.92–12.58 interval the measurement's own 1σ band allowed there. All four now move it the same way, downward. In relative terms that is −2.1 to −0.8 per cent, and it is the relative figure that carries over to the present crossing. It is also no larger than the spread of the procedure itself: re-solving the *same* profile from a different starting state moves the equivalent width by 1.09 per cent at $\mathrm{He/H} = 11.11$ and 0.38 per cent at 2.09, against the 0.32–0.83 per cent the eddy form produces. The signal is buried in the spread of the procedure that would have to measure it, by a narrower margin than before: 1.09 against 0.83 per cent where the previous branching gave 1.29 against 0.79.

One objection to the test itself has to be stated. The deep half of Eq. (9) is not the right law. The $P^{-1/2}$ growth is the saturated gravity-wave scaling of a *stratified* region (Lindzen 1981), and the climate solve puts this planet's tropopause at 2.636 bar: below it the column is convective, where the mixing-length form of Gierasch & Conrath applies and K_{zz} should flatten or grow with depth, not collapse. Nor is the amplitude defensible there. At 1 bar Eq. (9) gives 1.05×10^5 cm^2 s^{-1}, which is at or below the floor of the modeling literature, one and a half to two and a half decades under the 3×10^6–3×10^7 Charnay et al. (2015) use for GJ 1214 b, 3.7 decades under the 5×10^8 of Parmentier et al. (2013) for HD 209458 b, and equal to the 10^5 floor Ackerman & Marley (2001) impose by hand. The physically correct construction is two-branch, a wave law above the radiative-convective boundary joined to a mixing-length law below it; this test ran one branch all the way down to 16.73 bar. Read as interpretation, that is very likely why the carriers moved as far as they did, since the 0.1–10 bar interval the single branch over-quenches is the one that fixes where CH_4 and NH_3 quench. The objection does not reach the verdict, though: a form this wrong in the deep column still failed to move an elemental ratio at the match by more than 1.2 per cent.

At the other end the same equation is on firmer ground than the adopted constant. Its value at the match, 1.05×10^8 cm^2 s^{-1}, falls inside the 2.8×10^7–1.4×10^8 that the Arfaux & Lavvas homopause scaling gives when it is evaluated with *this* planet's gravity and temperature: the one estimate in the survey derived from LHS 1140 b itself, and the direction in which our own notes on the eddy coefficient already record the adopted 10^9 as having no planet-specific basis.

Verdict. The constant assumption was carrying a great deal of the lower atmosphere's carrier chemistry, up to four decades of it, and none of this memo's conclusions. What EXHALE consumes from the profile is the elemental ratios and the thermodynamics, and those agree at the match to 1.2 per cent under a form that differs from the constant by a

factor 9.5 there and four decades at the bottom. The match is still well mixed under either form: 1.05×10^8 is 200 times the molecular coefficient of $\simeq 4.6 \times 10^5\,\mathrm{cm^2\,s^{-1}}$ at the base. What the results carry is therefore the magnitude of K_{zz}, the 10^8–10^{10} bracket across which the equivalent width moves by a factor 9, rather than its pressure form. One limit: the verdict is tied to this anchor. Anchoring Eq. (9) at 1 μbar instead would raise by 9.5 the value the wind inherits over its whole domain, which is a change of magnitude and therefore a different test; it was not run.

15.4 The cold trap the climate solve puts in

Solving the temperature rather than prescribing it puts a cold trap into the handoff, at the level the climate solution places it rather than at the level a closed form assumes.

Table 17: The climate solve over its one free parameter, the deep boundary. f_{H2O} is the water mixing ratio above the tropopause, that is, what crosses into the photochemical column. The stratospheric temperature is robust and the tropopause pressure is not. The trapped water follows the pressure, since at fixed T_{trop} the trapped ratio is $p_{\mathrm{sat}}(T_{\mathrm{trop}})/P_{\mathrm{trop}}$.

deep boundary [bar]	T_{deep} [K]	P_{trop} [bar]	T_{trop} [K]	f_{H2O}
1	242.4	0.488	186.0	3.33×10^{-7}
3	308.5	0.765	185.8	2.06×10^{-7}
10	439.9	0.974	185.4	1.49×10^{-7}
20 (adopted)	**555.9**	**1.031**	**185.0**	$\mathbf{1.32 \times 10^{-7}}$
30, 50	refused: the pseudoadiabat leaves the thermodynamic data			

T_{trop} moves by 1 K over a factor of 20 in the deep boundary, against the skin temperature $T_{\mathrm{eq}}/2^{1/4} = 190$ K. P_{trop} moves by a factor of two, and the water moves with it. The scan itself gives the sensitivity, since $f_{\mathrm{H2O}}P_{\mathrm{trop}}$ is $p_{\mathrm{sat}}(T_{\mathrm{trop}})$: across the rows it comes out at 19% per K at 185 K. The trapped water is therefore conditional on the tropopause temperature at about a fifth per kelvin, and linearly on the tropopause pressure.

Against the published estimate. Cherubim et al. (2026) quote $f_{\mathrm{H2O}} \sim 7$ ppm from a closed-form cold trap at a 0.1 bar tropopause with $T_{\mathrm{skin}} = 194$ K. The climate solution gives 0.13–0.33 ppm at a tropopause of 0.5–1.0 bar: a factor 20–50 less water at a tropopause 5–10× deeper. The saturation physics agrees: $p_{\mathrm{sat}}(185\,\mathrm{K})/1\,\mathrm{bar}$ is 1.3×10^{-7} and $p_{\mathrm{sat}}(194\,\mathrm{K})/0.1\,\mathrm{bar}$ about 5×10^{-6}, so each number follows from its own tropopause, and what differs is where the tropopause is put. In our case it is a solved level in a helium-dominated column ($\mu = 3.6$ at $\mathrm{He/H} = 2.09$); in theirs it is assumed. We report this as a result of the climate model, and it was not tuned toward the published number.

In the handoff the trap shows up as a depletion with height rather than as a condensate sitting at the base. Gas-phase oxygen falls from $\mathrm{O/H} = 6.062 \times 10^{-4}$ at the deep boundary to 4.957×10^{-7} at the match, a factor 1223, while C/H (2.778×10^{-4}), N/H (8.187×10^{-5}) and He/H cross the tropopause unchanged. The carriers of carbon and nitrogen in this column are CH_4 and NH_3 (Sect. 15.2), and this solution keeps both of them in the gas phase across the trap (Figs. 26c and 22a–c).

15.5 The closure, and what it returns

Every number in this subsection is the 2026-08-30 measurement. The mechanism it establishes does not depend on the physics that has changed since (Sect. 5.), but its magnitudes have not been re-measured on the current code.
Three runs were made, from the reference trial pair $(\Phi_{\rm H}, \Phi_{\rm He}) = (1.8, 1.3) \times 10^7\,{\rm g\,s^{-1}}$ and from $0.3\times$ and $3\times$ it, all started from the same converged wind and all starting from the same ${\rm He/H} = 2.09$ reservoir.

Table 18: Making the base consistent with the elemental fluxes on LHS 1140 b. k is the iteration at which the 5% criterion was met. The equivalent width is re-measured here on this memo's own red-pair window (10832.60–10834.20 Å vacuum), so it is directly comparable with the observed 1.108 ± 0.030 % Å; integrated instead over the whole synthesized band, as the earlier record gives them, the same curves give 5.99–6.03 mÅ. Every column comes from one iteration re-run end to end on the current build and on Photochem 0.9.0, with the interface to it re-pinned. As on the sequence of Sect. 15.7, the wind of each converged iterate is then re-solved on its own output with the profile held fixed, so that runs which stopped at different k are read at the same place. Both line columns are at $R = 68{,}000$, the resolving power of the measurement and of the rest of this memo. An earlier version of this table printed its red column from the $R = 80{,}000$ synthesis while its EW column and the observed row were at $R = 68{,}000$, and that mixture is corrected here.

start	k	$F_{\rm H}$ [g s^{-1}]	$F_{\rm He}$ [g s^{-1}]	He/H at match	$\log_{10}\dot{M}$	red [%]	EW [% Å]
$1.0\times$	0	1.8017×10^7	1.2969×10^7	2.09235	7.500	2.022	0.532
$0.3\times$	4	1.8161×10^7	1.3436×10^7	2.09234	7.500	2.030	0.534
$3.0\times$	6	1.8147×10^7	1.3364×10^7	2.09236	7.500	2.019	0.531
observed	-	-	-	-	-	$1.254^{+0.22}_{-0.23}$	1.108 ± 0.030

The three runs agree on He/H to 7.8×10^{-6}, on $\log_{10}\dot{M}$ to better than the 0.005 the log prints, and on the fluxes to 0.8% and 3.5%, over a factor of ten in where they started, so the iteration is single-valued on this planet. The damping was never exercised, since the mismatch fell on every iteration, but the $3\times$ run used six of its eight allowed steps, so a start further than an order of magnitude out would need more steps.

One systematic effect on the profile route is larger than anything the iteration resolves, and it is stated here rather than corrected. Solving the same run again in four cycles (identical input, identical handoff column, identical build, each cycle started from the previous one and each converging at $\|R\| \simeq 10^{-4}$) lands on different fixed points depending on which converged wind it started from: on the $1.0\times$ run, $EW = 0.5185$, 0.5318 and 0.5423 % Å from three starting states, a spread of 4.6 per cent. Four further cycles move each line of descent by at most 0.23 per cent and leave the flux spread and the base temperature unmoved, so these are three fixed points of the procedure rather than three truncations of one. The diffusion step at fixed wind reports three different composition drifts (5.9, 6.2 and 7.3×10^{-4}), which is where the history dependence lives. Every line number taken from the profile route therefore carries a systematic effect of that order from its starting state, larger than the 1.8 per cent the secondary-ionization correction is worth on these runs. The comparisons here are internally consistent because each holds the procedure fixed. The history dependence itself is not diagnosed.

The composition is an output for the first time, and at this eddy coefficient it is the well-mixed value. He/H at the match comes back at 2.09235 against the 2.09 the reservoir was started from, a change of 1.1×10^{-3}, of which only 2×10^{-4} comes from the escape flux itself. The reason is measured: at $K_{zz} = 10^9$ cm^2 s^{-1} eddy mixing makes the column uniform all the way to 1 μbar, so a 1.8×10^7 g s^{-1} hydrogen escape imposed at the model top moves He/H by 1.8×10^{-2} at that top and by 2×10^{-4} at the match. At the matching level the composition is set by the eddy coefficient rather than by the escape flux. Closing the loop therefore does not retire the conditionality of Sect. 12.4, but moves it onto the same adopted K_{zz}.

Above the match the wind separates the two elements as strongly as before. The elemental fluxes carry $F_{\rm He}/F_{\rm H} = 0.73$ by mass against the base reservoir's 8.31, and He/H falls from 2.09 at the base to 0.183 at $30\,R_p$. Making the base consistent with the fluxes does not weaken that; it places the fractionation in the wind rather than below the match. One caveat on the budget, stated rather than corrected: the diffusion term uses the mass fraction of the H + He mixture and the trace metals ride inside ρ, so the measured $F_{\rm H}$ is hydrogen plus metals and overstates elemental hydrogen by 4.8×10^{-4}, two decades below the tolerance.

15.6 What the base metals do, and what the closed solution does not reproduce

The profile brings something the scalar base never had: the carbon, nitrogen and oxygen of the photochemical column. Isolating it takes a sequence of controls in which the scalar-base run is moved one setting at a time onto the profile's own base state, every step a converged solution. That sequence was run on the 2026-08-30 code and its magnitudes are superseded. The paragraph after it gives the current measurement, and Sect. 5.4 says what moved. Setting the base helium ratio to the profile's 2.0921 moves $\log_{10}\dot{M}$ by less than 0.01. Setting the base temperature to the solved 185.05 K, from the 226 K the scalar runs used, moves it by less than 0.01 more, so the base temperature is not the lever. Adding the base radius moves it to 7.85; the molecular fraction does not enter, since it is inert unless a molecular base is switched on, which neither route does. Adding the profile's C/H, N/H and O/H takes it to **7.49** and brings the red-pair depth down by a factor **2.9** at that step (5.59 → 1.93 per cent), that is −0.36 dex in $\dot{M}$ and a correspondingly shallower helium line, from metal-line cooling of a 185 K base at $\mathrm{C/H} = 2.8 \times 10^{-4}$.

Over the sequence end to end the same pair is −0.32 dex and a factor 2.7. The profile route reproduces that step to 0.00 dex in $\dot{M}$ and 0.5 per cent in the line, inside the spread the procedure itself carries, which is the consistency check the two routes exist to give. The six steps were re-solved at a residual tolerance of 1.0×10^{-4} on the four without metals and 4.0×10^{-4} and 5.0×10^{-4} on the two with them. The scalar step and the scalar column of Table 19 are now the same configuration at the same tolerance and give 1.420 and 1.424 % Å, 0.3 per cent apart, which is the spread this procedure carries from one starting state to another.

Every number in the two paragraphs above is the 2026-08-30 measurement, and the C, N and O cooling that carries it has since been replaced (Sect. 5.4). The decomposition of the handoff into its five terms (reservoir, base temperature, base radius, molecular fraction, elemental abundances) stands as an account of which term does what; the size of the last term does not. Re-measured on the current code at $K_{zz} = 10^9$ and $\mathrm{He/H} = 2.13$, adding the column's C, N and O to the scalar base moves $\log_{10}\dot{M}$ from 7.87 to 7.87 and the red-pair equivalent width from 1.3749 to 1.3721 % Å, 0.20 per cent, where the archived pair moved by −0.36 dex and a factor 2.9 in depth. The base metals are no longer the lever that

separates the two lower boundaries. The flux-consistent solution at He/H = 2.09 now gives EW = 1.4765 % Å against the metal-free base's 1.3749 at 2.13, above the measurement rather than a factor 2.0 below it.

Making the base consistent with the elemental fluxes and matching the equivalent width are still two different constraints, but they no longer pull in opposite directions. Fixing the reservoir by the line, as Sects. 11. and 12.4 do, was done above a base carrying no metals; with the photochemical column beneath it the same reservoir now absorbs *more*, so the composition that satisfies both conditions at once is smaller than the metal-free crossing, and smaller than every reservoir the flux-consistent sequence was run at (Sect. 15.7). The lower boundary is still an axis of its own, of the same standing as the assumed spectrum, the assumed mixing and composition itself. What changed is its sign and its size.

The mechanism: helium as a thermostat, and what is left of it

Why the same reservoir absorbs as it does over the two bases can be read off the solved structures. Table 19 measures the four solutions that bracket the question: the metal-free scalar base at He/H = 2.13; the same scalar base carrying the column's C, N and O reservoirs at the same composition, which isolates the metals from everything else the profile changes; the flux-consistent solution on the photochemical column at 2.0924; and the flux-consistent solution at 10.01. All four run with H/He element diffusion active at $K_{zz} = 10^9$ cm^2 s^{-1} (constant through the profile in the two profile runs, the same single value in the other two), and with EXHALE's molecular chemistry off, so no molecule is transported in the wind. Mean mass per particle is $(n_{\rm H}+4n_{\rm He})/(n_{\rm H}+n_{\rm He}+n_{\rm met}+n_e)$; nuclei are counted as H I + H II and He I + He II + He III. The values are from the advection-corrected profiles.

Table 19: The two directions the thermostat runs, measured on the solved structures of the set of record solved on 2026-09-16. EW is on this memo's red-pair window against the observed 1.108 ± 0.030 % Å. The C, N and O line cooling of all four is the density-resolved CHIANTI treatment of Sect. 5.4, which is what makes columns one and two nearly identical where the 2026-08-30 solutions put a factor 2.6 between them.

	scalar base no metals	scalar base C, N, O	photochem base C, N, O	photochem base C, N, O
reservoir He/H	2.13	2.13	2.0924	10.011
$T_{\rm max}$ [K]	5730	5566	5415	7682
$T(2R_p)$ [K]	4488	4423	4324	6780
mean mass per particle at $2R_p$ [$m_{\rm H}$]	2.14	2.14	2.15	3.17
n(H I) at $2R_p$ [cm^{-3}]	5.69×10^7	5.71×10^7	5.45×10^7	1.47×10^7
He/H by nuclei at $2R_p$	0.682	0.681	0.696	3.587
max $n(2^3S)$ [cm^{-3}]	34.6	35.5	35.8	88.8
$\int n(2^3S)\,dr$, 1–10 R_p [cm$^{-3}R_p$]	68.7	69.4	70.2	112.0
of which 1–2 R_p	25.3	25.9	26.2	52.7
of which 2–10 R_p	43.0	43.1	43.6	58.8
EW [% Å]	1.3749	1.3721	1.4765	2.2224

The base metals are no longer a thermostat setting. With the C, N and O line cooling evaluated at the electron density

of each cell rather than in the coronal limit (Sect. 5.4), adding the column's elemental reservoirs to the scalar base at fixed composition costs the wind 164 K in $T_{\max}$ and the line 0.20 per cent of its equivalent width. The comparison the 2026-08-30 solutions made between columns one and three ($T_{\max}$ 5447 $\rightarrow$ 3955 K and EW 1.424 $\rightarrow$ 0.545 % Å, a factor 2.6) was a consequence of the coronal coefficient, which stood up to four decades above the statistical equilibrium of the same ion at these densities. What is left between the scalar base and the photochemical column at the same reservoir is 315 K in $T_{\max}$ and +7.4 per cent in the line, and it is no longer dominated by the metals. Columns two and three still differ by the temperature at the matching level, the base radius and the elemental profile the column carries.

The cost to the wind. With no metals, radiative cooling is 4.1–6.9 per cent of the heating over 1.2–2 R_p and the wind is limited by expansion rather than by radiation. With the column's C, N and O beneath it, cooling is 5.1–7.9 per cent, of which the metal lines are 23 per cent at 1.2 R_p and 3 per cent at 2 R_p, and $T_{\max}$ falls from 5730 to 5415 K. Both the metastable peak and the metastable column rise slightly (34.6 $\rightarrow$ 35.8 cm^{-3} and 68.7 $\rightarrow$ 70.2 cm$^{-3}R_p$ over 1–10 R_p), and the line rises with them. The 2026-08-30 draft stated here that the metals take the outer column, because the cooler wind has the smaller scale height and does not carry the metastable population out to where it adds absorbing area. That is a real mechanism at a cooling rate that large, but it is not what these solutions do at the cooling rate the density-resolved coefficient gives.

Raising the reservoir reheats the wind, by two terms that are both measured. From 2.09 to 10.01 on the same base, $T_{\max}$ goes 5415 $\rightarrow$ 7682 K. Two things change together. First, there are fewer particles to share the deposited energy. The mean mass per particle at 2 R_p rises from 2.15 to 3.17 m_{H}, a factor 1.47; the heating rate at 2 R_p divided by the particle density rises by 1.50; and $T(2R_p)$ rises by 1.57. Here n_{tot} counts one particle per species with electrons excluded, as elsewhere in this section. The three factors agree to within 6 per cent, which is a consistency check on that reading rather than a proof of it, since the cooling term is not held fixed between the two runs. Second, the coolant is diluted along with the hydrogen it is referred to. The profile states C, N and O per hydrogen nucleus, and the code carries them that way, so when the neutral hydrogen density at 2 R_p falls by 3.7 the carbon density falls with it by the same factor, the two profiles carrying C/H within 0.1 per cent of each other. The volumetric metal cooling nonetheless *rises*, 2.36 $\rightarrow$ 4.65 $\times$ 10^{-11} erg cm^{-3} s^{-1} at 1.5 R_p, because the cooling function is steeply temperature dependent. What the dilution does is let the gas settle at the higher temperature at which that happens.

The metastable balance. The 2 ^3S population in these solutions is fed almost entirely by recombination and drained by Penning ionization on neutral hydrogen and by electron collisional de-excitation. Reconstructing the balance EXHALE solves apportions the budget: the source is $n_e n_{\mathrm{He\,II}} \alpha_{2^3S}$ plus collisional excitation from the singlet ground state, against radiative decay A_{31}, photoionization, destruction on H I, electron collisional de-excitation to 2 ^1S and 2 ^1P, and electron collisional ionization. At the metastable peak of the advection-corrected profiles, collisional excitation supplies a negligible part of the source and radiative decay plus collisional ionization take 0.02–0.03 per cent of the sink. The rest of the loss is Penning ionization and electron de-excitation: 72% and 28% at He/H = 2.09, 54% and 46% at 10.01. Between the two the source rate rises by 2.76 (more He II and more electrons in the hotter gas) while the metastable lifetime is essentially unchanged, 1.09 against 0.98 s. The neutral hydrogen the Penning term feeds on is thinned by only 1.20 at the peak, and the faster electron de-excitation of the hotter gas takes back what that gains. The product, 2.48, is the peak ratio 35.8 $\rightarrow$ 88.8 cm^{-3}. There is no charge-exchange term on the metastable in this code, since the

He/H charge exchange acts on ground-state He and He^+, so it is not part of this balance.

Read as an interpretation, the thermal structure of the 10.01 solution (T_{max} = 7682 K, column 112.0 over 1–10 R_p) now stands well above that of the metal-free 2.13 solution (5730 K, 68.7), and its equivalent width is correspondingly above the measured value rather than back at it. The sequence therefore still looks like a thermal trade between the metal content of the base and the helium content above it, but the metal side of the trade has become small (Sect. 5.4) and what is left is the helium. Helium in these solutions is not only the absorber; it is also the thermostat that sets how far the absorber is carried.

Composition and ionization. Fig. 23 carries the elemental fractions and the ionization state of the same three solutions, on the same advection-corrected profiles as Table 19. Two things follow from it that the thermostat argument rests on.

Diffusion keeps the wind less helium-rich than the reservoir. In both solutions the helium fraction falls through the line-forming region and then flattens: He/H = 2.09 at the base gives 0.696 by nuclei at 2 R_p and 0.554 at 5 R_p; 10.01 gives 3.587 and 2.747. That is a factor 3.0 and 2.8 between the reservoir and the wind at 2 R_p. In the composition's own terms, the flux-consistent solution at 2.09, a reservoir that is 68 per cent helium by number, has a *hydrogen-dominated* line-forming region, n_H/n_{tot} = 0.59 against n_{He}/n_{tot} = 0.41 at 2 R_p. At 10.01 helium leads there, 0.78 against 0.22, but by a factor 2.8 less than the reservoir states. A composition quoted for this planet therefore has to say at which level it is quoted; the reservoir and the absorbing layer are not the same number, and the memo's crossings are reservoir values.

The hotter solution is also the more ionized one. At 2 R_p, x(H II) = 0.031 on the photochemical base at 2.09 and 0.057 at 10.01: a factor 1.9, in the run whose neutral hydrogen density is 3.7 times lower. Both layers are optically thin at 13.6 eV, so the reading that thinner hydrogen shields itself less does not account for it. What grows is the ionization of hydrogen *by helium*, through Penning ionization He($2\,^3S$)+H, charge exchange He^++H and the He-recombination continuum above 24.6 eV. The split between those channels cannot be settled from the outputs: the individual rates (direct $\Gamma_{H\,I}$, the He-recombination-driven addition to it, and the secondary-ionization yield) are not written to any output file. One channel can be excluded, photoionization from n = 2, since the heating it would deposit is zero in all cells of all three runs. Helium ionization itself tracks the temperature more simply: x(He II) is 0.046, 0.046 and 0.055 at 2 R_p across the three, and x(He III) is at most 1.3×10^{-4} at 2 R_p (9.1×10^{-5}, 9.0×10^{-5} and 1.3×10^{-4}).

Fig. 19 shows the same kind of structure for the single well-mixed solution that reproduces the line with no diffusion and no base metals, while Fig. 23 is the set the lower-boundary axis is built on. The two are complementary rather than alternative views.

15.7 The composition the line needs above a photochemical base

Sect. 15.6 left unsolved the composition that makes the base consistent with the elemental fluxes *and* reproduces the line. This section solves the sequence, at the adopted eddy coefficient, and finds that on the current code there is nothing to solve for: every flux-consistent solution stands above the measured equivalent width. The iteration is run at nine starting reservoirs between 2.09 and 12, each cycling Photochem and EXHALE until the elemental fluxes are continuous across the 1 μbar match, with the climate step solved, $K_{zz} = 10^9$, the GJ 1132 proxy spectrum, a 5% tolerance on the mismatch and the same stationary solve inside every iteration. Each reservoir is read at its last iterate.

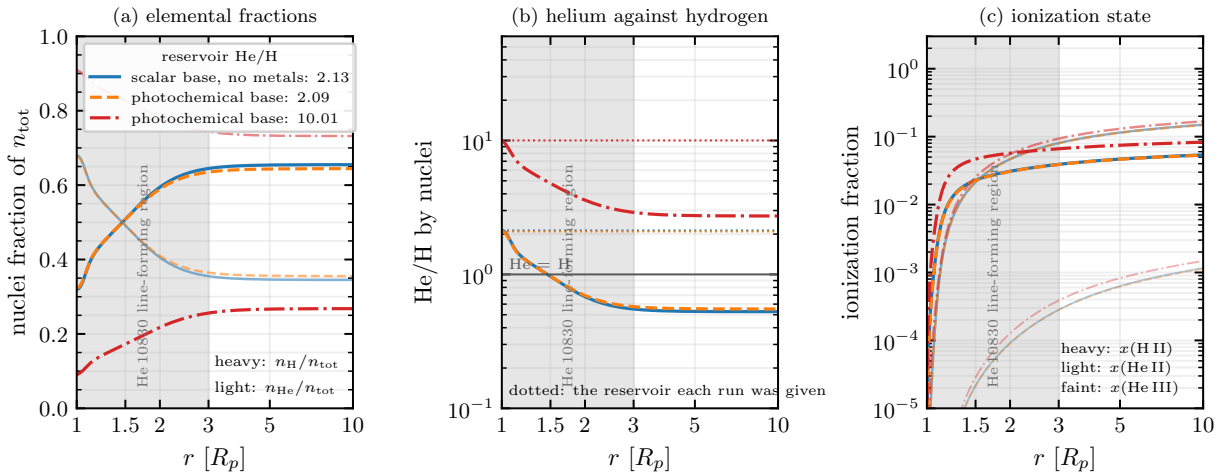


Figure 23: Composition and ionization against radius, for the three solutions of Table 19, from the same advection-corrected profiles. **(a)** the hydrogen and helium nuclei fractions of $n_{\rm tot}$; **(b)** He/H by nuclei, with the reservoir each run was given as a dotted line, so the vertical drop from dotted line to curve is what element diffusion does; **(c)** the ionization fractions. $n_{\rm tot}$ counts one particle per species, H I, H II, He I, He II, He III and every metal ion stage, with electrons excluded, and He $2\,^3S$ not added because it is an excited level inside He I, so (a) does not sum to one on the photochemical base. The metals it misses are negligible: their largest share of $n_{\rm tot}$ anywhere in 1–30 R_p is 2.3×10^{-4} at He/H = 2.09 and 9.7×10^{-5} at 10.01, and zero on the metal-free base. Nuclei are counted as H I + H II and He I + He II + He III. The flux-consistent sequence does not cross the measurement anywhere (Sect. 15.7), so no crossing is marked between the two profile runs drawn here. The He 10830 line-forming region 1–3 R_p is shaded.

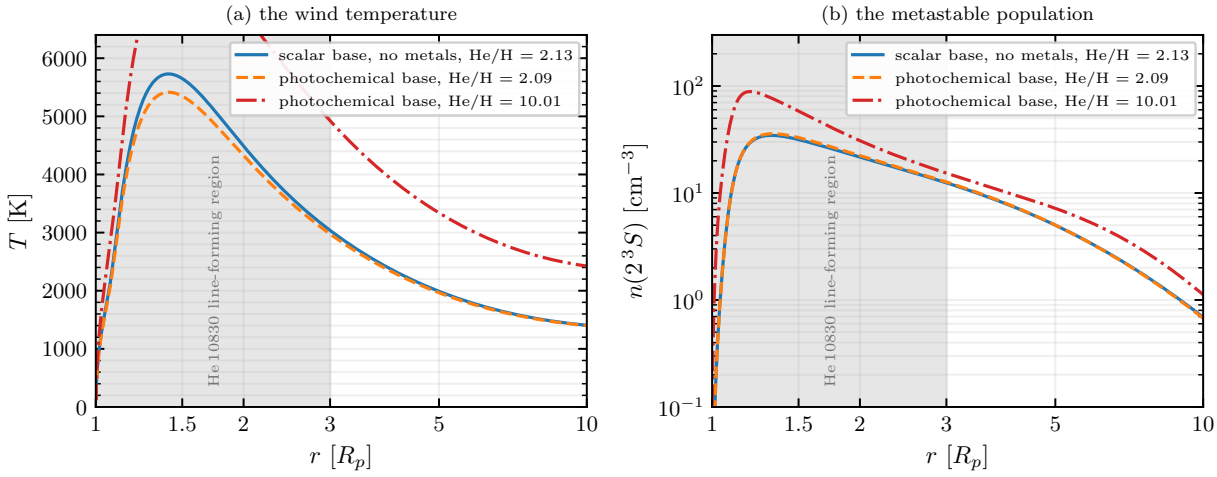


Figure 24: The thermostat, on the three solutions of Table 19. **(a)** temperature and **(b)** the metastable density, against radius, with the He 10830 line-forming region 1–3 R_p shaded. Adding the photochemical column's C, N and O at fixed reservoir (solid to dashed) now costs the wind 315 K in $T_{\rm max}$ and leaves the metastable population where it was, to within 2 per cent in the peak and in the column. Raising the reservoir on that same base (dashed to dot-dashed) reheats it by 2267 K and lifts the metastable peak by 2.5.

Table 20: The sequence of reservoirs made consistent with the elemental fluxes, 2026-09-16. One row per reservoir, at the iteration k where the 5% criterion was met; fluxes in 10^6 g s^{-1}, the cold-trap oxygen in 10^{-7} nuclei per H interpolated to the matching level. The equivalent width is on this memo's red-pair window, directly comparable with the observed 1.108 ± 0.030 % Å, and no row reaches it from below. ε is the larger of the two elemental mismatches. The first column is the reservoir the run was started from, the sixth the one the iteration returned.

start	k	ε	$F_{\rm H}$	$F_{\rm He}$	He/H at match	$\log_{10}\dot{M}$	red [%]	FWHM [Å]	EW [% Å]	O/H trap
2.09	5	0.026	26.34	54.42	2.0924	7.910	5.321	0.2603	1.4765	4.97
3	4	0.048	19.89	60.52	3.0034	7.900	6.156	0.2684	1.7640	5.52
5	4	0.043	13.37	66.32	5.0057	7.900	6.933	0.2805	2.0745	6.53
7	4	0.044	9.17	70.06	7.0080	7.900	7.102	0.2871	2.1768	7.77
8	4	0.045	8.11	70.98	8.0091	7.900	7.115	0.2896	2.2004	7.99
9	4	0.045	7.37	71.66	9.0103	7.900	7.111	0.2916	2.2136	8.86
9.7	4	0.045	6.88	72.10	9.7111	7.900	7.101	0.2931	2.2206	9.03
10	4	0.045	6.70	72.28	10.0114	7.900	7.094	0.2936	2.2224	9.06
12	4	0.045	5.56	73.36	12.0137	7.900	7.037	0.2966	2.2289	10.51
observed	-	-	-	-	-	-	$1.254^{+0.22}_{-0.23}$	0.841	1.108 ± 0.030	-

Every run returns a matching-level He/H equal to the reservoir it was started from to four digits, for the reason Sect. 15. measured at 2.09: at $K_{zz} = 10^9$ the eddy-mixed column reaches the microbar match, so the escape flux does not set the composition there. What this sequence adds is the line.

There is no crossing. The lowest reservoir solved, He/H = 2.0924, already makes $EW = 1.4765$ % Å against the measured 1.108 ± 0.030: 33 per cent, that is 12σ, above it. The sequence rises from there and flattens near 2.22 % Å above He/H $\simeq 8$, so the measurement is not reached anywhere on it. Read against its own local slope $\mathrm{d}\ln EW/\mathrm{d}\ln(\mathrm{He/H}) = 0.49$ between the two lowest rows, the measured equivalent width would be reached near He/H ≈ 1.2; that is an extrapolation below every solved reservoir and is not offered as the answer. What the sequence does say is that the photochemical lower boundary no longer makes the line harder to reach than the metal-free one does; it makes it easier. The comparison at one composition is Table 19, and its cause is the density-resolved C, N and O cooling of Sect. 5.4.
The sequence is shallow and becomes shallower: the local slope runs 0.49, 0.32, 0.14, 0.08, 0.05, 0.04, 0.03 and 0.02 over the successive intervals, so above He/H $\simeq 7$ the line stops responding to composition. The saturation is a result in its own right: on this base the equivalent width is a weak lever on the reservoir at the helium-rich end, and a composition read there would carry a large error bar even if the sequence did cross.
At the lowest reservoir the escape rate is $\log_{10}\dot{M} = 7.91$, that is 8.1×10^7 g s^{-1}, a factor 2.5 below the $2.03^{+0.58}_{-0.67} \times 10^8$ g s^{-1} the p-winds retrieval imposes, and the elemental fluxes are $F_{\rm H} = 2.6 \times 10^7$ and $F_{\rm He} = 5.4 \times 10^7$ g s^{-1}.

Two separate statements follow.

- Consistency with the elemental fluxes by itself does not select a reservoir on this planet. Every row of Table 20 satisfies it, at elemental mismatches of 2.6–4.8% against the 5% tolerance. What used to select one was the line, as in Sects. 11. and 12.4; on the current code the line selects none of them, and the composition this base would

need lies below the sequence.

- The escape rate is flat along the whole sequence, 7.90 to 7.91 in $\log_{10}\dot{M}$ over a factor 5.7 in reservoir, so here too the wind does not fix the composition. It agrees with the 7.84–7.87 of the prescribed-base branch to within 0.05 dex, so what is placed at the lower boundary is no longer visible in the escape rate either.

The elemental mismatch ε of Table 20 is the test applied; the radial spread of the flux over the window it is measured on is 3.2×10^{-3} in hydrogen and 6.5×10^{-4} in helium.

The sequence does not fix the width. The line is still too narrow: FWHM 0.26–0.30 Å against the measured 0.841, unchanged from Sect. 12.4 and from the solutions without diffusion. The winds these numbers come from also carry the caveats of Sect. 16. in full: their domain stops inside the critical point, and nothing in their profiles above about $10\,R_p$ is usable. The He 10830 layer is not affected: it sits at $\mathrm{Kn} \leq 0.011$.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

15.8 The outer region a converged wind inherits

Every number in this subsection is the 2026-08-30 measurement. The mechanism it establishes does not depend on the physics that has changed since (Sect. 5.), but its magnitudes have not been re-measured on the current code.

The effect was found on the earlier sequence, at its $\mathrm{He/H} = 12.01$ step, and it is why the sequence of Table 20 is climbed from one common parent. That step was first reached in one jump from 8.0, a factor 1.50 in reservoir; a control run reached the same reservoir from 11.11, a factor 1.08. The two solutions agree on everything the iteration and the chemistry set, and disagree on the line:

- Same reservoir: He/H at the match 12.013522 against 12.013520, seven digits.
- Same lower atmosphere: cold-trap O/H 1.0656×10^{-6} against 1.0660×10^{-6} nuclei per H.
- Both converged: $\|R\| = 1.866 \times 10^{-4}$ for the one reached in one jump and 1.287×10^{-4} for the control, both inside the 4.0×10^{-4} target the step was run to, and both inside the tolerance on the flux spread.
- Different line: $EW = 1.4293$ against 1.2094 % Å and a red depth of 4.789 against 3.956 per cent, an 18 per cent difference in equivalent width at fixed composition.

The difference sits entirely in the outer wind. Two diagnostics that do not depend on where a spectral window is placed separate the two solutions, and both vary monotonically along the sequence except at the discarded point:

r_{drop} moves inward monotonically as the reservoir rises, from 30.0 to $19.5\,R_p$, and the outer share of the metastable column stays in a narrow band, 0.0083–0.0117, across the whole sequence. The discarded solution departs from both: $25.8\,R_p$ where its own reservoir gives 19.5, and about twice the outer column share of any other step (2.05 times

Table 21: Two measures of the outer wind that do not depend on a spectral window, along the earlier sequence, from the advection-corrected profiles. $r_{\rm drop}$ is the first radius outside $1.5\,R_p$ at which T falls below half of $T(12\,R_p)$, and the last column is the share of the He $2\,^3S$ radial column that lies outside $10\,R_p$; every entry was recomputed for this table from the runs, which hold these solutions re-solved one step at a time on the current build. On the single-parent sequence of Table 20 the same two diagnostics give $r_{\rm drop} = 20.46$ falling to $19.47\,R_p$ and an outer column share of 0.0091–0.0095, flat across every step, with none standing out.

case	$r_{\rm drop}$ [R_p]	$T(12\,R_p)$ [K]	He $2\,^3S$ column outside $10\,R_p$
2.09	30.00	899	0.0117
3.0	27.63	1026	0.0096
5.0	23.76	1299	0.0084
8.0	20.80	1630	0.0083
9.7	19.79	1794	0.0090
10.3	19.79	1839	0.0091
10.7	19.47	1870	0.0092
11.1	19.47	1898	0.0092
12.0	19.47	1959	0.0095
12.0 discarded	**25.83**	1846	**0.0195**

its control's, 1.67 times the largest any other step reaches). Its $T(12\,R_p)$ does *not* match its control's (1846 K against 1959, 5.8 per cent apart), so that quantity no longer shows that the two agree on the base. The agreement rests on the matching-level composition, identical to seven digits, and on the profile ratio below. The discarded solution is a hotter, more extended outer atmosphere sitting on an identical base.

The mechanism appears to be that the outer state is inherited from the starting state and the Newton finish does not re-solve it. The density there is low enough that it contributes almost nothing to the norm of the residual, so $\|R\|$ cannot tell the two states apart. Both converge at residuals inside the target, and which of the two reports the smaller one says nothing about which is right: on the current version of the code, as on the one before it, the discarded solution reports the larger residual, and only on the retired rate coefficient did it report the smaller. He 10830 is optically thick at these columns, so a hot, extended outer region fills the line wings and raises the equivalent width even though it carries about one per cent of the metastable column.

Measured on the metastable profiles, the two solutions agree to 2.3 per cent out to $11.9\,R_p$ and then separate abruptly (the departure passes 10 per cent at $11.95\,R_p$ and 50 per cent at 12.14, reaching a factor 5.1), so the region in which they differ is 12–$30\,R_p$ rather than the 20–$30\,R_p$ an earlier draft of this paragraph gave. The exobase of these two runs is also not the 18.8–$25.9\,R_p$ of Sect. 16.: that is the three-case range of Table 23, whose cases are two runs with a prescribed base and the flux-consistent run at 2.09. For this pair it is $11.81\,R_p$ for the one reached in one jump and 14.73 for the control.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\rm bulk} = 0.1$ radii and exobase radii in this passage were computed be-

fore the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $Kn_{bulk} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $Kn(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

Putting the corrected radii together: for the solution being discarded the whole differing region lies outside its own exobase, where $Kn > 1$; for its control the innermost $2.6\,R_p$ of that region is nominally still collisional. The fluid solution is therefore not valid in most of the region that makes the difference, and in the discarded solution it is not valid anywhere in it. The ground for discarding that solution is continuity of the trend rather than a physical criterion, and by the same token the absolute equivalent widths of the whole sequence carry a systematic error that a 5% tolerance on the flux spread does not catch. Its size is small, because the outer region is about one per cent of the metastable column, but it is not zero. The exobase itself does not move with either correction: across the sequence it is 11.8–18.9 R_p on the current version of the code against 11.8–18.8 on the one before it.

Independence of the path was checked directly at $He/H = 3.0$, from a starting state 3.7× away in composition and in the downward direction the sequence never takes: it returns the same r_{drop} to the digit, with everything else inside tolerance. The 5.0, 8.0 and 2.09 reservoirs were not checked this way. Separately, the 10.31 one was re-run from the same starting state in its own directory and reproduced identically, which fixes determinism but says nothing about the dependence on the starting state. The rule the episode leaves is that a scan over reservoirs must be climbed one step at a time.

15.9 The 2025 non-detection

Cherubim et al. detect the line in 2024 and not in 2025, at a limit of 0.6 per cent in depth. The grid below asks what fraction of the fiducial XUV reproduces that. The spectrum is the fiducial GJ 1132 proxy with its flux column multiplied by the stated factor; the code recomputes L_X and L_{EUV} from the file it reads, so the luminosities stated in the input are a record of provenance only. The lower atmosphere is held fixed in every case, since neither the flux iteration nor the chemistry that made the profile is re-run, so the grid isolates the wind's response to the irradiation.

It is run from both lower boundaries, since the two are not interchangeable: the scalar-base solution at $He/H = 2.13$ ($EW = 1.3749$ % Å) and the flux-consistent solution at $He/H = 9.71$ ($EW = 2.2206$). Every point is solved by the same route as the fiducial, apart from the numerical flux of the entries marked in Table 22 and the starting state of the scalar case at 0.01.

The entries marked † belong to the seven lowest-XUV atomic cases that are, since 2026-09-17, solved with a different choice for the numerical flux the solver uses to move mass, momentum and energy between cells, on the same 500-cell grid and by the same recipe in every other respect. The original flux does not reach the stationary state there at all, and the reason is the grid and not the wind: the same case solved with the original flux on a grid of twice the cells does reach it, and the state it reaches is the one the other flux reaches on the grid of record, the two agreeing to 1.0 per cent in $\dot{M}$ and 0.4 per cent in the equivalent width. What the change of flux costs, where both fluxes solve and the comparison is therefore possible, is 0.3 to 0.6 per cent in $\dot{M}$ and in the equivalent width, both downward, and 6 to 8 per cent in the first cell, which comes out colder and denser; the spread of the mass flux does not move. Those shifts are far smaller

Table 22: The He 10830 line against the XUV scaling, from the two lower boundaries. Bold marks the row at which each model is nearest the 0.6 per cent depth limit of the 2025 epoch. Entries marked † are solved with the Roe numerical flux in place of HLLC, for the reason and with the systematic difference given after the table: the flux-consistent cases at 0.15, 0.20 and 0.25 of the fiducial spectrum, which stopped solving under HLLC on the corrected lower boundary of Sect. 5.8, and the scalar case at 0.01, reached by a continuation in XUV through 0.07, 0.05, 0.03, 0.02 and 0.015 on 2026-09-21 (the scalar depths at those five scalings are below 10^{-3} per cent and are not tabulated). All four certify. Their certification and $\log_{10}\dot{M}$ are READ from `LHS1140b/MODELS.md`, sections 3.1, 7 and 8, as of 2026-09-23; their red depth is the case's own `tpm_He10830_metrics.txt` and their EW is measured here through the operator of Sect. 11., both equal to that record to the digits shown. The flux-consistent case at 0.01 is not certified: its last run was stopped at the campaign's 30-minute ceiling with the hydrodynamic equations refusing, and it is not tabulated. The 0.05 case the previous draft carried for the flux-consistent branch was solved on the earlier boundary and is not quoted here.

	scalar base, He/H = 2.13			flux-consistent, He/H = 9.71		
$F_{\rm XUV}$ / fiducial	red [%]	EW [% Å]	$\log_{10}\dot{M}$	red [%]	EW [% Å]	$\log_{10}\dot{M}$
1.00	4.958	1.3749	7.87	7.101	2.2206	7.90
0.33	1.020	0.2555	7.36	3.821	1.0979	7.39
0.30	0.760	0.1894	7.32	3.542	1.0097	7.35
0.25	**0.279**	0.0700	7.24	3.002†	0.8428	7.26
0.20	0.007	0.0022	7.13	2.433†	0.6711	7.16
0.15	0.001	0.0003	7.00	1.783†	0.4822	7.02
0.10	0.000	0.0001	6.82	**1.001**	0.2674	6.84
0.01	0.000†	0.0000	5.79	not certified		

than the spacing of the grid, so they do not change what the grid is read for, and they are quoted wherever such a row appears.

The scalar model meets the 0.6 per cent limit at 0.28 of the fiducial XUV; the flux-consistent one has not reached it on any scaling at which it certifies. In the scalar model the depth crosses 0.6 between 0.25 (0.279 per cent) and 0.30 (0.760), at 0.283 read linearly in both variables. On the paper's fiducial $F_{\mathrm{XUV}} = 33$ erg cm^{-2} s^{-1} that puts the 2025 epoch at about 9 erg cm^{-2} s^{-1} or less. The flux-consistent branch is still at 1.001 per cent at 0.10, the faintest scaling at which it certifies; from 0.30 down to 0.10 its depth falls monotonically (3.542, 3.002, 2.433, 1.783 and 1.001 per cent), so *its crossing lies below* 0.10, and since its case at 0.01 is not certified *the crossing is not bracketed*. The figure draws that uncertified point as well, and the 0.06 it marks on this branch is a linear reading between it and 0.10, not a measured crossing.

The two branches no longer agree on this, as the 2026-08-30 grids did (both then at about 0.33). The flux-consistent branch makes a much stronger line at every scaling it solves, which is the same statement Table 20 makes at the fiducial, so its limit is reached at a smaller fraction of the fiducial XUV than the scalar one's: below 0.10 against 0.28, a factor of at least 2.8. How much smaller is what a certified flux-consistent state below 0.10 would settle.

The decline is also far steeper in the scalar model than in the flux-consistent one: over the 0.33 to 0.10 interval the scalar depth falls by a factor 3×10^3 and the other by 3.8. The *shape* of the XUV dependence is therefore a property of the lower boundary rather than a result, and it is a much larger difference than the 2026-08-30 grids showed.

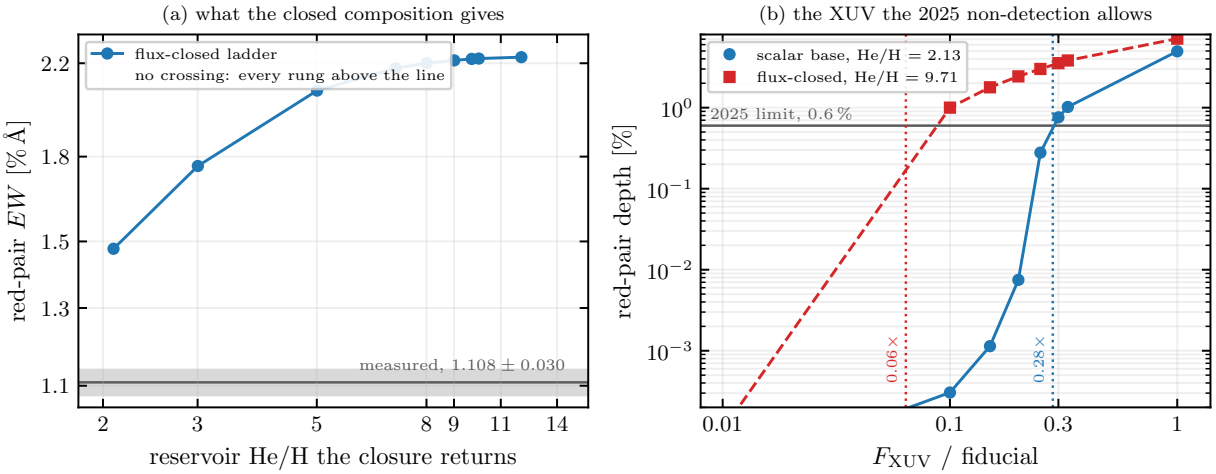


Figure 25: (a) The red-pair equivalent width of the flux-consistent solutions against the reservoir He/H returned (Table 20), with the measured 1.108 ± 0.030 % Å as the band at the bottom of the panel; every point stands above it, the lowest by 33 per cent, so no crossing is drawn. Every point satisfies the elemental-flux condition, so that condition never picked a reservoir, and on the current code the line does not pick one either. (b) The red-pair depth against the XUV scaling (Table 22), from the scalar base at He/H = 2.13 and the photochemical column at 9.71. The scalar model falls below the 0.6 per cent limit of the 2025 non-detection at about 0.28 of the fiducial irradiation (dotted). The other model is drawn from 0.10 to 1.00, where every point certifies, and continued to its case at 0.01, which does not certify; the 0.06 marked where that last segment crosses the limit is a linear reading between an uncertified point and 0.10, not a measured crossing. On certified states its limit lies below 0.10 and is not bracketed.

15.10 The width

Measured the way Sect. 12.2 measures it, by convolving the solved profile with a Gaussian and asking which one reproduces the observed width, the lowest of the flux-consistent solutions asks for 22.15 km s^{-1} FWHM ($\sigma = 9.40$ km s^{-1}) and the highest for 21.78, against 22.22 in the well-mixed limit, 22.22 with diffusion at the adopted K_{zz}, and 21.97 on the 5.1× brighter proxy. Those solutions span a factor 40 in composition and two different lower boundaries, and the Gaussians they ask for differ by 2 per cent. As solved, the red-pair FWHM along the sequence is 0.260–0.297 Å against the measured 0.841.

The lower-boundary axis therefore behaves like the other three: it moves the composition inference and the escape rate, and it does not move the width. It does not reach the measurement here either: at its matched Gaussian the lowest flux-consistent solution sits at a red depth of 1.648% against the observed $1.254^{+0.22}_{-0.23}$ and an equivalent width of 1.428 against 1.108 ± 0.030, both above the band, because broadening conserves absorbing power and this solution has too much of it. That is the mirror image of the p-winds case of Sect. 12.5, where the curve has too little: a convolution moves absorption in velocity and cannot change the amount.

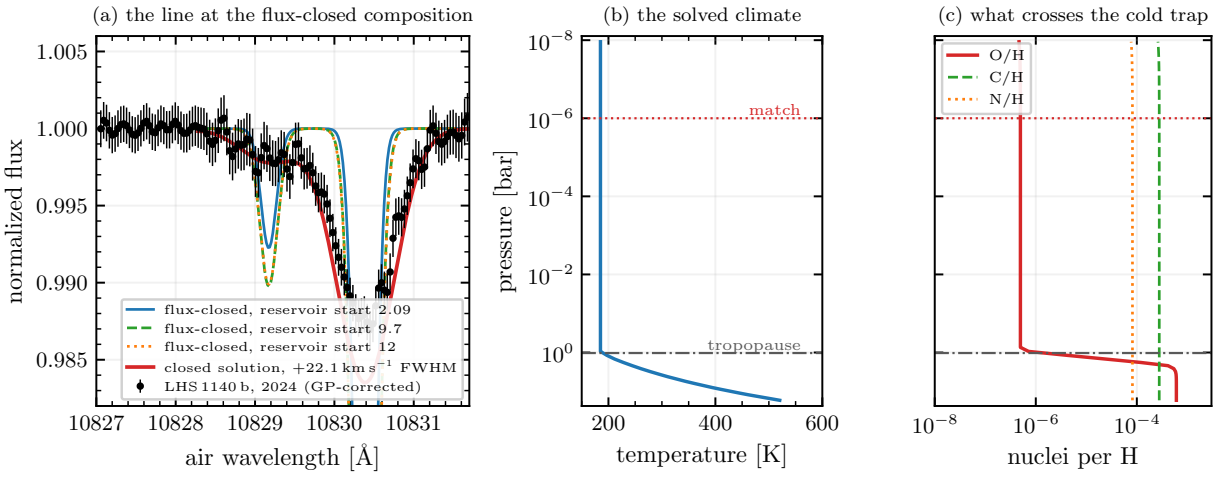


Figure 26: The flux-consistent solutions and the column they stand on. (a) Three of them (started at reservoirs 2.09, 9.7 and 12) against the measurement; the thick curve is the lowest of them convolved with the 22.1 km s^{-1} FWHM Gaussian that reproduces the observed width, which then lands *below* the observed flux because the equivalent width was already 33 per cent too large. Model curves are shifted by the fitted $v_{\rm wind} = +2.26$ km s^{-1}, as elsewhere in this memo. (b) The solved climate of the column handed over, with the tropopause at 1.031 bar and the 1 μbar matching level marked; the stratosphere sits at the skin temperature, 185 K. (c) The elemental abundances the profile carries. Oxygen is cut by a factor 1223 at the tropopause by the water cold trap, while carbon and nitrogen cross it unchanged. On the current C, N and O cooling those elements no longer weaken the line (Sect. 5.4); on the 2026-08-30 solutions they made it 2.6× weaker than the metal-free base at the same reservoir. Panels (b) and (c) place the column in the iteration; every column of the file itself is in Figs. 21 and 22.

16. Is the wind collisional where it is set?

Note added 2026-10-04. Changed. The 2026-09 solutions had the critical point at $40\,R_p$, outside the domain, and the line-forming layer at Kn ≤ 0.012. The 2026-10 states have the sonic point inside the domain (25.7–$28.8\,R_p$), and 31.4 per cent (He/H = 0.65, width matched) to 39.8 per cent (2.09, unbroadened) of the equivalent width forms beyond $\mathrm{Kn_{bulk}} = 0.1$ (Sect. 1.4).

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

Everything above reads the solutions as continuum solutions. A flow is a hydrodynamic wind only where the gas is collisional on the scale over which it varies, and in particular through its critical point, where the topology and with it the mass flux are fixed. This section measures that. The tool reads an existing run and changes nothing; the definitions, the tables and the record are listed in Appendix A. The numbers below were measured on the outputs the rest of this memo uses, and nothing was re-run.

16.1 What is measured

The collisions. The four momentum-transfer limits are the ones the binary element diffusion of Sect. 12.4 already uses, in the same Chapman–Enskog first approximation: rigid-sphere neutral-neutral, non-resonant ion-neutral with the induced-dipole and rigid-core frictions added, and screened Coulomb with the Spitzer $\ln\Lambda$ for ion-ion and ion-electron. Sharing them keeps a mean free path quoted here and a diffusion coefficient used by the wind from drifting apart. The step from a binary diffusion coefficient to a collision frequency is the definition of that coefficient itself, $\nu_{st} = n_t kT/(\mu_{st}\hat{D}_{st})$ with $\hat{D} = nD$, in which the total carrier density cancels. Species are H I, H II, He I, He II, He III and e, plus the molecules where a run tracks them; He I 2^3S is a level inside He I and is not counted twice, and metals are dropped as trace ($< 10^{-3}$ of the particles) except in the electron budget. Two channels are left out, resonant charge exchange and electron-neutral momentum transfer. Both lengthen the mean free path, so the verdict here is the conservative one.

The numbers. $\lambda_s = \bar{v}_s/\nu_s$ with the harmonic sum over partners; the local structure scale is the length over which the state vector of the local Maxwellian changes,

$$\frac{1}{L^2} = \left(\frac{\mathrm{d}\ln\rho}{\mathrm{d}r}\right)^2 + \left(\frac{\mathrm{d}\ln T}{\mathrm{d}r}\right)^2 + \left(\frac{|\mathrm{d}v/\mathrm{d}r|}{c_s}\right)^2 + \frac{1}{r^2}, \qquad L = \max\left(\tfrac{1}{\sqrt{\cdot}},\ \Delta r_{\mathrm{cell}}\right), \tag{10}$$

and $\mathrm{Kn}_s = \lambda_s/L$. The pressure is deliberately left out of it. The continuum closure expands about a local Maxwellian, which is fixed by (n, T, u); $p = nkT$ is a derived field, so including it would count the density and temperature gradients twice wherever they do not cancel and contribute nothing where they do. The second half of that is what matters here: across a heating front ρ falls and T rises with nearly the same logarithmic slope, so $|\mathrm{d}\ln p/\mathrm{d}r|$ passes through a broad near-zero while the gas is changing fast, and the earlier definition $L = \min(H_p, L_v, r)$ read a scale height fifteen times longer than the density and temperature scales there. The sum is taken in quadrature rather than as a minimum because a minimum jumps to the next length whenever the field it is taken from flattens; in quadrature a flat field contributes

zero, and L still cannot exceed the shortest individual scale. The velocity enters as $|\mathrm{d}v/\mathrm{d}r|/c_s$ and not $|(1/v)\mathrm{d}v/\mathrm{d}r|$ because the term the closure drops is the viscous stress, whose size relative to the pressure is $\sim \lambda|\mathrm{d}v/\mathrm{d}r|/\bar{v}$: a change in bulk velocity distorts the distribution in proportion to the thermal speed, not to the local bulk speed. That removes the Mach-number floor the old definition needed. The floor at the local cell width says that a gradient claiming structure below one cell is measuring the mesh; it binds in three cells of one of the four runs, all inside $1.002\,R_p$, and in no critical region. The *exobase* is the first crossing of $\mathrm{Kn}_{\mathrm{bulk}} = 1$ from below, never extrapolated beyond the domain. The *critical point* is the sonic point taken with the code's own sound speed, $c_s = \sqrt{\gamma p/\rho}$ at $\gamma = 5/3$, and not a redefinition. The *critical region* runs from the peak of volumetric heating to the sonic point. "Collisional" means $\max \mathrm{Kn} < 0.1$ across that region, over the bulk and over every species carrying at least 10^{-3} of the mass. Where it fails the solution is reported as unvalidated rather than as overestimated, because a continuum solution that assumed collisionality away cannot decide which way a collisionless critical region moves the answer.

16.2 The result

Table 23: Collisional validity of three representative LHS 1140 b solutions and of an HD 209458 b control, read from the advection-corrected profiles. Radii in R_p; the LHS 1140 b domain ends at $29.03\,R_p$, the HD 209458 b one at $4.15\,R_p$. "Max Kn" is over the critical region, that is, from peak heating to the sonic point, or to the outer boundary where there is no sonic point inside the domain. The HD 209458 b control has no counterpart among the LHS 1140 b models and is the archived 2026-08-30 run.

case	critical point	exobase	Kn = 0.1 at	max Kn	verdict
well mixed, He/H = 0.55	none in domain, $\mathcal{M}_{\max} = 0.34$	above 29.0	7.30	1.03	unvalidated
diffusion, He/H = 2.13	none in domain, $\mathcal{M}_{\max} = 0.34$	above 29.0	7.44	1.00	unvalidated
flux-consistent, He/H = 2.09	none in domain, $\mathcal{M}_{\max} = 0.37$	above 29.0	7.30	1.04	unvalidated
HD 209458 b control	4.085	above 4.15	-	**0.018**	validated

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

The wind of LHS 1140 b does reach a critical point, and the domain stops short of it. All three solutions stay subsonic out to $29\,R_p$, at $\mathcal{M}_{\max} = 0.34$–$0.37$, so inside the domain the mass flux is set by the outer boundary condition and not by a critical point. Extending the domain settles what that costs: at 45 and $60\,R_p$ the same case crosses its critical point at 40.06 and $40.08\,R_p$ and leaves the domain supersonic, and $\log_{10}\dot{M}$ is 7.8737, 7.8740 and 7.8735 at 30, 45 and $60\,R_p$, a spread of 5×10^{-4} dex. The escape rate is therefore not an artifact of where the boundary was put, but the profiles above $10\,R_p$ are: they move by 25 per cent in temperature, 29 in density and 22 in velocity at $28\,R_p$, and nothing is quoted from them.

What remains unvalidated is the *continuum* description out there. $\mathrm{Kn}_{\mathrm{bulk}}$ passes 0.1 at 7.30–7.44 R_p and reaches 0.68–0.71 at the top of the 30 R_p domain, and the exobase of the extended solutions sits at 29.0 R_p, *below* the critical point at 40.06. The continuum equations therefore fix their own topology in gas they cannot describe. The escape rates quoted throughout this memo, $\log_{10}\dot{M} \simeq 7.8$–$7.9$, are unvalidated in the collisional sense, and their insensitivity to the domain size is a statement about the discrete problem rather than a physical validation.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

The p-winds retrieval sits in the same regime. Its isothermal Parker solution places its sonic point at 8–9.5 R_p, where $\mathrm{Kn}_{\mathrm{bulk}}$ is 0.12–0.17, past the threshold. The critical region of LHS 1140 b is therefore *transitional*: it is not collisional, and it does not lie above the exobase either. The scale separation the continuum equations need has been lost by the radius at which either code sets the mass flux, while the gas is not yet free molecular.

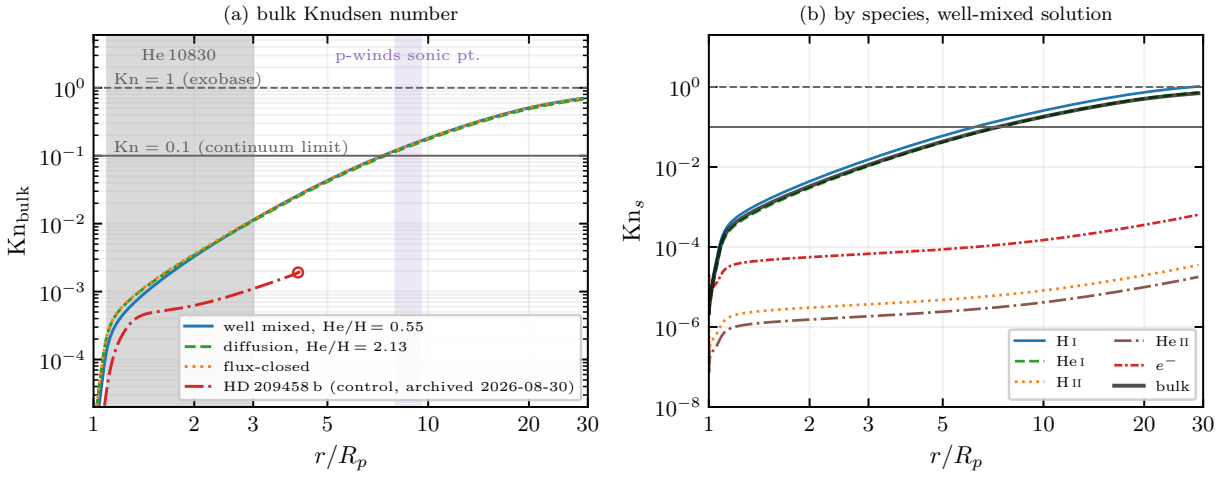


Figure 27: Where the continuum description holds. (a) The bulk Knudsen number of the three representative LHS 1140 b solutions and of the HD 209458 b control, whose sonic point at 4.085 R_p is the open circle; the shaded bands are the He 10830 line-forming region, 1.1–3 R_p, and the 8–9.5 R_p at which the p-winds retrieval places its sonic point. The LHS 1140 b solutions cross Kn = 0.1 at 7.30–7.44 R_p and reach 0.68–0.71 at the top of the domain, while the control stays below $\mathrm{Kn}_{\mathrm{bulk}} = 3 \times 10^{-3}$ everywhere and is collisional through its critical point, over which its largest Kn is 0.018 against 1.00–1.04 for the LHS 1140 b solutions, a factor of about 55. (b) The species split of the well-mixed solution: neutral hydrogen and neutral helium set the number, while the ions are held by Coulomb collisions four to five orders of magnitude more tightly and the electrons three.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn}_{\mathrm{bulk}} = 0.1$ radius moves out by 1.0–

5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; Kn(H_2) and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

Which species sets it. Neutral hydrogen decides the bulk number, and the line-forming layer is unaffected. In the critical region of the well-mixed solution the species split is H I 1.03, He I 0.73, H II 3.6×10^{-5}, He II 1.8×10^{-5}, He III 4.7×10^{-6}, e 6.5×10^{-4} (Fig. 27b). The neutrals carry the mass and the long mean free path, and the charged species are held by Coulomb collisions. That matters for reading the rest of this memo, because the He 10830 metastable population forms at 1.1–3 R_p, where $\mathrm{Kn_{bulk}}$ is 1.5–2.4 $\times 10^{-4}$ at 1.1 R_p, 0.0033–0.0036 at 2 R_p and 0.0110–0.0115 at 3 R_p in every case here. The line-forming layer is collisional by two orders of magnitude. The unvalidated quantity is the mass flux of the wind solution. The absorption that Sects. 11.–12. measure, and the 22 km s^{-1} of missing width, are asked for at radii where the continuum description is secure.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; Kn(H_2) and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

The energy equation. The single-temperature energy equation is not where the description fails. Electron-ion energy coupling is fast wherever it matters: the largest $\tau_{ei}^{E}/\tau_{\mathrm{heat}}$ over the critical region is 0.027, 0.024 and 0.026 in the three LHS 1140 b solutions and 8.3×10^{-5} in the control. Momentum coupling times are likewise short compared with the flow time. That is a different test, not a contradiction: a strongly subsonic flow gives hundreds of collisions per flow time even at Kn $\sim$ 1. The Knudsen number is what decides continuum validity, because the terms the closure drops (viscous stress, heat flux, the non-Maxwellian tail) scale with the gradient over a mean free path rather than with the collision count.

The kinetic scale. Jeans escape through the top of the domain (Chamberlain & Hunten 1987, eq. 7.2.5) gives 4.5–4.7×10^7 g s^{-1}, a factor 1.5–1.7 below the continuum $\dot{M}$. That is a *scale*, not a bound. Jeans escape bounds a continuum rate from below only when the exobase is well bound, and here $\lambda_J = 0.46$–0.48: the outer atmosphere is barely gravitationally bound, the Jeans integral is most of the Maxwellian rather than its escaping tail, and the atmosphere is in blow-off. What the number does say is that the continuum result sits within a factor of two of what a collisionless outer atmosphere at the same density and temperature would lose, so the failure of the assumption is not hiding an order of magnitude. On HD 209458 b, where $\lambda_J = 10.6$ and the solution is validated through its critical point, the hydrodynamic rate exceeds Jeans by 10^4: the signature of a driven wind rather than an evaporating atmosphere.

16.3 What this changes, and what it leaves open

- The domain stops inside the critical point, and the region it stops in is not collisional. The extended-domain runs put the critical point at 40.06 R_p and the exobase at 29.0, so the topology that sets the mass flux is fixed in gas the continuum equations do not describe. That the flux itself moves by 5×10^{-4} dex between 30, 45 and 60 R_p

bounds the discretization error and not the physical one. Any $\dot{M}$ quoted for LHS 1140 b from these solutions has to carry that warning, and so does the p-winds number, whose sonic point sits at Kn = 0.12–0.16.

- The outer profiles are not usable, and the residual cannot say so. At $28\,R_p$ the $30\,R_p$ solution stands 25 per cent too hot, 29 per cent too dense and 22 per cent too slow against the converged answer, at a residual that passes every convergence test. Nothing above about $10\,R_p$ in these profiles is quoted.
- The rate itself is not settled here, and cannot be. Settling it would take a kinetic or transitional-flow calculation (DSMC, or a 13-moment closure carried through the transitional region), and this repository contains neither. It is recorded as open, not estimated.
- The line analysis stands. The He 10830 layer is collisional by two orders of magnitude, so the equivalent-width matching, the composition inference conditional on it, and the invariant $22\,\mathrm{km\,s^{-1}}$ width deficit are unaffected by this section. It constrains what the escape rate may be quoted as, not what the line says.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; $\mathrm{Kn}(H_2)$ and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

17. Status against the goal (2026-09-16)

Note added 2026-10-04. The status of 2026-09-16. The current status is Sects. 1.–3.: the amount of absorption is not matched at any He/H on the present equations, and the width demand is unchanged.

The line is not reproduced. What stands:

- Amount of absorption: matched over a prescribed metal-free scalar base, and no longer matched over a solved photochemical column. Over the scalar base the answer is $\mathrm{He/H} = 1.5161$ at the adopted $K_{zz} = 10^9\,\mathrm{cm^2\,s^{-1}}$, and inside that branch it is further conditional: 0.0526 well mixed on the GJ 699 proxy, 0.4044 well mixed on the GJ 1132 proxy, 3.10 wherever $K_{zz} \leq 10^5$ leaves the molecular coefficient alone. Every one of those is read between two solved compositions. Over the photochemical column every flux-consistent solution from $\mathrm{He/H} = 2.09$ to 12.01 stands *above* the measurement, the lowest by 33 per cent, so that branch has no crossing (Sect. 15.7). The match, where it exists, is conditional: it constrains the composition only as far as the assumptions behind it are secure.
- Everything here was re-solved twice, and one of the two branches reversed. Against 2026-08-30 the crossings over the prescribed base moved by 2 to 19 per cent and $\dot{M}$ by about 0.1 dex, and the branch with the solved column went from making the line 2.6× weaker than the metal-free base at the same composition to making it 7.4 per cent stronger; the cause is the C, N and O line cooling, which was evaluated in the coronal limit above the critical density of the levels it excites and is now the CHIANTI statistical equilibrium at the electron density of each cell (Sect. 5.4). The first re-solve was needed because the 2026-08-30 states were never stationary

below $1.1\,R_p$, and it was possible because of the discretization that keeps a hydrostatic atmosphere hydrostatic. The second was needed because the lower boundary chose its entropy branch on the velocity in the first cell rather than on the mass flux through its face, so the campaign of 2026-09-14 carried a 418 K first cell where the prescribed reservoir was 226 K (Sect. 5.8). That correction takes the base cell down by a median 46 per cent in temperature, $\dot{M}$ down by 2 per cent everywhere, and the crossings *up* by 1.5 to 14 per cent with diffusion, and by 0.8 and 0.002 per cent in the two well-mixed groups.

- Line shape: not matched, by a factor 3.3 in depth and width alike, and not fixable by composition. The missing ingredient is $22.22\,\mathrm{km\,s^{-1}}$ FWHM of velocity spread ($\sigma = 9.44\,\mathrm{km\,s^{-1}}$), measured in Sect. 12.2 and Fig. 12; supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, but its physical origin is unidentified. The demand is unchanged under diffusion and the adopted K_{zz}: measured the same way on the diffusive solution matched on equivalent width it is the same 22.22 to four digits (Sect. 12.4); it is $22.15\,\mathrm{km\,s^{-1}}$ at the lowest flux-consistent solution (Sect. 15.7); and it is $21.97\,\mathrm{km\,s^{-1}}$ on the $5.1\times$ brighter proxy spectrum (Sect. 14.), 1.1 per cent of spread over four solutions that differ by a factor 40 in composition. The adiabatic metastable bump, which would have supplied it without a free parameter, is present and measured (Sect. 12.6) but sits too slow to contribute.

- The width shortfall is not a measurement artifact, and it is not this code's alone. Audited term by term, the instrument profile, the triplet splitting, stellar rotation, planetary rotation and orbital smearing account for at most $0.4\,\mathrm{km\,s^{-1}}$ of it; deconvolving both sides from their own instrument kernels puts the like-for-like requirement at $22.00\,\mathrm{km\,s^{-1}}$ FWHM (Sect. 12.3). The paper's own p-winds fit, measured the same way, is itself 1.5 times narrower than the width printed in the same paper, so the shortfall appears to be a property of one-dimensional steady-state wind models rather than of this one.

- Escape rate: an output, not a fit, and set by the assumed spectrum. $\log_{10}\dot{M} = 7.84$–7.87 on the GJ 1132 proxy, a factor 2.7 below the value the retrieval imposes, and nearly independent of composition, of diffusion, of the eddy coefficient and now also of what is placed at the lower boundary (the flux-consistent sequence gives 7.90–7.91). On the GJ 699 proxy, $5.1\times$ brighter in XUV, it is 8.55–8.57: within 0.06 dex of the value the paper retrieves on that same proxy (Sect. 14.).

- That escape rate is unvalidated in the collisional sense; the line is not. The solutions are the subsonic part of a transonic escape whose critical point sits at $40.06\,R_p$, outside the $30\,R_p$ domain, and whose exobase sits at $29.0\,R_p$, below the critical point. Moving the boundary to 45 and $60\,R_p$ changes $\log_{10}\dot{M}$ by 5×10^{-4} dex and the equivalent width by at most 1.7 per cent, but changes the profiles at $28\,R_p$ by 22–29 per cent, so nothing above about $10\,R_p$ is quoted. The p-winds sonic point at 8–$9.5\,R_p$ sits at Kn = 0.12–0.17, so the critical region of this planet is transitional for both codes. The He 10830 layer at 1.1–$3\,R_p$ is collisional by two orders of magnitude ($\mathrm{Kn} \leq 0.012$), and an HD 209458 b control validates cleanly at a factor 56 lower Kn (Sect. 16.). What the true rate is needs a kinetic or transitional-flow calculation, which this repository does not contain.

- Composition: limited by the base and by the assumed spectrum, and weakly limited by K_{zz}. In the well-mixed limit the equivalent-width crossing is $\mathrm{He/H} = 0.4044$ on one proxy spectrum and 0.0526 on the other, helium-

rich against helium-poor, with the measurement unable to separate them. When helium and hydrogen are allowed to separate by diffusion the crossing rises, and the whole K_{zz} axis has been walked: He/H = 3.10 on the plateau at $K_{zz} \leq 10^5$, then 2.89, 2.54, 1.97, 1.52, 1.16 and 0.89 at 10^6 through 10^{11} cm^2 s^{-1} (Table 9, Fig. 13), a factor 1.7 across the range the literature leaves open and 3.5 across the entire axis, against the factor 7.7 the spectrum carries. So any composition read off this line, the paper's included, is strongly conditional on the assumed XUV spectrum, and weakly conditional on the eddy coefficient, as $K_{zz}^{-0.115}$ wherever the eddy term acts at all.

- The lower boundary is still an axis, and it now points the other way. With the lower atmosphere solved as a photochemical column and the elemental fluxes iterated to continuity across the 1 μbar match, He/H at the match is a solution rather than an input, and at $K_{zz} = 10^9$ that solution is the well-mixed value because the eddy term reaches the match (Sect. 15.). What the column brings with it is C, N and O, and on the current cooling those elements cost the wind 164 K in $T_{\max}$ and the line 0.20 per cent of its equivalent width at fixed composition (Table 19). The flux-consistent sequence therefore sits above the measurement everywhere, and the composition this base would need lies below every reservoir solved.

- The 2025 non-detection sits at about 0.28 of the fiducial XUV over the scalar base; over the photochemical column it is not yet bracketed. Scaling the irradiation with the lower atmosphere held fixed, the red-pair depth falls below the 0.6 per cent limit of the 2025 epoch at $\simeq 0.283$ of fiducial in the scalar-base model, about 9 erg cm^{-2} s^{-1} (Sect. 15.9, Table 22). The flux-consistent grid is still at 1.001 per cent at 0.10, the faintest scaling at which it certifies, so its limit lies below 0.10 and is not bracketed: the three scalings between 0.10 and 0.30 certify with the Roe numerical flux, and its case at 0.01 does not. The scalar case at 0.01 certifies, reached by a continuation in XUV on 2026-09-21.

- Not verified: the cross-check with a second chemistry code. VULCAN did not converge on this cold helium-rich column, its convergence measure rose against the criterion with the timestep pinned, the limiting cell being H_2O at the cold trap, so no composition from a second code, and no spread between the two codes, is quoted (Sect. 15.).

- Partly solved: the molecular lower boundary. Three groups of models solve the molecular layer in EXHALE itself rather than taking it from the column, and two of their cases now converge, both with element diffusion at $K_{zz} = 10^9$: He/H = 2.13 at $\log_{10}\dot{M} = 7.91$ with a red-pair EW = 1.5758 % Å, and 9.7 at 7.95 with 2.4212 % Å. Both stand above the measurement, as the flux-consistent solutions do. These are the spectra of 2026-09-16; both cases have been run again since, and the spectra now on disk give 1.5604 and 2.4060 % Å (Sect. 5.12). The other cases of those groups have not converged, are a marching stop, or are still running at the time of writing, so no crossing is read from a molecular group here. What they wait on is the step that lets the transported H_2 settle.

Stale (2026-10-07). The Knudsen numbers, $\mathrm{Kn_{bulk}} = 0.1$ radii and exobase radii in this passage were computed before the resonant $H^+ + H$ and $He^+ + He$ momentum-transfer coefficients entered `collisional_validity.py` (Update_EXHALE_stage3 section 110). On the 102 stored LHS 1140 b states the $\mathrm{Kn_{bulk}} = 0.1$ radius moves out by 1.0–5.9 %, the exobase out by 0.4–5.7 % and the largest Kn down by 0–16 %; Kn(H_2) and the sonic point do not move and no verdict changes. Not re-measured for the states quoted here.

Four axes have now been walked: the assumed XUV spectrum, the mixing assumption with its eddy coefficient, composition itself, and what is placed at the lower boundary. Each separates the two questions rather than answering them together. All four move the composition the equivalent width implies, two of them by a large factor, and none of them moves the width. The possible sources of the missing $22\,\mathrm{km\,s^{-1}}$ are therefore unchanged: velocity fields that are three-dimensional, or heating a 1D hydrodynamic energy equation does not carry. The adiabatic-cooling metastable bump, the candidate that would have needed no free parameter, has been looked for on both spectra and does not supply it for this host (Sect. 12.6).

A Reproducing the runs: settings, files and records

This appendix carries the material the body deliberately leaves out: the input keys that switch each physical option on, the directories and scripts that hold the runs, the environment variables the analysis reads, and the working notes each measurement is recorded in. Everything below lives under the repository root, and the model directories under `LHS1140b/`.

A1 The 2026-10 campaign

The runs of Sects. 1.–3. live under `LHS1140b/models/.P2/ew_fit/`, each directory with its own `NOTES.md`: `phase3/he23s_transport/` (the He/H = 2.09 reference and its acceptance), `phase4/` (outer domain, Roche geometry), `phase5/` (the He/H scan, `HeH<v>/`; the width kernel, `nonthermal_kernel/`; the column transfer, `column_transfer/`; the decomposition, `dEW_decomposition/`) and `phase6/` (grid, tolerance, closure, domain, Roche, drift, NH_3 + H_2, observation). The accepted He/H = 0.65 state is `phase5/HeH0.65/k02`, the tight closure `phase6/closure_direct/k01`. Binary `LHS1140b/models/.P2/EXHALE_99a76cb0.x`, manifest `.P2/BINARY_MANIFEST_99a76cb05e56.txt`. Equivalent widths come from `.P1/fiducial_conduction/ew_of_state.sh` with `EW_NONTHERMAL_KERNEL=width_match`. The records are `md/PLAN_20261003_lhs1140b_ew.md` (sections 8 and 9) and `md/Update_EXHALE_stage3.md` sections 63–92. The four figures of those sections are copies of the run-directory PNGs named in their captions.

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Table 24: The input keys that decide the physics of these runs, and what each one does.

key	what it does
`Base BC: pressure 1.0`	Anchors the lower boundary on a pressure of 1 μbar and derives the base density from it (Sect. 6.2).
`Log10 lower boundary number density`	The base number density, which the pressure anchor re-derives and enforces.
`Well balanced: True`	Selects the discretization that keeps a hydrostatic atmosphere exactly hydrostatic. Without it no case here converges.
`Low-Mach damping`	A fourth difference in the velocity; tested and insufficient at this base.
`Secondary_ionization: Immediate`	Applies the Shull & van Steenberg (1985) photoelectron channel from the first evaluation instead of waiting for a converged march.
`Restart intent: stationary`	Solves directly for the steady state instead of marching in time.
`Spectrum file:`	The stellar spectrum; its integral overrides the luminosities stated elsewhere in the input.
`He_diffusion`, `He_Kzz`	Turn binary H/He element diffusion on and set the eddy coefficient, when it is a single number.
`Lower atmosphere profile:`	Hands the lower atmosphere over as a full profile, including its eddy coefficient, in place of the single base numbers; a scalar `He_Kzz` is then inert.
`Molecular base`	Lets the H_2 fraction of the handoff enter the equation of state at the base. Off in every run of this memo.
`Numerical flux:`	Chooses the flux between cells; the seven lowest XUV cases use Roe, the rest HLLC (Sect. 15.9).
`EXHALE_PTC_DTAU0`	The artificial time step the Newton solver starts from; 10^8 for the low-XUV cases.
`EXHALE_DIFFUSION_CHECK=1`	Writes the table of fluxes through cell faces that Fig. 14 reads the homopause from.
`EXHALE_TRANSIT_TURB=1`	Adds the sonic turbulence term to the transit synthesis; off by default.

Table 25: Where the runs, the tools and the records live.

path	contents
`LHS1140b/models/`	The models of record, solved 2026-09-16 on the build whose md5 begins `c2e9c9990b9f`. Group names carry the spectrum, the eddy coefficient and the base: `atomic_scalar_gj1132_wellmixed`, `atomic_scalar_gj1132_kzz*`, `atomic_scalar_gj699_wellmixed`, `atomic_photochem_gj1132_kzzprofile`, `gj1132x*` for the XUV grid.
`LHS1140b/models_20260914_preL21/`	The superseded campaign of 2026-09-14, kept unchanged; `models_20260915_db87/` is the intermediate generation.
`LHS1140b/archive_20260830/`	The 2026-08-30 runs, from which the subsections marked as archived are read.
`LHS1140b/lower_profile/lower_atmosphere_profile.dat`	The photochemical column handed over at iteration 0, read column by column in Sect. 15.2.
`LHS1140b/lower_profile_vulcan/`	The VULCAN attempt that did not converge.
`LHS1140b/LHS1140b_analysis.ipynb`	Regenerates every number and figure in this memo.
`LHS1140b/make_memo_figures.py`	The convolution and the three-Gaussian extractor used for every model spectrum here.
`LHS1140b/he10830_equivalent_width.py`	The window and the measurement operator of every equivalent width here, observed and modeled alike (Sect. 11.; since 2026-09-23).
`he_line_metrics.py`	The three-Gaussian extractor applied to the released spectrum and to every model.
`LHS1140b/winered_hires_y.sh`	Sets the resolving power to WINERED's 68,000 for the transit synthesis.
`LHS1140b/kzz_decision.md`	The literature range for K_{zz} at this base and the adoption of 10^9 cm^2 s^{-1}.
`LHS1140b/bump_analysis.py`	Reconstructs the metastable balance and measures the adiabatic-cooling bump.
`src/utils/collisional_validity.py`	Reads a run and computes the Knudsen numbers of Sect. 16..
`EXHALE_transit.py`	The transit synthesis; its built-in resolving power is CARMENES, overridden here.

Table 26: The working notes each measurement is recorded in.

record	what it holds
`docs/well_balanced_and_hydrostatic_20260913.tex`	The hydrostatic-exact discretization, term by term.
`md/collisional_validity.md`	The definitions and tables behind Sect. 16..
`docs/eddy_diffusion_kzz.tex`	The survey of eddy coefficients and the note that the adopted 10^9 has no planet-specific basis.
`docs/photochem_solver_modification_*.md`	The local corrections applied to Photochem 0.9.0.
`docs/Update_EXHALE_stage1.pdf`	The running update log; Sects. 72, 78, 79, 81, 83, 85, 87, 96 and 99 carry the changes cited in this memo.
`references/`	The 43 papers swept for the broadening review of Sect. 8.1.

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